

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\Hot Exchanger-AK.pvdb

Date of Analysis : Sep 3,2026 4:25pm

PV Elite 28, January 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 1700.000 mm. OD
 Vessel Design Length, Tangent to Tangent 12910.00 mm.
 Distance of Bottom Tangent above Grade 0.00 mm.
 Distance of Base above Grade 0.00 mm.
 Specified Datum Line Distance 0.00 mm.
 Shell Side Design Temperature 260 °C
 Channel Side Design Temperature 260 °C
 Shell Side Design Pressure 191.000 bars
 Channel Side Design Pressure 191.000 bars
 Shop Shell Side Test Pressure 99.164 bars
 Field Shell Side Test Pressure 98.564 bars
 Shop Channel Side Test Pressure 93.244 bars
 Field Channel Side Test Pressure 92.564 bars
 Wind Design Code ASCE/SEI 7-22
 Earthquake Design Code ASCE/SEI 7-22

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-387 11	2	...	K11789	+IT
Head	SA-387 11	2	...	K11789	+IT
Skirt	SA-387 11	2	...	K11789	
Nozzle	SA-336 F11	3	...	K11572	+IT
Tubes	SA-213 T11	...	...	K11597	
Tubesheet	SA-336 F11	3	...	K11572	
Basing	SA-387 11	2	...	K11789	
Base Bolting	SA-36	...	...	K02600	+IT

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (bars & mm.):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Bottom Dish end	191.122	0.00	193.000	1.5000	N/A	No
Main Shell-2	191.039	0.00	193.000	1.5000	N/A	No
Skirt-1	...	...	...	1.5000	N/A	No
Main Shell-1	191.041	0.00	193.000	1.5000	N/A	No
Top Dish End	191.005	0.00	193.000	1.5000	N/A	No

Liquid Level: 14374.00 mm. Dens.: 0.000 kg./cm³ Sp. Gr.: 0.060

Element Types and Properties:

Element Type	"To" Elev mm.	Element Length mm.	Nominal Thickness mm.	Finished Thickness mm.	Reqd Thk Internal mm.	Reqd Thk External mm.	Long Eff	Circ Eff
Sphere	0.0	0.0	71.0	60.0	59.3	...	1.00	1.00
Cylinder	6700.0	6550.0	122.0	122.0	120.6	...	1.00	1.00
Skirt	6700.0	4000.0	10.0	10.0	...	...	1.00	1.00
Cylinder	12910.0	6060.0	122.0	122.0	120.7	...	1.00	1.00
Sphere	12910.0	0.0	71.0	60.0	59.3	...	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
A1-Tube inlet	WRC-107/537	0.978	n/a	Passed
B2-Shell outle	WRC-297	0.980	Shell	Passed
B3-Shell outle	WRC-297	0.980	Shell	Passed
A2-Shell inlet	WRC-297	0.980	Shell	Passed
A3-Shell inlet	WRC-297	0.980	Shell	Passed
B1-Tube outlet	WRC-107/537	0.977	n/a	Passed

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm.	Cumulative Wind Shear Kg	Earthquake Shear Kg	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	3247.98	...	...	...	...
20	30	725	199.259	0.0016835	109.191	0.00092251
30	40	2000	5680.29	16619.4	19721.9	81388.9
40	50	7030	2156.18	10355.5	7366.68	34214.4
50	60	10464	242.658	825.999	98.0492	333.756

Abs Max of the all of the Stress Ratio's : 0.9487

Basing Data : Continuous Top Ring W/Gussets

Thickness of Basing	13.0000	mm.
Inside Diameter of Basing	3276.0000	mm.
Outside Diameter of Basing	3582.0000	mm.
Nominal Diameter of Bolts	24.0000	mm.
Diameter of Bolt Circle	3492.0000	mm.
Number of Bolts	24	
Thickness of Gusset Plates	10.0000	mm.
Average Width of Gusset Plates	96.0000	mm.
Height of Gussets	117.0000	mm.
Distance between Gussets	78.0000	mm.
Thickness of Top Plate or Ring	42.0000	mm.
Radial Width of the Top Plate	117.0000	mm.

Loads for Foundation/Support Design:

Total Wind Shear on Support	5680.	Kgf
Total Earthquake Shear on Support	16619.	Kgf
Wind Moment on Support	19722.	Kg-m.
Earthquake Moment on Support	81389.	Kg-m.
Wind Load Case Test Moment on Support	6508.	Kg-m.
Test Case Wind Shear on Support	1874.	Kgf

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	151005.2	kg.
Shop Test - Fabricated + water (full)	182827.8	kg.
Shipping - Fab. + rem. intls.+ shipping app.	151030.2	kg.
Erected - Fab. + rem. intls.+ insul. (etc)	151030.2	kg.
Empty - Fab. + intls. + details + wghts.	151030.1	kg.
Operating - Empty + operating liquid (no ca)	152939.4	kg.
Field Test - Empty weight + test medium (full)	182852.8	kg.

Nozzle Calculation Summary (bars & mm.):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
A1-Tube inlet	193	...	...	OK	36.05	OK	Passed
B2-Shell outlet	193	...	...	OK	24.48	OK	Passed
B3-Shell outlet	193	...	...	OK	24.48	OK	Passed
A2-Shell inlet	192	...	...	OK	24.48	OK	Passed
A3-Shell inlet	192	...	...	OK	24.48	OK	Passed
B1-Tube outlet	193	...	...	OK	36.03	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 193 Nozzle : B2-Shell outlet [Shellside]
 Minimum MAWP Nozzles : 192 Nozzle : A2-Shell inlet [Tubeside]

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm.	Layout Angle deg	Dia. Limit mm.
10	A1-Tube inlet	0.000	0.000	987.151
20	B2-Shell outlet	950.000	180.000	656.951
20	B3-Shell outlet	950.000	360.000	656.951
40	A2-Shell inlet	12349.999	180.000	656.951
40	A3-Shell inlet	12349.999	360.000	656.951
50	B1-Tube outlet	0.000	0.000	987.151

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia mm	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
B2-Shell outlet	400 mm	160	None	406.4	40.462	...	...	388	...
B3-Shell outlet	400 mm	160	None	406.4	40.462	...	...	388	...
A2-Shell inlet	400 mm	160	None	406.4	40.462	...	...	388	...
A3-Shell inlet	400 mm	160	None	406.4	40.462	...	...	388	...
A1-Tube inlet	600 mm	160	None	609.6	59.512	...	...	353	...
B1-Tube outlet	600 mm	160	None	609.6	59.512	...	...	353	...

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm.):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
B2-Shell ou	SA-336 F11	122.000	10.000	...	...	...
B3-Shell ou	SA-336 F11	122.000	10.000	...	...	...
A2-Shell in	SA-336 F11	122.000	10.000	...	...	...
A3-Shell in	SA-336 F11	122.000	10.000	...	...	...
A1-Tube inl	SA-336 F11	58.000	10.000	...	...	...
B1-Tube out	SA-336 F11	58.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm.	Layout Angle deg	Proj Outside mm.	Proj Inside mm.	Installed in Component
B2-Shell outlet	950.000	180.0	200.00	0.00	Main Shell-2
B3-Shell outlet	950.000	360.0	200.00	0.00	Main Shell-2
A2-Shell inlet	12349.999	180.0	200.00	0.00	Main Shell-1
A3-Shell inlet	12349.999	360.0	200.00	0.00	Main Shell-1
A1-Tube inlet	...	0.0	200.00	0.00	Bottom Dish end
B1-Tube outlet	...	0.0	200.00	0.00	Top Dish End

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	SPHERICAL HEAD: 71.0mm. THK X 1764.0mm. ID X 0.0mm.	SA-387 11
1	INSULATION: 850mm X 10mm THK	Hot Insulation
2	TUBESHEET: 1944.0mm X 150.0mm THK	SA-336 F11
5139	TUBES: 12910.0mm X 12.7mm DIA X 1.6mm THK	SA-213 T11
1	INSULATION: 882mm X 10mm THK	Hot Insulation
1	BASERING: 3582.0mm. OD X 3276.0 ID X 13.0mm.	SA-387 11
24	BASERING BOLTS: 24.0mm. DIA	SA-36
24	BASERING BOLT NUTS: 24.0mm. DIA	...
1	BASERING TOP RING: 2197.7mm. OD X 1963.7 ID X 42.0mm.	...
48	BASERING GUSSET PLATES: 96.0mm. X 96.0mm. X 117.0mm. X 10.0mm.	...
1	CYLINDRICAL SEGMENT 122.0mm. THK X 3048.0mm. X 6550.0mm.	SA-387 11
1	CYLINDRICAL SEGMENT 122.0mm. THK X 2292.7mm. X 6550.0mm.	SA-387 11
1	CON SKIRT SEGMENT 10.0mm. THK X 3484.9mm. X 4000.0mm. X 6041.5mm	SA-387 11
1	CON SKIRT SEGMENT 10.0mm. THK X 2621.3mm. X 4000.0mm. X 4544.4mm	SA-387 11
1	CYLINDRICAL SEGMENT 122.0mm. THK X 3041.2mm. X 6060.0mm.	SA-387 11
1	CYLINDRICAL SEGMENT 122.0mm. THK X 2299.4mm. X 6060.0mm.	SA-387 11
1	NAMEPLATE	...

Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm.	E*	PWHT reqd
Notes								
Main Shell-2 [20]	!	-10	-53		0.436	122.000	1.00	Yes
B2-Shell outle[1]	!	-10	-11		0.989	122.000	1.00	Yes
B3-Shell outle[1]	!	-10	-11		0.989	122.000	1.00	Yes
Bottom Dish en[9]	!	-10	-75		0.368	60.000	1.00	Yes
Main Shell-1 [20]	!	-10	-54		0.430	122.000	1.00	Yes
Top Dish End [9]	!	-10	-75		0.368	60.000	1.00	Yes
A1-Tube inlet [1]	!	-10	-11		0.988	60.000	1.00	Yes
A2-Shell inlet[1]	!	-10	-11		0.989	122.000	1.00	Yes
A3-Shell inlet[1]	!	-10	-11		0.989	122.000	1.00	Yes
B1-Tube outlet[1]	!	-10	-11		0.987	60.000	1.00	Yes
Tubesheet: CS[14]	B	24	24		1.000	71.000	1.00	Yes
Warmest MDMT:		24	24					

Exchanger Side	Computed MDMT	Required MDMT	Pass/Fail
Channel/Tube	24.0	-10.0	Fail

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.
 UCS-66(g) was not considered.
 UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
 UCS-66(i) applies to impact tested materials not by specification and
 UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
 The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (bars):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm.
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0689	bars
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm.
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm.
Thickness	inches	25.4000	mm.

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	191	bars
Channel Side Design Pressure	191	bars
Shell Side Design Temperature	260.0	°C
Channel Side Design Temperature	260.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	Lethal	
Service Type, Channel Side	Lethal	
MDMT (CET), Shell Side	-10.0	°C
MDMT (CET), Tube Side	-10.0	°C
User defined MAWP, Shell Side	0	bars
User defined MAWP, Channel Side	0	bars
User defined MAPnc, Shell Side	0	bars
User defined MAPnc, Channel Side	0	bars
User defined Test Pres., Shell Side	0	bars
User defined Test Pres., Channel Side	0	bars

Projection of Nozzle from Vessel Top	0	mm.
Projection of Nozzle from Vessel Bottom	0	mm.
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Vertical
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes



Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-22
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	180 Km/hr
Surface Roughness Category	D: Flat, unobstructed
Type of Surface	0
Base Elevation	0 mm.
Percent Wind for Hydrotest	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Seismic Design Code	ASCE/SEI 7-22
Importance Factor	1.500
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Component resonance duct. fact. [CAR]	0.000
Component strength factor [Rpo]	0.000
Overstrength factor [Omega0]	0.000
Response Acceleration Factor [SMS]	0.326
Response Acceleration Factor [SM1]	0.139
Building Importance Factor [Ie]	1.000
Site Latitude	0.000 deg
Site Longitude	0.000 deg

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
--	-----



Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/5
Element From Node	10
Element To Node	20
Element Type	Spherical
Description	Bottom Dish end
Distance "FROM" to "TO"	0 mm.
Inside Diameter	1764 mm.
Element Thickness	60 mm.
Internal Corrosion Allowance	1.5 mm.
Nominal Thickness	71 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	191 bars
Design Temperature Internal Pressure	260 °C
Design External Pressure	0 bars
Design Temperature External Pressure	260 °C
Effective Diameter Multiplier	1.2
Material Name	SA-387 11 +IT
Allowable Stress, Ambient	148 N./mm ²
Allowable Stress, Operating	148 N./mm ²
Allowable Stress, Hydrotest	278.99 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	16.002 mm.
Yield Stress, Operating	251.4 N./mm ²
UCS-66 Chart Curve Designation	Impact Tested
External Pressure Chart Name	CS-3
UNS Number	K11789
Class / Thickness / Grade	2 ::
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	10
Detail Type	Liquid
Detail ID	Liquid: 10
Dist. from "FROM" Node / Offset dist	-882 mm.
Height/Length of Liquid	882 mm.
Liquid Density	0.59900E-04 kg./cm ³
Element From Node	10
Detail Type	Insulation
Detail ID	Hot Insulation
Dist. from "FROM" Node / Offset dist	-850 mm.
Height/Length of Insulation	850 mm.
Thickness of Insulation	10 mm.
Density	0.0001918 kg./cm ³
Element From Node	10
Detail Type	Nozzle
Detail ID	A1-Tube inlet
Dist. from "FROM" Node / Offset dist	0 mm.
Nozzle Diameter	600 mm
Nozzle Schedule	160
Nozzle Class	None



Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	329.49	Kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-336 F11 +IT	

Element#	2/5
Element From Node	20
Element To Node	30
Element Type	Cylinder
Description	Main Shell-2
Distance "FROM" to "TO"	6550 mm.
Inside Diameter	1700 mm.
Element Thickness	122 mm.
Internal Corrosion Allowance	1.5 mm.
Nominal Thickness	122 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	191 bars
Design Temperature Internal Pressure	260 °C
Design External Pressure	0 bars
Design Temperature External Pressure	260 °C
Effective Diameter Multiplier	1.2
Material Name	SA-387 11 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	20
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	6550 mm.
Liquid Density	0.59900E-04 kg./cm³

Element From Node	20
Detail Type	Nozzle
Detail ID	B2-Shell outlet
Dist. from "FROM" Node / Offset dist	800 mm.
Nozzle Diameter	400 mm
Nozzle Schedule	160
Nozzle Class	None
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	588.81 Kgf
Grade of Attached Flange	None
Nozzle Matl	SA-336 F11 +IT

Element From Node	20
Detail Type	Nozzle
Detail ID	B3-Shell outlet
Dist. from "FROM" Node / Offset dist	800 mm.
Nozzle Diameter	400 mm
Nozzle Schedule	160
Nozzle Class	None
Layout Angle	360.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	588.81 Kgf
Grade of Attached Flange	None
Nozzle Matl	SA-336 F11 +IT

Element From Node	20
Detail Type	Weight
Detail ID	Datasheet diffe
Dist. from "FROM" Node / Offset dist	3500 mm.
Miscellaneous Weight	12770 Kgf
Offset from Element Centerline	0 mm.

Element From Node	20
-------------------	----



Detail Type	Weight
Detail ID	Design Margin
Dist. from "FROM" Node / Offset dist	0 mm.
Miscellaneous Weight	5000 Kg
Offset from Element Centerline	0 mm.

Element#	3/5
Element From Node	30
Element To Node	40
Element Type	Skirt Support
Description	Skirt-1
Distance "FROM" to "TO"	4000 mm.
Skirt Outside Diameter	1964 mm.
Diameter of Skirt at Base	3390 mm.
Skirt Thickness	10 mm.
Internal Corrosion Allowance	1.5 mm.
Nominal Thickness	10 mm.
External Corrosion Allowance	0 mm.
Design Temperature Internal Pressure	260 °C
Design Temperature External Pressure	260 °C
Effective Diameter Multiplier	1.2
Material Name	SA-387 11
Efficiency, Longitudinal Seam	1.0
Efficiency, Head-to-Skirt or Circ. Seam	1.0
Weld is pre-Heated	No

Element#	4/5
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	Main Shell-1
Distance "FROM" to "TO"	6060 mm.
Inside Diameter	1700 mm.
Element Thickness	122 mm.
Internal Corrosion Allowance	1.5 mm.
Nominal Thickness	122 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	191 bars
Design Temperature Internal Pressure	260 °C
Design External Pressure	0 bars
Design Temperature External Pressure	260 °C
Effective Diameter Multiplier	1.2
Material Name	SA-387 11 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	40
Detail Type	Liquid
Detail ID	Liquid: 50
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	6060 mm.
Liquid Density	0.59900E-04 kg./cm ³

Element From Node	40
Detail Type	Nozzle
Detail ID	A2-Shell inlet
Dist. from "FROM" Node / Offset dist	5500 mm.
Nozzle Diameter	400 mm
Nozzle Schedule	160
Nozzle Class	None
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	646.55 Kg
Grade of Attached Flange	None
Nozzle Matl	SA-336 F11 +IT



Element From Node 40
 Detail Type Nozzle
 Detail ID A3-Shell inlet
 Dist. from "FROM" Node / Offset dist 5500 mm.
 Nozzle Diameter 400 mm
 Nozzle Schedule 160
 Nozzle Class None
 Layout Angle 360.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 646.55 Kg
 Grade of Attached Flange None
 Nozzle Matl SA-336 F11 +IT

Element From Node 40
 Detail Type Weight
 Detail ID Datasheet diffe
 Dist. from "FROM" Node / Offset dist 3000 mm.
 Miscellaneous Weight 12770 Kg
 Offset from Element Centerline 0 mm.

Element From Node 40
 Detail Type Weight
 Detail ID Design Margin
 Dist. from "FROM" Node / Offset dist 0 mm.
 Miscellaneous Weight 5000 Kg
 Offset from Element Centerline 0 mm.

Element# 5/5
 Element From Node 50
 Element To Node 60
 Element Type Spherical
 Description Top Dish End
 Distance "FROM" to "TO" 0 mm.
 Inside Diameter 1764 mm.
 Element Thickness 60 mm.
 Internal Corrosion Allowance 1.5 mm.
 Nominal Thickness 71 mm.
 External Corrosion Allowance 0 mm.
 Design Internal Pressure 191 bars
 Design Temperature Internal Pressure 260 °C
 Design External Pressure 0 bars
 Design Temperature External Pressure 260 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-387 11 +IT
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 50
 Detail Type Liquid
 Detail ID Liquid: 60
 Dist. from "FROM" Node / Offset dist 0 mm.
 Height/Length of Liquid 882 mm.
 Liquid Density 0.59900E-04 kg./cm³

Element From Node 50
 Detail Type Insulation
 Detail ID Hot Insulation
 Dist. from "FROM" Node / Offset dist 0 mm.
 Height/Length of Insulation 882 mm.
 Thickness of Insulation 10 mm.
 Density 0.0001918 kg./cm³

Element From Node 50
 Detail Type Nozzle
 Detail ID B1-Tube outlet
 Dist. from "FROM" Node / Offset dist 0 mm.

PV Elite 28 Licensee: SL User
FileName : Hot Exchanger-AK
Input Echo:

Step: 1 4:25pm Sep 3,2026



HEXAGON

Nozzle Diameter	600	mm
Nozzle Schedule	160	
Nozzle Class	None	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	484.27	Kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-336 F11	+IT

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XY Coordinate Calculations:

From	To	X (Horiz.) mm.	Y (Vert.) mm.	DX (Horiz.) mm.	DY (Vert.) mm.
Bottom Dish end		...	...	...	...
Main Shell-2		...	6700	...	6550
Skirt-1		...	6700	...	-4000
Main Shell-1		...	12910	...	6060
Top Dish End		...	12910	...	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd bars	Nominal Thickness mm.	Total Corr Allowance mm.	Element Diameter mm.	Allowable Stress(SE) N./mm ²	Element#
Bottom Dish end		191.12	71	1.5	1764	148	1
Main Shell-2		191.04	122	1.5	1700	148	2
Skirt-1		...	10	1.5	1964	...	3
Main Shell-1		191.04	122	1.5	1700	148	4
Top Dish End		191.01	71	1.5	1764	148	5

Element Required Thickness and MAWP :

From	To	Design Pressure bars	M.A.W.P. Corroded bars	M.A.P. New & Cold bars	Minimum Thickness mm.	Required Thickness mm.	Element#
Bottom Dish end		191	193.295	198.642	60	59.2969	1
Main Shell-2		191	192.997	195.565	122	120.649	2
Skirt-1		...	No Calc	No Calc	10	No Calc	3
Main Shell-1		191	192.994	195.565	122	120.65	4
Top Dish End		191	193.411	198.642	60	59.2612	5

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

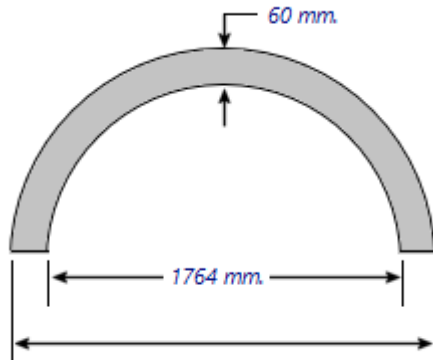
Shell Side MAWP = 76 bars
 Shell Side MAPnc = 76 bars
 Channel Side MAWP = 72 bars, < design pressure: 191.000 bars
 Channel Side MAPnc = 72 bars

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Bottom Dish end

Spherical Head Design Information:					
Design Pressure	P	191.122 bars	Int. Design Temperature		260 °C
Material		SA-387 11	Ext. Design Temperature		260 °C
Length		0.000 mm.	External Pressure Chart		CS-3
			UNS Number		K11789
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	148.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	148.0 N./mm ²
Inside Diameter	Di	1764.000 mm.	Actual Stress		146.2 N./mm ²
Spec. Min. Thk t,e, or sa		60.0000 mm.	Spec. Nominal Thk.		71.000 mm.
Surface Area		57064.5 cm ²	SG of Contents		0.060
Element Volume		1437282.9 cm ³	Weight of Contents		86.2 Kg
Empty Weight		2912.0 Kg	Operating Weight		2998.2 Kg
			Comp. All. Ope. Stress		118.5 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 22.050 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 11.025 mm.

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:
 $= (P \cdot R) / (2 \cdot S \cdot E - 0.2 \cdot P)$ per UG-27 (d)
 $= (191.122 \cdot 883.5001) / (2 \cdot 148.0 \cdot 1.0 - 0.2 \cdot 191.122)$
 $= 57.7969 + 1.5000 = 59.2969 \text{ mm.}$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.122 bars
 $= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t)$ per UG-27 (d)
 $= (2 \cdot 148.0 \cdot 1.0 \cdot 58.5) / (883.5001 + 0.2 \cdot 58.5)$
 $= 193 - 0 = 193 \text{ bars}$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t)$ per UG-27 (d)
 $= (2 \cdot 148.0 \cdot 1.0 \cdot 60.0) / (882.0 + 0.2 \cdot 60.0)$
 $= 199 \text{ bars}$



Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned} &= (P \cdot (R + 0.2 \cdot t)) / (2 \cdot E \cdot t) \\ &= (191.122 \cdot (883.5001 + 0.2 \cdot 58.5)) / (2 \cdot 1.0 \cdot 58.5) \\ &= 146.241 \text{ N./mm}^2 \end{aligned}$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 5.804 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Spherical Portion:

Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 22.050 mm.

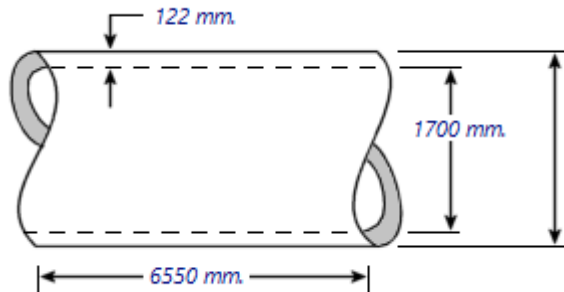
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 11.025 mm.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Main Shell-2

Cylindrical Shell Design Information:					
Design Pressure	P	191.039 bars	Int. Design Temperature	260 °C	
Material		SA-387 11	Ext. Design Temperature	260 °C	
Length		6550.000 mm.	External Pressure Chart	CS-3	
			UNS Number	K11789	
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	148.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	148.0 N./mm ²
Inside Diameter	Di	1700.000 mm.	Actual Stress	146.5 N./mm ²	
Spec. Min. Thk t,e, or sa		122.0000 mm.	Spec. Nominal Thk.	122.000 mm.	
Surface Area		400025.3 cm ²	SG of Contents	0.060	
Element Volume		10605096.0 cm ³	Weight of Contents	635.9 Kg	
Empty Weight		35451.1 Kg	Operating Weight	36086.9 Kg	
			Comp. All. Ope. Stress	130.1 N./mm ²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (191.039 \cdot 851.5001) / (148.0 \cdot 1.0 - 0.6 \cdot 191.039)$$

$$= 119.1489 + 1.5000 = 120.6489 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.039 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (148.0 \cdot 1.0 \cdot 120.5) / (851.5001 + 0.6 \cdot 120.5)$$

$$= 193 - 0 = 193 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (148.0 \cdot 1.0 \cdot 122.0) / (850.0 + 0.6 \cdot 122.0)$$

$$= 196 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (191.039 \cdot (851.5001 + 0.6 \cdot 120.5)) / (1.0 \cdot 120.5)$$

$$= 146.466 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 6.696 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.



Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

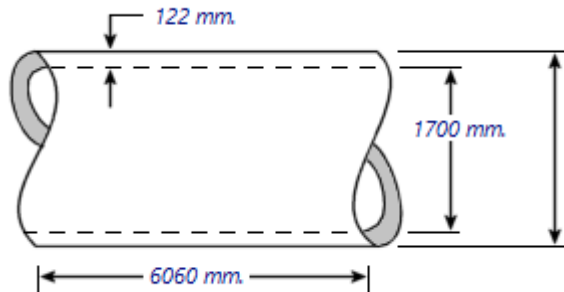
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Main Shell-1

Cylindrical Shell Design Information:					
Design Pressure	P	191.041 bars	Int. Design Temperature		260 °C
Material		SA-387 11	Ext. Design Temperature		260 °C
Length		6060.000 mm.	External Pressure Chart		CS-3
			UNS Number		K11789
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	148.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	148.0 N./mm ²
Inside Diameter	Di	1700.000 mm.	Actual Stress		146.5 N./mm ²
Spec. Min. Thk t _e , or s _a		122.0000 mm.	Spec. Nominal Thk.		122.000 mm.
Surface Area		370099.8 cm ²	SG of Contents		0.060
Element Volume		13757460.0 cm ³	Weight of Contents		824.9 Kg
Empty Weight		32799.0 Kg	Operating Weight		33623.9 Kg
			Comp. All. Ope. Stress		130.1 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (191.041 \cdot 851.5001) / (148.0 \cdot 1.0 - 0.6 \cdot 191.041)$$

$$= 119.1504 + 1.5000 = 120.6504 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.041 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (148.0 \cdot 1.0 \cdot 120.5) / (851.5001 + 0.6 \cdot 120.5)$$

$$= 193 - 0 = 193 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (148.0 \cdot 1.0 \cdot 122.0) / (850.0 + 0.6 \cdot 122.0)$$

$$= 196 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (191.041 \cdot (851.5001 + 0.6 \cdot 120.5)) / (1.0 \cdot 120.5)$$

$$= 146.468 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 6.696 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.



Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

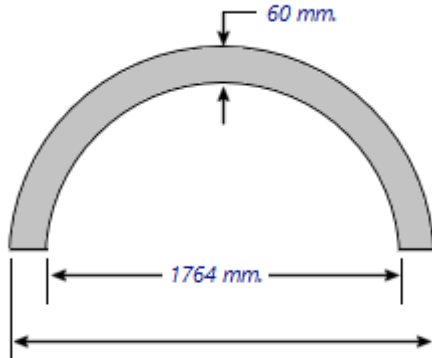
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Top Dish End

Spherical Head Design Information:					
Design Pressure	P	191.005 bars	Int. Design Temperature		260 °C
Material		SA-387 11	Ext. Design Temperature		260 °C
Length		0.000 mm.	External Pressure Chart		CS-3
			UNS Number		K11789
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	148.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	148.0 N./mm ²
Inside Diameter	Di	1764.000 mm.	Actual Stress		146.2 N./mm ²
Spec. Min. Thk t,e, or sa		60.0000 mm.	Spec. Nominal Thk.		71.000 mm.
Surface Area		57064.5 cm ²	SG of Contents		0.060
Element Volume		1437282.9 cm ³	Weight of Contents		86.2 Kg
Empty Weight		2912.0 Kg	Operating Weight		2998.2 Kg
			Comp. All. Ope. Stress		118.5 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 22.050 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 11.025 mm.

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:
 $= (P \cdot R) / (2 \cdot S \cdot E - 0.2 \cdot P)$ per UG-27 (d)
 $= (191.005 \cdot 883.5001) / (2 \cdot 148.0 \cdot 1.0 - 0.2 \cdot 191.005)$
 $= 57.7612 + 1.5000 = 59.2612 \text{ mm.}$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.005 bars
 $= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t)$ per UG-27 (d)
 $= (2 \cdot 148.0 \cdot 1.0 \cdot 58.5) / (883.5001 + 0.2 \cdot 58.5)$
 $= 193 - 0.01 = 193 \text{ bars}$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t)$ per UG-27 (d)
 $= (2 \cdot 148.0 \cdot 1.0 \cdot 60.0) / (882.0 + 0.2 \cdot 60.0)$
 $= 199 \text{ bars}$



Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned} &= (P \cdot (R + 0.2 \cdot t)) / (2 \cdot E \cdot t) \\ &= (191.005 \cdot (883.5001 + 0.2 \cdot 58.5)) / (2 \cdot 1.0 \cdot 58.5) \\ &= 146.152 \text{ N./mm}^2 \end{aligned}$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 5.804 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Spherical Portion:

Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 22.050 mm.

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 11.025 mm.

Material treatment options: +IT

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Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	99.164 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	0.000 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	98.564 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	83.908 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	0.000 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	0.000 bars
App 27-4	= M.A.W.P.	76.280 bars

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	93.244 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	248.300 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	92.564 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	78.899 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	273.130 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	273.130 bars
App 27-4	= M.A.W.P.	71.726 bars

Vertical Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
Bottom Dish end	70.1	279.0	0.251	94.09
Main Shell-2	75.5	279.0	0.271	99.81
Main Shell-1	71.1	279.0	0.255	93.92
Top Dish End	69.5	279.0	0.249	93.33

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
A1-Tube inlet	Nozzle	148.00	148.00	1.000
B2-Shell outlet	Nozzle	148.00	148.00	1.000
B3-Shell outlet	Nozzle	148.00	148.00	1.000
A2-Shell inlet	Nozzle	148.00	148.00	1.000
A3-Shell inlet	Nozzle	148.00	148.00	1.000
B1-Tube outlet	Nozzle	148.00	148.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Bottom Dish end	148.00	148.00	1.000
Main Shell-2	148.00	148.00	1.000
Main Shell-1	148.00	148.00	1.000
Top Dish End	148.00	148.00	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	118.00	111.40	1.059
Tubesheet Material	148.00	148.00	1.000

Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
A1-Tube inlet	44.43	278.99	0.159
B2-Shell outlet	46.10	278.99	0.165
B3-Shell outlet	46.10	278.99	0.165
A2-Shell inlet	43.16	278.99	0.155
A3-Shell inlet	43.16	278.99	0.155
B1-Tube outlet	44.07	278.99	0.158

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	99.164 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	0.000 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	98.564 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	83.908 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	0.000 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	0.000 bars
App 27-4	= M.A.W.P.	76.280 bars

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	93.244 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	248.300 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	92.564 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	78.899 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	273.130 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	273.130 bars
App 27-4	= M.A.W.P.	71.726 bars

Vertical Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
Bottom Dish end	71.5	279.0	0.256	93.41
Main Shell-2	76.1	279.0	0.273	99.21
Main Shell-1	71.5	279.0	0.256	93.24
Top Dish End	70.9	279.0	0.254	92.65

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
A1-Tube inlet	Nozzle	148.00	148.00	1.000
B2-Shell outlet	Nozzle	148.00	148.00	1.000
B3-Shell outlet	Nozzle	148.00	148.00	1.000
A2-Shell inlet	Nozzle	148.00	148.00	1.000
A3-Shell inlet	Nozzle	148.00	148.00	1.000
B1-Tube outlet	Nozzle	148.00	148.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Bottom Dish end	148.00	148.00	1.000
Main Shell-2	148.00	148.00	1.000
Main Shell-1	148.00	148.00	1.000
Top Dish End	148.00	148.00	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	118.00	111.40	1.059
Tubesheet Material	148.00	148.00	1.000

Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
A1-Tube inlet	44.11	278.99	0.158
B2-Shell outlet	45.82	278.99	0.164
B3-Shell outlet	45.82	278.99	0.164
A2-Shell inlet	42.85	278.99	0.154
A3-Shell inlet	42.85	278.99	0.154
B1-Tube outlet	43.75	278.99	0.157

Elements Suitable for Test Pressure.

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MDMT Calculations for Stiffening Rings:

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Input Echo, Tubesheet Number 1, Description:

Shell Data:

Main Shell Description: Main Shell-2

Shell Test Pressure		98.56 bars
Shell Thickness	ts	122.0000 mm.
Shell Internal Corrosion Allowance	cas	0.0000 mm.
Shell External Corrosion Allowance	casext	0.0000 mm.
Inside Diameter of Shell	Ds	1700.000 mm.
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-387 11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K11789
Shell Allowable Stress at Temperature	Ss	148.00 N./mm ²
Shell Allowable Stress at Ambient		148.00 N./mm ²

Channel Data:

Channel Description: Bottom Dish end

Channel Type:		Hemispherical Head
Channel Test Pressure		92.56 bars
Channel Thickness	tc	71.0000 mm.
Channel Internal Corrosion Allowance	cac	0.0000 mm.
Channel External Corrosion Allowance	cacext	0.0000 mm.
Inside Diameter of Channel	Dc	1764.000 mm.
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-387 11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K11789
Channel Allowable Stress at Temperature	Sc	148.00 N./mm ²
Channel Allowable Stress at Ambient		148.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	5139
Tube Wall Thickness	et	1.6500 mm.
Tube Outside Diameter	D	12.7000 mm.
Total Straight Tube Length	Lt	12910.00 mm.
Straight Tube Length (bet. inner tubsht faces) L		12610.00 mm.
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-213 T11
Tube Material UNS Number		K11597
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		118.00 N./mm ²
Tube Allowable Stress At Ambient		118.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	205.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	19.0500 mm.
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	1.6500 mm.
Groove Weld Leg	ag	1.6500 mm.
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	733.019 mm.
Largest Center-to-Center Tube Distance	Ul	0.0000 mm.
Length of Expanded Portion of Tube	ltx	0.0000 mm.
Tube-side pass partition groove depth	hg	8.0000 mm.
Area of Tube Layout	Ap	15789.6 cm ²
Perimeter of Tube Layout	Cp	4706.0 mm.

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A



Tubesheet Design Metal Temperature T 21.11 °C
 Tubesheet Material SA-336 F11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number K11572
 Tubesheet Allowable Stress at Temperature S 148.00 N./mm²
 Tubesheet Allowable Stress at Ambient Tt 148.00 N./mm²
 Thickness of Tubesheet h 150.0000 mm.
 Tubesheet Corr. Allowance (Shell side) Cats 0.0000 mm.
 Tubesheet Corr. Allowance (Channel side) Catc 0.0000 mm.
 Tubesheet Outside Diameter A 1944.000 mm.

Area of the Untubed Lanes AL 0.0 cm²

Additional Data for Fixed/Floating Tubesheet Exchangers:

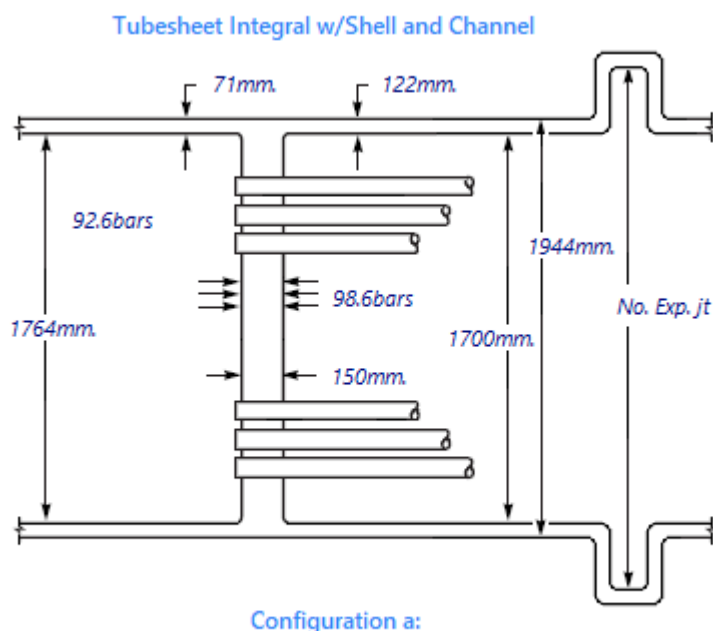
Unsupported Tube Span under consideration l 535.000 mm.
 Tube End condition corresponding to Span (l) k 1.00

Ignore Radial Thermal Exp. effects (4.18.8/4.18.9) Yes
 Perform Differential Pressure Design Yes
 Differential Design Pressure Pdif 20.0000 bars
 Run Multiple Load Cases Yes

Mean Shell Metal Temp. along Shell len. Tsm 21.11 °C
 Mean Tube Metal Temp. along Tube length Ttm 21.11 °C
 Junction Stress Reduction option Perform Plastic Calculation

Tubesheet Integral with Both
 Tubesheet Extended as Flange No

Is Exchanger in Creep range (skip EP, Use 3S for Sps) No



ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
 Shell - TM-1 Material Group C

Th. Exp. Coeff. Metal Temp. along Len 21.1 °C 0.0000115191 /°C



Elastic Mod. at Design Temperature	21.1 °C	0.20408E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20408E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
 Channel - TM-1 Material Group C

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20408E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
 Tubes - TM-1 Material Group C

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20408E+09 KPa.
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20408E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.20408E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
 TubeSheet - TM-1 Material Group C

Elastic Mod. at Design Temperature	21.1 °C	0.20408E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
 Please insure these properties are consistent with the
 type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Differential pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (20.0 \cdot (12.7/2 - 0)) / (118.0 \cdot 1.0 + 0.4 \cdot 20.0)$
 $= 0.1069 + 0.0000 = 0.1069 \text{ mm.}$

Tube Required Thickness under External Pressure (Differential pressure) :

External Pressure Chart CS-2 at 21.11 °C
 Elastic Modulus for Material 199943392.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.6500	12.70	12610.00	7.70	50.0000	0.0185675	122.73

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 243.2871 \text{ bars}$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.3034	12.70	12610.00	41.85	50.0000	0.0006280	62.78

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 62.7848) / (3 \cdot 41.8524) = 20.0008 \text{ bars}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.3034 mm.
Allowable Internal Pressure at Corroded thickness	342.15 bars
Required Internal Design Pressure	20.00 bars
Allowable External Pressure at Corroded thickness	243.29 bars
Required External Design Pressure	20.00 bars
Required Thickness due to Shell Side pressure	0.3034 mm.

Note: The tubes were evaluated using the specified differential pressure.

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Note: Shell/Channel Design Pressures modified by Input differential pressure:
 Pdif = 20.000 Ps = 98.564 Pt = 78.564 bars



Results for ASME Fixed Tubesheet Calculations for Configuration a:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio [ρ]:

$$= l_{tx} / h \text{ (modified for corrosion if present)}$$

$$= 0/150.0$$

$$= 0.0000 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Effective tube hole diameter [d^*]:

$$= \max(dt - 2 \cdot tt(Et/E) \cdot (St/S) \cdot (\rho); dt - 2 \cdot tt)$$

$$= \max(12.7 - 2 \cdot 1.65 (0.20408E+09/.20408E+09) \cdot (117/147) \cdot (0); 12.7 - 2 \cdot 1.65)$$

$$= 12.7000 \text{ mm.}$$

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot ro + dt$$

$$= 2 \cdot 733.019 + 12.7$$

$$= 1478.738 \text{ mm.}$$

Determine the basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (19.05 - 12.7) / 19.05$$

$$= 0.333$$

Equivalent outer tube limit radius [ao]:

$$= Do / 2$$

$$= 1478.738/2$$

$$= 739.3690 \text{ mm.}$$

Effective tube pitch [p^*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) }$$

$$= 19.05 / \sqrt{ (1 - 4 \cdot \min(0 \cdot 100.0 ; 4 \cdot 1478.738 \cdot 19.05) / (\pi \cdot 1478.738^2)) }$$

$$= 19.0500 \text{ mm.}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (19.05 - 12.7) / 19.05$$

$$= 0.3333$$

Ratio [ps]:

$$= as / ao$$

$$= 850.0/739.369$$

$$= 1.1496$$

Ratio [pc]:

$$= ac / ao$$

$$= 882.0/739.369$$

$$= 1.1929$$

Compute parameter [xt]:

$$= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot ao))^2$$

$$= 1 - 5139 ((12.7 - 2 \cdot 1.65) / (2 \cdot 739.369))^2$$

$$= 0.7923$$

Determine parameter [xs]:

$$= 1 - Nt(dt / (2 \cdot ao))^2$$

$$= 1 - 5139 (12.7 / (2 \cdot 739.369))^2$$

$$= 0.6209$$

Determine the value [$h'g$]:

$$= \max((hg - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((8.0 - 0); 0)$$

$$= 8.000 \text{ mm.}$$



ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$= \pi \cdot ts (Ds + ts) Es / L$$

$$= \pi \cdot 122.0 (1700.0 + 122.0) \cdot 20408E+09 / 12610.0$$

$$= 1152498.3750 \text{ Kg/mm.}$$

Axial Tube Stiffness [Kt]:

$$= \pi \cdot tt (Dt - tt) Et / L$$

$$= \pi \cdot 1.65 (12.7 - 1.65) \cdot 20408E+09 / 12610.0$$

$$= 94.5319 \text{ Kg/mm.}$$

Stiffness Factor [Ks,t]:

$$= Ks / (Nt \cdot Kt)$$

$$= 1152498 / (5139 \cdot 94.532)$$

$$= 2.37238$$

Rigidity Ratio [J]:

$$= 1 / (1 + Ks/Kt)$$

$$= 1 / (1 + 1152498/94)$$

$$= 1.0000 (= 1 \text{ if No Exp. Jt.})$$

Shell coefficient [β_s]:

$$= \left(\frac{12(1 - \nu_s^2)}{Ds + ts} \right)^{1/4} \sqrt{\frac{ts}{(Ds + ts)ts}}$$

$$= \left(\frac{12(1 - 0.3^2)}{1700.0 + 122.0} \right)^{1/4} \sqrt{\frac{122.0}{(1700.0 + 122.0)122.0}}$$

$$= 0.0039 \text{ 1/mm.}$$

Shell coefficient [ks]:

$$= \beta_s \cdot Es \cdot ts^3 / (6(1 - \nu_s^2))$$

$$= 0.004 \cdot 0.20408E+09 \cdot 122.0^3 / (6(1 - 0.3^2))$$

$$= 2616912128.0000 \text{ bars} \cdot \text{mm.}^2$$

Shell coefficient [λ_s]:

$$= 6(Ds + ts) \cdot ks / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2/2)$$

$$= 6(1700.0 + 122.0) \cdot 0.26169E+11 / 150.0^3 \cdot (1 + 150.0 \cdot 0.004 + (150.0 \cdot 0.004)^2/2)$$

$$= 14796515.0000 \text{ bars}$$

Shell coefficient [δ_s]:

$$= Ds^2 / (4 \cdot Es \cdot ts) \cdot (1 - Ds \cdot \nu_s / (2(Ds + ts)))$$

$$= 1700.0^2 / (4 \cdot 0.20408E+09 \cdot 122.0) \cdot (1 - 1700.0 \cdot 0.3 / (2(1700.0 + 122.0)))$$

$$= 0.24956E-01 \text{ mm./N./mm}^2$$

Channel coefficient [β_c]:

$$= \left(\frac{12(1 - \nu_c^2)}{Dc + tc} \right)^{1/4} \sqrt{\frac{tc}{(Dc + tc)tc}}$$

$$= \left(\frac{12(1 - 0.3^2)}{1764.0 + 71.0} \right)^{1/4} \sqrt{\frac{71.0}{(1764.0 + 71.0)71.0}}$$

$$= 0.0050 \text{ 1/mm.}$$

Channel coefficient [kc]:

$$= \beta_c \cdot Ec \cdot tc^3 / (6(1 - \nu_c^2))$$

$$= 0.005 \cdot 0.20408E+09 \cdot 71.0^3 / (6(1 - 0.3^2))$$

$$= 673738944.0000 \text{ bars} \cdot \text{mm.}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(Dc + tc / \cos(\theta)) \cdot kc / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2/2 \cdot CqDelta)$$

$$= 6(1764.0 + 71.0 / \cos(0)) \cdot 0.67374E+11 / 150.0^3 \cdot (1.0 + 150.0 \cdot 0.005 \cdot (1.0 + 1.0) / 2 + (150.0 \cdot 0.005)^2/2 \cdot 1.0)$$

$$= 4485411.0 \text{ bars}$$

Channel coefficient [δ_c]:

$$= Dc^2 / (4 \cdot Ec \cdot Tc) \cdot (1/2 - Dc \cdot \nu_c / (2(Dc + tc)))$$

$$= 1764.0^2 / (4 \cdot 0.20408E+09 \cdot 71.0) \cdot (1/2 - 1764.0 \cdot 0.3 / (2(1764.0 + 71.0)))$$

$$= 0.19101E-01 \text{ mm./N./mm}^2$$

Channel coefficient [zetac]:

$$\text{zetac} = 0$$



ASME Section VIII-2 Part 4.18.8.4 Step 3:

$h/p = 7.874016$; $\mu^* = 0.333333$; $E^* = 66344700$ KPa.
 $E^*/E = 0.325091$; $\nu^* = 0.334167$

Note: As h/p (7.874) is > 2 , data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor $[X_a]$:
 $= ((24(1 - \nu^{*2})Nt \cdot Et \cdot tt(dt - tt)ao^2) / (E^* \cdot L \cdot H^3))^{1/4}$
 $= ((24(1 - 0.334^2)5139 \cdot 0.20408E+09 \cdot 1.65$
 $(12.7 - 1.65)739.369^2) / (66344700 \cdot 12610.0 \cdot 150.0^3))^{1/4}$
 $= 2.9806$

Values from Table 4.18.3

$Z_d = 0.056202$; $Z_v = 0.108292$; $Z_m = 0.510753$
 $Z_a = 0.208360E+01$; $Z_w = 0.108292$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio $[K]$:

$= A / D_o$
 $= 1944.0/1478.738$
 $= 1.3146$

Compute coefficient $[F]$:

$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$
 $= (1 - 0.33) / 66344700 \cdot (14796515 + 4485411 +$
 $0.20408E+09 \cdot \ln(1.31))$
 $= 19.9116$

Compute parameter $[\phi]$:

$= (1 + \nu^*) F$
 $= (1 + 0.3342) 19.9116$
 $= 26.5653$

Compute parameter $[Q_1]$:

$= (p_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m)$
 $= (1.1496 - 1 - 26.5653 \cdot 0.1083) / (1 + 26.5653 \cdot 0.5108)$
 $= -0.187200$

Compute Parameter $[Q_{z1}]$:

$= (Z_d + Q_1 \cdot Z_w) / 2 \cdot X_a^4$
 $= (0.0562 + -0.1872 \cdot 0.10829) / 2 \cdot 2.98064^4$
 $= 1.4180$

Compute parameter $[Q_{z2}]$:

$= (Z_v + Q_1 \cdot Z_m) / 2 \cdot X_a^4$
 $= (0.10829 + -0.1872 \cdot 0.51075) / 2 \cdot 2.98064^4$
 $= 0.5004$

Compute parameter $[U]$:

$= (Z_w + (p_s - 1) Z_m) X_a^4 / (1 + \phi \cdot Z_m)$
 $= (0.1083 + (1.1496 - 1) 0.5108) 2.98064^4 / (1 + 26.5653 \cdot 0.5108)$
 $= 1.0008$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor $[\gamma_b]$:

$= 0$ (config a)

Compute Parameter $[\gamma]$:

$= 0.000$ mm. (For Pressure only cases)

Calculate parameter $[\omega_s]$:

$= p_s \cdot k_s \cdot \beta_s \cdot \delta S (1 + h \cdot \beta_s)$
 $= 1.1496 \cdot 0.26169E+1 \cdot 0.0039 \cdot 0.024956 (1 + 150.0 \cdot 0.0039)$
 $= 45693.0703$ mm.²

Calculate parameter $[\omega^*s]$:

$= ao^2 (p_s^2 - 1) \cdot (p_s - 1) / 4 - \omega_s$



$$= 739.369^2 (1.15^2 - 1) \cdot (1.15 - 1) / 4 - 45693.07$$

$$= -39115.6211 \text{ mm.}^2$$

Parameter [wc]:

$$= \rho_c \cdot k_c [\delta C (C_m \theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) -$$

$$\zeta c (C_m \theta + h \cdot C_q \theta + h \cdot \beta C / 2)]$$

$$= 1.193 \cdot 0.67374E+09 [0.019101 (1.0 + 0.005 \cdot 1.0 \cdot 150.0) -$$

$$0 (1.0/150.0 + 1.0 \cdot 0.005/2)]$$

$$= 13573.1904 \text{ mm.}^2$$

Parameter [ω*c]:

$$= a o^2 [(\rho_c^2 + 1) \cdot (\rho_c - 1) / 4 - (\rho_s - 1) / 2] - \omega c$$

$$= 739.36902^2 [(1.19291^2 + 1) \cdot (1.19291 - 1) / 4 - (1.14963 - 1) / 2] - 13573.19043$$

$$= 9409.6289 \text{ mm.}^2$$

Pressure [P*S]:

= 0 For Pressure only cases or Configurations d,e,f,A,B,C,D

Pressure [P*C]:

= 0 For Pressure only cases or Configurations b,c,d,B,C,D

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [P's]:

$$= P_s \{ x_s + 2(1 - x_s) v_t + [2/Kst \cdot (D_s/D_o)^2] v_s -$$

$$[(\rho_s^2 - 1)/(J \cdot Kst)] - [(1 - J)/(2J \cdot Kst)] [(D_j^2 - (D_s)^2)/D_o^2] \}$$

$$= 98.564 \{ 0.621 + 2(1 - 0.621) 0.3 + [2/2.372 \cdot$$

$$(1700.0/1478.738)^2] 0.3 - [(1.15^2 - 1)/(1.0 \cdot 2.372)] -$$

$$[(1 - 1.0)/(2 \cdot 1.0 \cdot 2.372)] [(0^2 - (1700.0)^2)/1478.738^2] \}$$

$$= 103.2020 \text{ bars}$$

Pressure [P't]:

$$= [x_t + 2(1 - x_t) v_t + 1/(J \cdot Kst)] P_t$$

$$= [0.792 + 2(1 - 0.792) 0.3 + 1/(1.0 \cdot 2.372)] 78.564$$

$$= 105.1543 \text{ bars}$$

Pressure [P_y]:

$$= N_t \cdot K_t \cdot \gamma / (\pi \cdot a o^2)$$

$$= 5139 \cdot 94.532 \cdot 0 / (3.142 \cdot 739.369^2)$$

$$= 0.0000 \text{ bars}$$

Pressure [P_w]:

$$= -\gamma b \cdot U \cdot W^* / (2 \cdot \pi \cdot a o^2)$$

$$= -0 \cdot 1.001 \cdot 0 / (2 \cdot \pi \cdot 739.369^2)$$

$$= -0.0000 \text{ bars}$$

Calculate the pressure [Prim]:

$$= -(U/a o^2) \cdot (\omega^* s \cdot P_s - \omega^* c \cdot P_t)$$

$$= -(1.001/739.369^2) \cdot (-39115.621 \cdot 98.564 - 9409.629 \cdot 78.564)$$

$$= 8.4113 \text{ bars}$$

Calculate the pressure [P_w]:

$$= (U/a o^2) \cdot (\omega s \cdot P^* s - \omega c \cdot P^* c)$$

$$= (1.001/739.369^2) \cdot (45693.0703 \cdot 0 - 13573.1904 \cdot 0)$$

$$= 0.0000 \text{ bars}$$

Determine the effective pressure [Pe]:

$$= J \cdot Kst / (1 + J \cdot Kst (Qz1 + (\rho_s - 1) Qz2)) \cdot$$

$$(P^* s - P^* t + P_y + P_w + P_w + P_{\text{Prim}})$$

$$= 0.1000E+01 \cdot 2.372 / (1 + 1.0 \cdot 2.372 (1.418 + (1.15 -$$

$$1) 0.5)) \cdot (103.202 - 105.154 + 0 + 0 + 0 + 8.411)$$

$$= 3.3740 \text{ bars}$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:

$$= [((\omega^* s \cdot P_s - \omega^* c \cdot P_t) - (\omega s \cdot P^* s - \omega c \cdot P^* c)) +$$

$$W^* \cdot \gamma b / (2 \cdot \pi)] / (1 + \phi \cdot Z_m)$$

$$= [((-39115.621 \cdot 98.564 - 9409.629 \cdot 78.564) -$$

$$(45693.07 \cdot 0 - 13573.19 \cdot 0)) \cdot 0.01 +$$



$$= 0 \cdot 0 / (2 \cdot \pi) / (1 + 26.56535 \cdot 0.51075) \\ = -3216.1675 \text{ Kgf}$$

Calculate factor [Q3]:

$$= Q1 + 2 \cdot Q2 / (Pe \cdot ao^2) \\ = -0.187 + 2 \cdot -3216.167 / (3.374 \cdot 739.369^2) \\ = -0.52918$$

Fm Value from Table 4.18.3 = 0.264590

Tubesheet Bending Stress - original thickness [σ]:

$$= (1.5 \cdot Fm / \mu \cdot) \cdot (2 \cdot ao / (H - h'g))^2 \cdot Pe \\ = (1.5 \cdot 0.2646 / 0.3333) \cdot (2 \cdot 739.369 / (150.0 - 8.0))^2 \cdot 3.37 \\ = 43.5679 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot \beta T \cdot Sy = 1.5 \cdot 0.95 \cdot 309.99 = 441.74 \text{ N./mm}^2, 4.18.233$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$= (1.5 \cdot Fm / \mu \cdot) \cdot (2 \cdot ao / (h - h'g))^2 \cdot Pe \\ = (1.5 \cdot 0.0284 / 0.3333) \cdot (2 \cdot 739.369 / (14.389 - 8.0))^2 \cdot 0.65 \\ = 441.7215 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + Cats + Catc \\ = 14.3891 + 0 + 0 \\ = 14.3891 \text{ mm.}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ limit]:

$$= (2 \cdot \mu \cdot h / ao) \cdot (0.8 \cdot \beta T \cdot Sy) \\ = (2 \cdot 0.333 \cdot 150.0 / 739.369) \cdot 235.596 \\ = 31.8644 \text{ N./mm}^2$$

Shear Stress is not required to be computed; [Pe] <= τ limit

Tubesheet Shear Stress is probably low, use the following req. thk:

$$\text{Tubesheet thickness (Incl. Corr.)} = 3.8100 \text{ mm.} \\ \text{Tubesheet Shear Stress} = 5.7240 \text{ N./mm}^2$$

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd] :

$$= \max(HreqB; HreqS) \\ = \max(14.3891; 3.81) \\ = 14.3891 \text{ mm.}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The Ftmin and Ftmax Coefficients from Table 4.18.4:

$$Ftmin = -0.0436, Ftmax = 1.4715$$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$= ((Ps \cdot xs - Pt \cdot xt) - Pe \cdot Ftmin) / (Xt - Xs) \\ = ((98.56 \cdot 0.6209 - 78.56 \cdot 0.7923) - (3.374) \cdot -0.044) / \\ (0.7923 - 0.6209) \\ = -0.5250 \text{ N./mm}^2$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$= ((Ps \cdot xs - Pt \cdot xt) - Pe \cdot Ftmax) / (Xt - Xs) \\ = ((98.56 \cdot 0.6209 - 78.56 \cdot 0.7923) - (3.374) \cdot 1.472) / \\ (0.7923 - 0.6209) \\ = -3.5077 \text{ N./mm}^2$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

$$= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 3.508 \text{ N./mm}^2$$

Allowable Tube Stress [$\sigma_{t,A}$]:

$$= \beta T \cdot Sy, t = 194.7465 \text{ N./mm}^2$$



Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:

$$\begin{aligned} &= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w)X_a^4); 1.25) \\ &= \max((3.25 - 0.25(0.056 + -0.529 \cdot 0.108)2.981^4); 1.25) \\ &= 1.7318 \text{ (Should be } \leq 2 \text{)} \end{aligned}$$

Determine the factor [rt]:

$$\begin{aligned} &= (\sqrt{ dt^2 + (dt - 2 \cdot tt)^2 }) / 4 \\ &= (\sqrt{ 12.7^2 + (12.7 - 2 \cdot 1.65)^2 }) / 4 \\ &= 3.9501 \text{ mm.} \end{aligned}$$

Determine the factor [Ct]:

$$\begin{aligned} &= \sqrt{ (2 \cdot \pi^2 \cdot Et / Syt) } \\ &= \sqrt{ (2 \cdot \pi^2 \cdot 0.20408E+09 / 204) } \\ &= 149.0780 \end{aligned}$$

Determine the factor [Ft]:

$$\begin{aligned} &= k \cdot L / r \\ &= 1.0 \cdot 535.0 / 3.95 \\ &= 135.4403 \end{aligned}$$

Buckling Allowable Stress [Stb]:

$$\begin{aligned} &= (Sy / t / Fs) \cdot (1 - Ft / (2 \cdot Ct)) \\ &= (204 / 1.73) \cdot (1 - 135.44 / (2 \cdot 149.078)) \\ &= 194.746 \text{ N./mm}^2 \text{ (Never greater than } Sot \text{)} \end{aligned}$$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

$$\begin{aligned} &= \sigma_{t,max} \cdot \text{Tube Area} \\ &= 3.51 \cdot 0.5728 \\ &= 20.49 \text{ Kg} \end{aligned}$$

Tube Metal Area [At]:

$$\begin{aligned} &= \pi (do - t) t \\ &= \pi (12.7 - 1.65) 1.65 \\ &= 0.573 \text{ cm}^2 \end{aligned}$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$\begin{aligned} &= \pi \cdot t (do - t) S \\ &= \pi \cdot 1.65 (12.7 - 1.65) 194.746 \\ &= 1137.462 \text{ Kg} \end{aligned}$$

Groove Weld Strength [Fg]:

$$\begin{aligned} &= 0.85 \cdot \pi \cdot ag (do + 0.67 \cdot ag) Sw \text{ (but not } > Ft \text{)} \\ &= 0.85 \cdot \pi \cdot 1.65 (12.7 + 0.67 \cdot 1.65) 194.75 \\ &= 1137.4623 \text{ Kg} \end{aligned}$$

Weld Strength Factor [fw]:

$$\begin{aligned} &= S / Sw \text{ where } Sw = \min(S \text{ and } St) \\ &= 194.746 / 194.746 \\ &= 1.000 \end{aligned}$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$\begin{aligned} &= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw \text{ (but not } > Ft \text{)} \\ &= 0.55 \cdot \pi \cdot 1.65 (12.7 + 0.67 \cdot 1.65) 194.75 \\ &= 781.6090 \text{ Kg} \end{aligned}$$

Minimum Length of Weld [ar]:

$$\begin{aligned} &= 2 (\sqrt{ (0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) fw \cdot fd } - 0.75 \cdot do) \\ &= 2 (\sqrt{ (0.75 \cdot 12.7)^2 + 1.07 \cdot 1.65 \cdot (12.7 - 1.65) 1.0 \cdot 1.0 } - 0.75 \cdot 12.7) \\ &= 1.949 \text{ mm.} \end{aligned}$$

Minimum Required Fillet Weld Leg [afr]: 0.9900 mm.
 Minimum Required Groove Weld Leg [agr]: 0.9900 mm.

Tube-Tubesheet Jt allowable, 1137.46 is >= tube strength 1137.46 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:
 $= \beta T \cdot S_{y,s} = 0.95 \cdot 309.99 = 441.74 \text{ N./mm}^2$

Axial Membrane Stress in Shell [σ_s, m]:
 $= a o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a s^2 \cdot P_t / ((D_s + t_s) t_s)$
 $= 739.369^2 / ((1700.0 + 122.0) 122.0) \cdot [3.37 + (1.15^2 - 1)$
 $(98.56 - 78.56)] + 850.0^2 \cdot 78.56 / ((1700.0 + 122.0) 122.0)$
 $= 27.9495 \text{ N./mm}^2$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:
 For a given Shell thickness of 122.0 mm., the minimum Shell length adjacent to the tubesheet should be 819.741 mm.

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	43.6 <=	441.7 N./mm ²	Pass
Tubesheet Shear Stress	2.3 <=	235.6 N./mm ²	Pass
Maximum Tube Stress	3.5 <=	194.7 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-3.5 <=	-194.7 N./mm ²	Pass
Maximum Force on any one Tube	20.5 <=	1137.5 Kgf	Pass
Axial Membrane Stress in Shell	27.9 <=	441.7 N./mm ²	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm.)	Required	Actual	P/F
Tubesheet Thickness :	14.389	150.000	Ok
Tube-Tubesheet Fillet Weld Leg :	0.990	1.650	Ok
Tube-Tubesheet Groove Weld Leg :	0.990	1.650	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses Bend Allwd Shear Allwd	Case Type	Pass/Fail
D3uc	14.389 ...	44	442 2 236	Ps+Pt-Th D3	Ok
Max:	14.3891 ... mm.	0.099	0.010 (Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+)-Th]:
 With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures



[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:
 Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Shell Axial Membrane Stress Summary:

---- Shell Stresses ----					---- Shell Band Stress ----					Pass
Case#	Ten	Allwd	Cmp	Allwd	Ten	Allwd	Cmp	Allwd	Fail	
D3uc	28	441	...	...	...	...	...	...	...	Ok
Max RATIO	0.063		...		...		...		...	

Tube, Shell and Channel Stress Summary:

Case#	---- Tube Stresses				Tube Loads		Shell Stress		Channel Stress		Pass
	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	
D3uc	4	194	-4	-194	20	1137	...	441	...	441	Ok
Max RATIO	0.018		0.018		0.018		...		...		

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm.)	Required	Actual	P/F
Tubesheet thickness :	14.389	150.000	Ok
Tube thickness :	0.303	1.650	Ok
Tube-tubesheet fillet weld leg :	0.990	1.650	Ok
Tube-tubesheet groove weld leg :	0.990	1.650	Ok

Min Shell length of thk, (122.000) adj. to tubesheet: 819.741 mm.

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm.
 Due to Pressure Shell Compresses by : -0.644 mm.
 Due to Pressure + Thermal Shell Compresses by : -0.644 mm.

Summary of Exchanger Pressure and Thermal Displacements: (mm.)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.1184	...	...	...	...	...
2	-0.6438	...	...	...	...	...
3	-0.5433	...	...	...	...	...
4	...	...	...	...	...	...
5	...	...	...	...	...	...
6	...	...	...	...	...	...
7	...	...	...	...	...	...
8	...	...	...	...	...	...



External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm.	Outside Diameter mm.	Corroded Thickness mm.	Factor A	Factor B N./mm ²
10	Tsht	No Calc	1884	58.5	0.0077627	98.7138
Tsht	20	No Calc	1884	58.5	0.0077627	98.7138
20	Tsht	6549.97	1944	120.5	0.0057506	94.4256
Tsht	30	No Calc	...	120.5	No Calc	No Calc
30	40	No Calc	...	...	No Calc	No Calc
40	50	6354.03	1944	120.5	0.0058821	94.75
50	60	No Calc	1884	58.5	0.0077627	98.7138

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	Tsht	60	No Calc	...	61.2996
Tsht	20	60	No Calc	...	61.2996
20	Tsht	122	No Calc	...	78.0358
Tsht	30	122	No Calc	...	No Calc
30	40	...	No Calc	...	No Calc
40	50	122	No Calc	...	78.3039
50	60	60	No Calc	...	61.2996

Minimum

61

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	Tsht	No Calc	No Calc	No Calc	No Calc
Tsht	20	No Calc	No Calc	No Calc	No Calc
20	Tsht	6549.97	No Calc	No Calc	No Calc
Tsht	30	No Calc	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	6354.03	No Calc	No Calc	No Calc
50	60	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	2912.04	1437282	2855.12	1444628	...
20	30	35451.1	10605096	35044	10657624	...
30	40	3265.77	...	3050.98	...	...
40	50	32799	13757460	32422.4	13806058	...
50	60	2912.04	1437282	2855.12	1444628	...
Total		77339	27237120.00	76227	27352940.00	0



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm.	Y Offset, Dtl. Cent. mm.	Z Offset, Dtl. Cent. mm.	Description
10	Liqd	86.1786	...	-441	...	Liquid: 10
10	Insl	12.4849	...	-425	...	Hot Insulation
10	Nozl	329.486	...	-952	...	A1-Tube inlet
20	Liqd	635.875	...	3275	...	Liquid: 20
20	Nozl	588.814	-950	800	...	B2-Shell outlet
20	Nozl	588.814	950	800	...	B3-Shell outlet
20	Wght	12770	...	3500	...	Datasheet diffe
20	Wght	5000	...	...	...	Design Margin
40	Liqd	824.889	...	3030	...	Liquid: 50
40	Nozl	646.546	-950	5500	...	A2-Shell inlet
40	Nozl	646.546	950	5500	...	A3-Shell inlet
40	Wght	12770	...	3000	...	Datasheet diffe
40	Wght	5000	...	...	...	Design Margin
50	Liqd	86.1786	...	441	...	Liquid: 60
50	Insl	12.4849	...	441	...	Hot Insulation
50	Nozl	484.27	...	952	...	B1-Tube outlet
10	FTsh	2693.84	...	75	...	
10	Tube	29453.1	...	6455	...	Tubes
10	RTsh	2693.84	...	12835	...	Tubesheet

Total Weight of Each Detail Type:

Liquid	1633.1
Insulation	25.0
Nozzles	3284.5
Weights	35540.0
Exchanger Components	34840.8
Liquid in Tubes	276.1
Sum of the Detail Weights	75599.5 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	77339.9	77339.9	77339.9	77339.9	77339.9	77339.9
Nozzles	3284.5	3284.5	3284.5	3284.5	3284.5	3284.5
Wld Weights	35540.0	35540.0	35540.0	35540.0	35540.0	35540.0
Exchanger	34840.8	34840.8	34840.8	34840.8	34840.8	34840.8
Insulation	...	...	25.0	25.0	25.0	25.0
Ope. Liquid	...	...	...	...	...	1633.1
Tube Ope Lqd	...	...	...	...	...	276.1
Test Liquid	...	27220.5	...	...	...	...



Tube Tst Lqd	...	4602.2	...	...	...	...
Totals	151005.2	182827.8	151030.2	151030.2	151030.1	152939.4

Field Installation Options:

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	151005.2 kg.
Shop Test Wt.	- Fabricated weight + water (full)	182827.8 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	151030.2 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	151030.2 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	151030.1 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	152939.4 kg.
Field Test Wt.	- Empty weight + test medium (full)	182852.8 kg.

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	41110.8 kg.
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	28705.9 kg.
Mass of the Upper 1/3 of the Vertical Vessel (empty)	40763.2 kg.

Exchanger Tube Data

Volume of Exchanger tubes :	4604985.5 cm ³
Weight of Ope Liq in tubes :	276.1 kg.
Weight of Water in tubes :	4602.2 kg.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	57064.5	48878.4
20	30	400025	349816
30	40	246764	244251
40	50	370100	323647
50	60	57064.5	48878.4

Total	1131018.0	1015470.9 cm ²
Total	1217.4	1093.0 ft ²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kg.	Total. Ele. Oper. Wgt. kg.	Total. Ele. Hydro. Wgt. kg.	Total Dtl. Offset Mom. Kg-m.	Oper. Wgt. No Liquid kg.
10	20	5947.85	6034.03	7384.26	...	5947.85
20	30	83851.8	84763.8	99052.6	...	83851.8
30	40	5959.61	5959.61	5959.61	...	5959.61
40	50	51862.1	52687	65611.2	...	51862.1
50	60	3408.79	3494.97	4845.2	...	3408.79

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kg.	Cumulative Oper. Wgt. kg.	Cumulative Hydro. Wgt. kg.
10	20	...	...	...
20	30	-5947.84	-6034.03	-7384.25
30	40	151030	152939	76416
40	50	55270.9	56182	70456.4
50	60	3408.79	3494.97	4845.2

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.



Cumulative Vessel Moment

From	To	Cumulative Empty Mom. Kg-m.	Cumulative Oper. Mom. Kg-m.	Cumulative Hydro. Mom. Kg-m.
10	20	...	...	...
20	30	...	...	...
30	40	...	...	...
40	50	...	...	...
50	60	...	...	...

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Nozzle Flange MAWP Results: (bars & °C)

Nozzle	Flange Rating		Des.			Equiv.	- - - -	Max Pressure	
Description	Ope.	Ambient	Temp	Class	Grade/Group	Press	UG-44	50%	DNV

Shellside Flange Rating

Channelside Flange Rating

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 5.60556 Hz.

The Natural Frequency for the Vessel (Ope...) is 5.55911 Hz.

The Natural Frequency for the Vessel (Filled) is 5.13807 Hz.

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-22
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 180 Km/hr
Surface Roughness Category	D: Flat, unobstructed
Type of Surface	0
Base Elevation	0 mm.
Percent Wind for Hydrotest	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.157 \cdot 0.943) / (1 + 1.7 \cdot 3.4 \cdot 0.157)))$
 $= \min(0.85, 0.9)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	5.559 Hz
Natural Frequency of Vessel (Empty)	5.606 Hz
Natural Frequency of Vessel (Test)	5.138 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.771
Structure Height to Diameter ratio	5.244
Height to top of Structure	13862.000 mm.

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
 Exposure Category: D from Table 26.11-1
 $\alpha = 11.5$; $Zg = 589788$ mm.

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 547.975 + 0.0 = 547.975$ mm.
 $= 1.798$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z $[Kz]$:
 Because z (1.798 ft.) < 15 ft.
 $= 2.41(15 / Zg)^{2/\alpha}$
 $= 2.41(15 / 1935.0)^{2/11.5}$
 $= 1.035$

Type of Hill: No Hill

Wind Directionality Factor $[Kd]$:
 $= 1$, per Table 26.6-1

As there is No Hill Present: $[Kzt]$:

K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$$

$$= 1.0$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 1.035 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 111.85^2)$$

$$= 33.1 \text{ psf } [161.849 \text{ Kgs/m}^2]$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 33.149 \cdot 0.85 \cdot 0.771 \cdot 18.389$$

$$= 399.3 \text{ lbs. } [181.1 \text{ Kg}]$$

Element	Hgt (z) mm.	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
Bottom Dish end	548.0	0.000	0.000	0.000	1.035	1.000	161.849
Main Shell-2	4307.0	0.000	0.000	0.000	1.035	1.000	161.849
Skirt-1	5403.1	0.000	0.000	0.000	1.066	1.000	166.620
Main Shell-1	10762.0	0.000	0.000	0.000	1.201	1.000	187.832
Top Dish End	14196.0	0.000	0.000	0.000	1.260	1.000	197.100

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

- Cf - Correction factor for natural frequency
- D - Average outer insulated diameter of vessel, mm.
- Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
- Dr - Average outer insulated of top half of vessel, mm.
- f - Natural frequency of vibration (Hertz)
- f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(10^4)$
- L - Total height of structure mm.
- Lc - Total length of conical section(s) of vessel, mm.
- tb - Uncorroded plate thickness at bottom of vessel, mm.
- V30 - Design Wind Speed provided by user, Km/hr
- Vc - Critical wind velocity, Km/hr
- Vw - Maximum wind speed at top of structure, Km/hr
- W - Total corroded weight of structure, Kg
- Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. Kg
- Z - Maximum amplitude of vibration at top of vessel, mm.
- Dl - Logarithmic decrement (taken as 0.03 for Welded Structures)
- Vp - Vib. Chance, <= 0.320E-06 (High); 0.320E-06 < 0.400E-06 (Probable)
- P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5
 :: 0.0/12910.0 = 0.0

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)
 :: $[8.95/42.36^2] \cdot 10^4 = 49.892$ [Geometry Violation]

Compute the vibration possibility. If Vp > 0.400E-06 no chance. [Vp]:

$$= W / (L \cdot Dr^2)$$

$$= 151827 / (12910.0 \cdot 1948.741^2)$$

$$= 0.30968E-05$$

Since Vp is > 0.400E-06 no further vibration analysis is required !

Wind Loads on Masses/Equipment/Piping

ID	Wind Area	Elevation	Pressure	Force
----	-----------	-----------	----------	-------

	cm ²	mm.	Kgs/m ²	Kgf
Datasheet dif	0.00	3650.00	161.85	0.00
Design Margin	0.00	150.00	161.85	0.00
Datasheet dif	0.00	9850.00	184.22	0.00
Design Margin	0.00	6850.00	172.35	0.00

The Natural Frequency for the Vessel (Ope...) is 5.55911 Hz.

Wind Load Calculation:

From	To	Wind Height mm.	Wind Diameter mm.	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	547.975	2284.8	17084.1	161.849	199.259
20	30	4307	2332.8	152798	161.849	1782.15
30	40	5403.08	3212.02	128481	166.62	1542.69
40	50	10762	2332.8	141368	187.832	1913.53
50	60	14196	2284.8	17084.1	197.1	242.658

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-22
Importance Factor	1.500
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio	z/h
Amplification Factor	Ap
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Component resonance duct. fact.	[CAR]
Component strength factor	[Rpo]
Overstrength factor	[Omega0]
Response Acceleration Factor	[SMS]
Response Acceleration Factor	[SM1]
Building Importance Factor	[Ie]
Site Latitude	0.000 deg
Site Longitude	0.000 deg

Seismic Analysis Results:

$$SDS = 2/3 * Sms = 2/3 * 0.326 = 0.217$$

$$SD1 = 2/3 * Sm1 = 2/3 * 0.139 = 0.093$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_{nx}$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(45.24930.75)$
 = 0.349 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = (1/Natural Frequency) = (1/5.559)
 = 0.180

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of ($1.7 \cdot 0.349$, 0.18) per 12.8.2
 = 0.180

Seismic Response Coefficient per equation 12.8-2 [Cs]:
 = $Sds / (R / I)$
 = $0.217 / (3.0 / 1.5)$
 = 0.109

Maximum value of C_s per equation 12.8-3 [Cs]:
 = $Sd1 / (T(R / I_e))$
 = $0.093 / (0.18(3.0 / 1.5))$
 = 0.258

Minimum value of C_s per equation 15.4-1 [Cs]:
 = $\max((0.044 \cdot SDS \cdot I_e), 0.030)$
 = $\max((0.044 \cdot 0.2173 \cdot 1.5), 0.030)$
 = 0.030

Total Base Shear [V]:
 = $C_s \cdot W$ (Equation 12.8-1):
 = $0.1087 \cdot 152939.4$
 = 16619.416 Kgf

Vertical load per 12.4-4, [Ev]:
 = $0.2 \cdot Sds \cdot W$
 = $0.2 \cdot 0.217 \cdot 152939$

= 6647.8 Kgf

Distribute the Base shear force to each element according to the equations
 $F_x = C_{vx} * V$ (eqn. 12.8-11) and the vertical distribution factor
 $C_{vx} = W_x * h_x^k / (\text{Sum of } W_i * h_i^k)$ and k is an exponent which is related
 to the period of Vibration.

In this case, the value of k was 1.0

The Natural Frequency for the Vessel (Ope...) is 5.55911 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height mm.	Earthquake Weight Kgf	Element Ope Load Kgf	Element Emp Load Kgf	Element Vertical Load Kgf
10	20	-0.01524	6034.03	0.0016835	0.0016614	-262.279
20	30	3425	84763.8	5314.73	5263.73	-3946.68
30	40	8700	5959.61	949.177	950.293	6647.77
40	50	9880	52687	9529.51	9391.34	2442.04
50	60	12910	3494.97	825.999	806.579	151.915

The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm.	Cumulative Wind Shear Kg-f	Earthquake Shear Kg-f	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	3247.98	...	...	...	...
20	30	725	199.259	0.0016835	109.191	0.00092251
30	40	2000	5680.29	16619.4	19721.9	81388.9
40	50	7030	2156.18	10355.5	7366.68	34214.4
50	60	10464	242.658	825.999	98.0492	333.756

Notes regarding vessels supported by Intermediate Skirts:

This model contains an intermediate skirt. The moment shown in the table above for the skirt, is the moment at the base of the skirt. It is computed by summing the moments framing into the top of the skirt and then by multiplying the total shear times the skirt length. The force on the skirt, times its moment arm is added to the result.

The moment at the top of the skirt is also computed. The stress at the skirt top is also computed and the worst of the stresses is displayed in the stress report.

The cumulative shear force on the skirt is generally the sum of the shear forces in the wind/seismic report including the shear force on the skirt and possibly other applied forces.

Bending Moments at top of Intermediate Skirt (Wind condition):

= [-Moment at Skirt top (below) + Moment at Skirt top (above)]
 = [-7280.9 + 7366.7]
 = 85.8 Kg-m.

Moment at Skirt Top, operating case : 85.8 Kg-m.

Bending Moments at top of Intermediate Skirt (Earthquake condition):

= [-Moment at Skirt top (below) + Moment at Skirt top (above)]
 = [-17406.1 + 34214.4]
 = 16808.3 Kg-m.

Moment at Skirt Top, operating case : 16808.3 Kg-m.

Moment at Skirt Top, empty case : 16431.1 Kg-m.

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

Note:

Review the Vessel Design Summary for the cumulative shear force and bending moment on the support.

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Wind Deflection Calculations:

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Longitudinal Stress Constants:

From	To	Metal Area New cm ²	Metal Area Corroded cm ²	Section Modulus New mm. ³	Section Modulus Corroded mm. ³
10	20	3438.16	3354.96	151951408	148510208
20	30	6983.26	6903.08	299461760	296476544
30	40	623.493	530.383	30298436	25813864
40	50	6983.26	6903.08	299461760	296476544
50	60	3438.16	3354.96	151951408	148510208



Longitudinal Allowable Stresses:

From	To	Tensile N./mm ²	Hydrotest Tensile N./mm ²	Compressive N./mm ²	Hydrotest Compressive N./mm ²
10	20	177.596	334.794	-118.457	-146.724
20	30	177.596	334.794	-130.145	-151.797
30	40	177.597	334.794	-83.8587	-129.829
40	50	177.596	334.794	-130.145	-151.797
50	60	177.596	334.794	-118.457	-146.724

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Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure N./mm ²	Longitudinal Stress External Pressure N./mm ²	Longitudinal Stress Hydrotest Pressure N./mm ²
10	20	140.418	...	68.0501
20	30	63.6677	...	32.8552
30	40	...	...	...
40	50	63.6677	...	30.8551
50	60	140.418	...	68.0501

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty N./mm ²	Wght. Str. Operating N./mm ²	Wght. Str. Hydrotest N./mm ²	Wght. Str. Emp. Mom. N./mm ²	Wght. Str. Opr. Mom. N./mm ²
10	20	...	...	...	...	...
20	30	0.084498	0.085722	0.084498	...	...
30	40	-10.2196	-10.3881	-13.0275	...	...
40	50	-0.7852	-0.7852	-0.7852	...	...
50	60	-0.099642	-0.099642	-0.099642	...	...

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. N./mm ²	Bend. Str. Oper. Wind N./mm ²	Bend. Str. Oper. Equ. N./mm ²	Bend. Str. Hyd. Wind N./mm ²	Bend. Str. Hyd. Equ. N./mm ²
10	20	...	...	...	...	...
20	30	...	0.0036118	0.30514E-07	0.0011919	...
30	40	...	0.032581	6.38543	0.010752	...
40	50	...	0.24367	1.13172	0.080411	...
50	60	...	0.0064745	0.022039	0.0021366	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case N./mm ²	Vortex Shedding Empty Case N./mm ²	Vortex Shedding Test Case N./mm ²	Earthquake Empty Case N./mm ²
10	20	...	...	...	...
20	30	...	...	...	0.30114E-07
30	40	...	...	...	...
40	50	...	...	...	1.11372
50	60	...	...	...	0.021521

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind N./mm ²	Longitudinal Stress Y Forces Seismic N./mm ²
10	20	...	-0.0076666
20	30	...	-0.056068
30	40	...	1.22918

40	50	...	0.034693
50	60	...	0.0044406

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded N./mm ²	Earthquake For/Mom Corroded N./mm ²	Wind For/Mom No Corrosion N./mm ²	Earthquake For/Mom No Corrosion N./mm ²
10	20	...	...	...	...
20	30	...	...	...	...
30	40	...	6.24215	...	5.31823
40	50	...	...	...	...
50	60	...	...	...	...

Bending Stresses due to Blast Load:

From	To	Bending Stress N./mm ²
10	20	...
20	30	...
30	40	...
40	50	...
50	60	...

Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	177.60	0.00	118.46	0.0000	-0.0000
20	0.09	177.60		130.15	0.0005	
30		177.60	-10.25	83.86		0.1223
40		177.60	-1.03	130.15		0.0079
50		177.60	-0.11	118.46		0.0009

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		177.60	-0.01	118.46		0.0001
20	0.03	177.60		130.15	0.0002	
30		177.60	-15.23	83.86		0.1816
40	0.36	177.60	-1.86	130.15	0.0020	0.0143
50		177.60	-0.12	118.46		0.0010

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	177.60	0.00	118.46	0.0000	-0.0000
20	0.09	177.60		130.15	0.0005	
30		177.60	-10.42	83.86		0.1243
40		177.60	-1.03	130.15		0.0079



50 177.60 -0.11 118.46 0.0009

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		177.60	-0.01	118.46		0.0001
20	0.03	177.60		130.15	0.0002	
30	3.47	177.60	-21.79	83.86	0.0195	0.2598
40	0.38	177.60	-1.88	130.15	0.0021	0.0145
50		177.60	-0.12	118.46		0.0010

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	334.79	0.00	146.72	0.0000	-0.0000
20	0.09	334.79		151.80	0.0003	
30		334.79	-13.04	129.83		0.1004
40		334.79	-0.87	151.80		0.0057
50		334.79	-0.10	146.72		0.0007

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	334.79	0.00	146.72	0.0000	-0.0000
20	0.08	334.79		151.80	0.0003	
30		334.79	-13.03	129.83		0.1003
40		334.79	-0.79	151.80		0.0052
50		334.79	-0.10	146.72		0.0007

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	140.42	177.60		118.46	0.7907	
20	63.76	177.60		130.15	0.3590	
30		177.60	-10.42	83.86		0.1243
40	63.13	177.60		130.15	0.3554	
50	140.32	177.60		118.46	0.7901	

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	140.41	177.60		118.46	0.7906	
20	63.70	177.60		130.15	0.3587	
30	3.47	177.60	-21.79	83.86	0.0195	0.2598
40	64.05	177.60		130.15	0.3606	
50	140.34	177.60		118.46	0.7902	

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	177.60	0.00	118.46	0.0000	-0.0000
20	0.09	177.60		130.15	0.0005	
30		177.60	-10.42	83.86		0.1243
40		177.60	-1.03	130.15		0.0079
50		177.60	-0.11	118.46		0.0009

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		177.60	-0.01	118.46		0.0001
20	0.03	177.60		130.15	0.0002	
30	3.47	177.60	-21.79	83.86	0.0195	0.2598
40	0.38	177.60	-1.88	130.15	0.0021	0.0145



50 177.60 -0.12 118.46 0.0010

Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	68.05	334.79		146.72	0.2033	
20	32.94	334.79		151.80	0.0984	
30		334.79	-13.04	129.83		0.1004
40	30.15	334.79		151.80	0.0901	
50	67.95	334.79		146.72	0.2030	

Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	68.05	334.79		146.72	0.2033	
20	32.94	334.79		151.80	0.0984	
30		334.79	-13.03	129.83		0.1003
40	30.07	334.79		151.80	0.0898	
50	67.95	334.79		146.72	0.2030	

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	140.42	177.60		118.46	0.7907	
20	63.75	177.60		130.15	0.3590	
30		177.60	-10.22	83.86		0.1219
40	62.88	177.60		130.15	0.3541	
50	140.32	177.60		118.46	0.7901	

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	140.42	177.60		118.46	0.7907	
20	63.77	177.60		130.15	0.3591	
30		177.60	-8.84	83.86		0.1054
40	62.89	177.60		130.15	0.3541	
50	140.32	177.60		118.46	0.7901	

Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	140.42	177.60		118.46	0.7907	
20	63.75	177.60		130.15	0.3590	
30		177.60	-10.39	83.86		0.1239
40	62.88	177.60		130.15	0.3541	
50	140.32	177.60		118.46	0.7901	

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	140.42	177.60		118.46	0.7907	
20	63.75	177.60		130.15	0.3590	
30		177.60	-10.22	83.86		0.1219
40	62.88	177.60		130.15	0.3541	
50	140.32	177.60		118.46	0.7901	

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	177.60	0.00	118.46	0.0000	-0.0000
20	0.09	177.60		130.15	0.0005	
30		177.60	-10.39	83.86		0.1239
40		177.60	-0.79	130.15		0.0060



50 177.60 -0.10 118.46 0.0008

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	140.41	148.00		98.71	0.9487	
20	63.70	148.00		108.45	0.4304	
30		148.00	-15.40	69.88		0.2204
40	62.92	148.00		108.45	0.4251	
50	140.32	148.00		98.71	0.9481	

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		148.00	-0.01	98.71		0.0001
20	0.03	148.00		108.45	0.0002	
30		148.00	-15.40	69.88		0.2204
40		148.00	-0.75	108.45		0.0069
50		148.00	-0.10	98.71		0.0010

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	140.42	177.60		118.46	0.7907	
20	63.75	177.60		130.15	0.3590	
30		177.60	-10.39	83.86		0.1239
40	62.88	177.60		130.15	0.3541	
50	140.32	177.60		118.46	0.7901	

Absolute Maximum of the all of the Stress Ratio's 0.9487

Governing Element: Bottom Dish end
 Governing Load Case 18 : FS+BS+IP+OW

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
Bottom Dish end	140.4	...	...	...	140.4	+ Tens
Bottom Dish end	140.4	...	...	...	...	- Comp
Main Shell-2	63.7	...	...	...	63.7	+ Tens
Main Shell-2	63.7	...	...	...	...	- Comp
Skirt-1	...	-9.2	...	6.2	...	+ Tens
Skirt-1	...	-9.2	...	-6.2	-15.4	- Comp
Main Shell-1	63.7	-0.8	...	...	62.9	+ Tens
Main Shell-1	63.7	-0.8	...	...	...	- Comp
Top Dish End	140.4	-0.1	...	...	140.3	+ Tens
Top Dish End	140.4	-0.1	...	...	...	- Comp



Skirt Support Analysis Per EN 13445-3 Section 16.12:

The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

Item: Skirt-1

Mean skirt diameter at the critical height is computed later.
This is a Conical type Support Skirt , Wider at the Bottom

Half Apex Angle of the skirt	a	10.107 deg
Total vessel (empty) weight	Fg	148040.500 Kgf
Weight of vessel contents (liquid)	Ff	1633.121 Kgf
Vessel Gross Weight	Fg + Ff	149673.625 Kgf
Skirt Mean Diameter Top End	DzTop	1955.366 mm.
Skirt Mean Diameter Bottom End	DzBot	3381.366 mm.
Skirt Analysis Thickness (Corroded)	ez	8.500 mm.
Skirt Height	H	4000.000 mm.
Additional Axial Force at section 1-1	F1	0.000 Kgf
Moment at section 1-1	M1	34214.371 Kg-m.
Applied Moment - Bottom of Skirt	Mb	81388.859 Kg-m.
Weight at Bottom of Skirt	W	152939.406 Kgf

Structural Type B Skirt per EN 13445 Figure 16.12-2:



For a Type B Skirt, the next element must be a dished head and this is not the case. Please refer to Figure 16.12-2 of EN 13445-3 for more information.

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Shop/Field Installation Options :

Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Liquid	7005.199 mm.
Center of Gravity of the Insulation	6462.985 mm.
Center of Gravity of the Nozzles	7151.124 mm.
Center of Gravity of the Added Weights (Operating)	5835.537 mm.
Center of Gravity of the Added Weights (Empty)	5835.537 mm.
Center of Gravity of the Tubesheet(s)	6455.000 mm.
Center of Gravity of the Tubes	6455.000 mm.
Center of Gravity of Bare Shell New and Cold	6620.965 mm.
Center of Gravity of Bare Shell Corroded	6615.817 mm.
Vessel CG in the Operating Condition	6411.713 mm.
Vessel CG in the Fabricated (Shop/Empty) Condition	6409.358 mm.
Vessel CG in the Test Condition	6499.207 mm.

Rigging Analysis Results:

Because this model is supported by an intermediate skirt, the rigging analysis cannot be performed.

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Skirt Data :

Skirt Outside Diameter at Base	SOD, D6	3389.6846	mm.
Skirt Thickness	STHK	10.0000	mm.
Skirt Internal Corrosion Allowance	SCA	1.5000	mm.
Skirt External Corrosion Allowance		0.0000	mm.
Skirt Material		SA-387 11	

Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basing	TBA, ea4	13.0000	mm.
Design Temperature of the Basing		38.00	°C
Basing Material		SA-387 11	
Basing Ope. All. Stress	BASOPE, f4	148.00	N./mm ²
Basing Yield Stress		309.99	N./mm ²
Inside Diameter of Basing	DI, D4	3276.0000	mm.
Outside Diameter of Basing	DOU, D5	3582.0000	mm.
Nominal Diameter of Bolts	BND, d(b0)	24.0000	mm.
Bolt Corrosion Allowance	BCA	0.7500	mm.
Bolt Material		SA-36	
Bolt Operating Allowable Stress	SA, fB	150.01	N./mm ²
Number of Bolts	RN, nb	24	
Diameter of Bolt Circle	DC, DBC	3492.0000	mm.
Bolt Allowable Shear Stress		67.962	N./mm ²
Ultimate Comp. Strength of Concrete	FPC	35.0	N./mm ²
Allowable Comp. Strength of Concrete	FC	10.0	N./mm ²
Modular ratio Steel/Concrete		9.833	
Thickness of Gusset Plates	TGA, e7	10.0000	mm.
Width of Gussets at Top Plate	TWDT	96.0000	mm.
Width of Gussets at Base Plate	BWDT	96.0000	mm.
Gusset Plate Elastic Modulus	E	203112000.0	KPa.
Gusset Plate Yield Stress	SY	310.0	N./mm ²
Height of Gussets	HG, h1	117.0000	mm.
Distance between Gussets	RG, b6	78.0000	mm.
Dist. from Bolt Center to Gusset (Rg/2)	CG	39.0000	mm.
Number of Gussets per bolt	NG	2	
Thickness of Top Plate or Ring	TTA, e5	42.0000	mm.
Radial Width of the Top Plate	TOPWTH, b5	117.0000	mm.
Anchor Bolt Hole Dia. in Top Plate	dia,H	27.0000	mm.
External Corrosion Allowance	CA	1.5000	mm.
Dead Weight of Vessel	DW	151030.1	Kgf
Operating Weight of Vessel	ROW	152939.4	Kgf
Test Weight of Vessel	TW	182852.8	Kgf
Earthquake Moment on Basing	EQMOM	81388.9	Kg-m.
Wind Moment on Basing	WIMOM	19721.9	Kg-m.
Test Moment on Basing	TM	6508.2	Kg-m.
Percent Bolt Preload	ppl	100.0	
Use AISC A5.2 Increase in Fc and Bolt Stress		No	
Use Allowable Weld Stress per AISC J2.5		No	
Factor for Increase of Allowables	Fact	1.0000	

Results for Basing Analysis : Analyze Option

Basing Thickness Calculation method used : Neutral Axis Shift

Calculation of Bolt Loads for Multiple Loading Conditions:

Bolt Load, Earthquake Operating Case [FB,e]:

$$= ((4 \cdot M/DC) - \text{abs}(W - \text{vertical force}))/nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 81389/3492.0) - 146292)/24$$

$$= -2211.0 \text{ Kgf}$$

Bolt Load, Windload Empty Case [FB,w]:
 $= ((4 \cdot M/DC) - W)/n_b$ per Jawad & Farr, Eq. 12.3
 $= ((4 \cdot 19722/3492.0) - 151030)/24$
 $= -5351.7 \text{ Kgf}$

Bolt Load, Windload Vortex Shedding Case [FB,v]:
 $= ((4 \cdot M/DC) - W)/n_b$ per Jawad & Farr, Eq. 12.3
 $= ((4 \cdot 0/3492.0) - 152939)/24$
 $= -6372.5 \text{ Kgf}$

Bolt Load, Test Moment Case [FB,t]:
 $= ((4 \cdot M/DC) - W)/n_b$ per Jawad & Farr, Eq. 12.3
 $= ((4 \cdot 6508/3492.0) - 182853)/24$
 $= -7308.2 \text{ Kgf}$

Calculation of Maximum Load per Bolt [W/Bolt],
 $= ((4 \cdot M/DC) - W)/n_b$ per Jawad & Farr, Eq. 12.3
 $= ((4 \cdot 81389/3492.0) - 152939)/24$
 $= -2211.0 \text{ Kgf} \text{ [•• No Uplift ••]}$

Area Available in all the Bolts (Corr) 64.1993 cm²

Shear Stress in a Single Bolt [rb]:
 $= \text{Shear Force}/(\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 16619.4/(2.675 \cdot 24)$
 $= 25.4 \text{ N./mm}^2$. Must be less than 68.0 N./mm².

Concrete Contact Area of Base Ring CCA 16481.96 cm²
 Concrete Contact Section Modulus of Base Ring 0.1355E+10 mm.³

Concrete Load (Simplified method), Earthquake in Operating Condition [Sc]:
 $= ((ppl/100 \cdot (Abt \cdot Sa) + W)/Cca) + M/CZ$ per Jawad & Farr Eq. 12.1
 $= (1.0(64.1993 \cdot 150 + 159587)/16481.96) + 81389/0.13553E+10$
 $= 2.12 \text{ N./mm}^2$

Note: N. A. Shift Calculations not performed due to low bending moment.

Required Thickness of Top Plate in Tension:
 (Calculated as a fixed beam per Megyesy)
 $F_t = (S_a \cdot A_{bss})$, Bolt Allowable Stress * Area
 $R_m = (F_t \cdot 2 \cdot C_g)/8$, Bending Moment
 S_b Allowable Bending Stress
 $W_t = (\text{Topwth} - B_{nd})$, Width of Section

Required Thickness of Top Plate [T]:
 $= \sqrt{(6 \cdot R_m / (S_b \cdot W_t))} + C_A$
 $= \sqrt{(6 \cdot 40 / (207 \cdot 93.0))} + 1.5$
 $= 12.551 \text{ mm.}$

Required Thickness of Continuous Top Ring per Moss:
 $a = (D_c - \text{SkirtOD})/2$ Skirt Distance to Bolt Circle
 $P = S_a \cdot A_{bss}$ Bolt Allowable Stress * Area
 $l = \text{Avgwdt}$ Average Gusset Width
 $g_1 = \text{Gamma } 1$ Constant Term f(b/l)
 $g_2 = \text{Gamma } 2$ Constant Term f(b/l)
 $g = \text{Flat distance} / 2$ Nut 1/2 Dimension (from Tema)
 $F_b =$ Allowable Bending Stress
 $M_o = P/(4\pi)[1.3(\ln((2l \sin(\pi a/l)/(\pi g))) + 1] - [(0.7 - g_2)P/(4\pi)]$ Moment Term

Required Thickness of Top Ring [T]:
 $= \sqrt{(6 \cdot \text{Abs}(M_o) / F_b)} + C_A$ Required Thickness
 $= \sqrt{(6 \cdot 15/207)} + 1.5$
 $= 14.483 \text{ mm.}$

Required Thickness of Gusset in Compression, per AISC E2-1:

1. Allowed Compression at Given Thickness:
 Factor K_1/r Per E2-1 40.5293



Factor Cc Per E2-1	113.7283	
Allowable Buckling Str. per E2-1	161.76	N./mm ²
Actual Buckling Str. at Given Thickness	20.90	N./mm ²

Required Gusset/Support Plate thickness, + CA 4.7249 mm.

2. Allowed Compression at Calculated Thickness:

Factor K1/r Per E2-1	125.6763	
Factor Cc Per E2-1	113.7283	
Allowable Buckling Str. per E2-2	66.22	N./mm ²
Act. Buckling Str. at Calculated Thickness	64.81	N./mm ²

Summary of Basing Thickness Calculations:

Required Basing Thickness (N.A. Shift)	0.0000	mm.
Actual Basing Thickness	13.0000	mm.

Required Top Ring/Plate Thickness as a Fixed Beam	12.5514	mm.
Required Thickness of Continuous Top Ring (Moss)	14.4827	mm.
Actual Top Plate/Ring Thickness	42.0000	mm.

Required Gusset/Support Plate thickness, + CA	4.7249	mm.
Actual Gusset/Support Plate Thickness	10.0000	mm.

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:

$$= \text{SkirtStress} \cdot (\text{SkirtThickness} - \text{CA})$$

$$= 21.787 \cdot (10.0 - 1.5)$$

$$= 18.88 \text{ Kg/mm.}$$

Results for Computed Minimum Basing Weld Size [BWeld]:

$$= W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 19 / [(0.4 \cdot 251) \cdot 2 \cdot 0.707]$$

$$= 1.302 \text{ mm.}$$

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:

$$= \text{Bolt Load} / (\text{Cmwth} + 2 \cdot (\text{Hg} + \text{Tta}))$$

$$= 4091.6 / (98.0 + 2 \cdot (117.0 + 42.0))$$

$$= 9.836 \text{ Kg/mm.}$$

Horizontal Plate Load [Wh]:

$$= \text{Bolt Load} \cdot e / (\text{Cmwth} \cdot (\text{Hg} + \text{Tta}) + 0.6667 \cdot (\text{Hg} + \text{Tta})^2)$$

$$= 4091.6 \cdot 51.0 / (98.0 \cdot (159.0) + 0.6667 \cdot (159.0)^2)$$

$$= 6.433 \text{ Kg/mm.}$$

Resultant Weld Load [Wr]:

$$= \sqrt{Wv^2 + Wh^2}$$

$$= \sqrt{9.84^2 + 6.43^2}$$

$$= 11.753 \text{ Kg/mm.}$$

Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:

$$= Wr / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 11.75 / [(0.4 \cdot 251) \cdot 2 \cdot 0.707]$$

$$= 0.811 \text{ mm.}$$

Results for Computed Minimum Gusset to Top Plate Weld Size

Weld Load [Wv]:

$$= \text{Bolt Load} / (2 \cdot \text{TopWth})$$

$$= 4091.6 / (2 \cdot 96.0)$$

$$= 21.311 \text{ Kg/mm.}$$

Weld Load [Wh]:

$$= \text{Bolt Load} \cdot e / (2 \cdot \text{Hgt} \cdot \text{TopWth})$$

$$= 4091.6 \cdot 51.0 / (2 \cdot 159.0 \cdot 96.0)$$

$$= 6.835 \text{ Kg/mm.}$$

Resultant Weld Load [Wr]:

$$\begin{aligned} &= \sqrt{(W_v^2 + W_h^2)} \\ &= \sqrt{(21.31^2 + 6.84^2)} \\ &= 22.38 \text{ Kg/mm.} \end{aligned}$$

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:
= $W_r / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$
= $22.38 / [(0.4 \cdot 251) \cdot 2 \cdot 0.707]$
= 1.544 mm.

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basing to Skirt Double Fillet Weld Size	4.7625	mm.
Required Gusset to Skirt Double Fillet Weld Size	4.7625	mm.
Required Top Plate to Skirt Weld Size	6.3500	mm.
Required Gusset to Top Plate Double Fillet Weld Size	1.5435	mm.

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: A1-Tube inlet

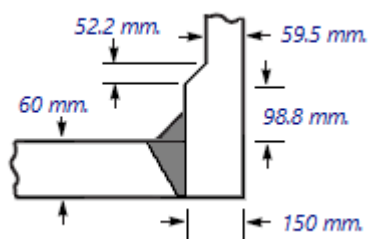
From: 10

Pressure for Reinforcement Calculations	P	191.122	bars
Temperature for Internal Pressure	Temp	260	°C
Parent Material		SA-387 11	+IT
Parent Allowable Stress at Temperature	Sv	148.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	148.00	N./mm ²
Inside Diameter of Hemispherical Head	D	1764.00	mm.
Head Finished (Minimum) Thickness	t	60.0000	mm.
Head Internal Corrosion Allowance	c	1.5000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	+IT
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	148.00	N./mm ²
Allowable Stress At Ambient	Sna	148.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		600.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Hub Height of Integral Nozzle	h	98.7569	mm.
Height of Beveled Transition	L'	52.2431	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		150.0000	mm.
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	58.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: A1-Tube inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	490.576	mm.
Actual Thickness Used in Calculation	59.512	mm.



Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
= $P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
= $191.12 \cdot 883.5001 / (2 \cdot 148 \cdot 1.0 - 0.2 \cdot 191.12)$
= 57.7969 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $191.12 \cdot 246.79 / (148 \cdot 1.0 - 0.6 \cdot 191.12)$
= 34.5489 mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / $\tan(30)$
= 90.488/0.5773
= 156.7295 mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

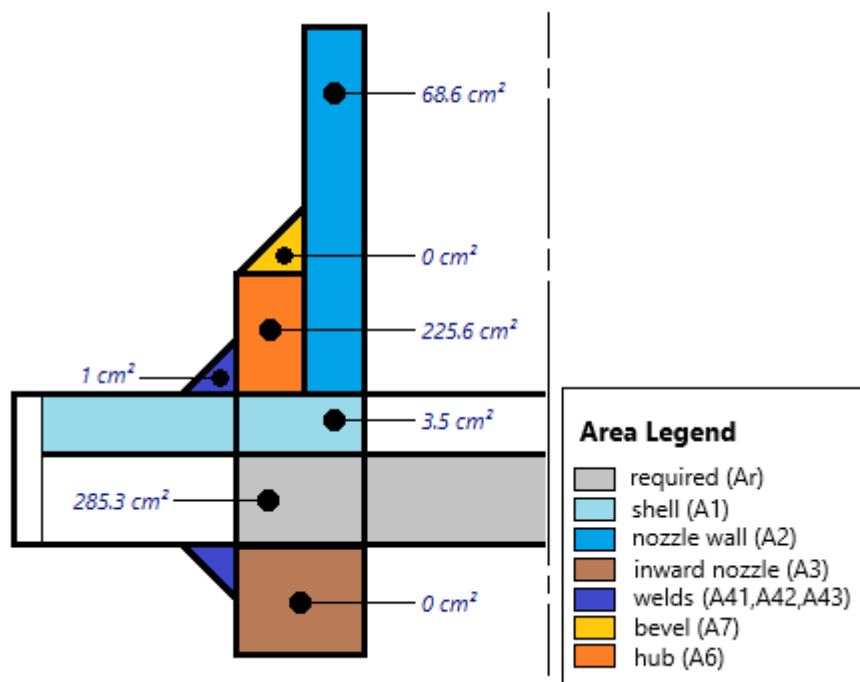
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	987.1512	mm.
Parallel to Vessel Wall, opening length	d	493.5756	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	146.2500	mm.

Weld Strength Reduction Factor [fr1]:
= min(1, S_n/S_v)
= min(1, 148.0/148.0)
= 1.000

Weld Strength Reduction Factor [fr2]:
= min(1, S_n/S_v)
= min(1, 148.0/148.0)
= 1.000

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(1.0, 1.0)
= 1.000



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	285.272	NA	NA
Area in Shell	A1	3.470	NA	NA
Area in Nozzle Wall	A2	68.630	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	225.609	NA	NA
TOTAL AREA AVAILABLE	Atot	298.709	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (493.5756 \cdot 57.7969 \cdot 1.0 + 2 \cdot 148.5 \cdot 57.7969 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 285.272 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 493.576 (1.0 \cdot 58.5 - 1.0 \cdot 57.797) - 2 \cdot 148.5$$

$$(1.0 \cdot 58.5 - 1.0 \cdot 57.7969) \cdot (1 - 1.0)$$

$$= 3.470 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp) (tn - trn) fr2$$

$$= (2 \cdot 146.25) (58.01 - 34.55) 1.0$$

$$= 68.630 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can / 0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$



Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lp}, h_o, H_{ubht})) \cdot (H_{ubt} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(146.3, 200.0, 98.8)) \cdot (150.0 - 59.5) \cdot 1.0$$

$$= 225.609 \text{ cm}^2 \text{ Includes Bevel Area of } 46.883 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 36.0489 \text{ mm.}$
Wall Thickness per UG16(b),	$tr_{16b} = 3.0000 \text{ mm.}$
Wall Thickness, shell/head, internal pressure	$trb_1 = 59.2969 \text{ mm.}$
Wall Thickness	$tb_1 = \max(trb_1, tr_{16b}) = 59.2969 \text{ mm.}$
Wall Thickness	$tb_2 = \max(trb_2, tr_{16b}) = 3.0000 \text{ mm.}$
Wall Thickness per table UG-45	$tb_3 = 9.8312 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[9.831, \max(59.2969, 3.0)]$$

$$= 9.8312 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(36.0489, 9.8312)$$

$$= 36.0489 \text{ mm.}$$

Available Nozzle Neck Thickness = $0.875 \cdot 59.512 = 52.073 \text{ mm.} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 34.5,	Allowable	: 148.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 335.5 N./mm ²	Passed
Occasional	: 36.4,	Allowable	: 196.8 N./mm ²	Passed
Shear	: 5.4,	Allowable	: 103.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -11 °C

Weld Size Calculations, Description: A1-Tube inlet

Intermediate Calc. for nozzle/shell Welds $T_{min} = 19.0000 \text{ mm.}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - tr)) \cdot S_v)$$

$$= \max(0, (285.2715 - 3.4702 + 2 \cdot 148.5 \cdot 1.0 \cdot (1.0 \cdot 58.5 - 57.7969)) \cdot 148)$$

$$= 428423.66 \text{ Kg}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A_2 + A_4 - (W_i - Can / .707)^2 \cdot fr_2) \cdot S_v$$

$$= (294.2388 + 1. - 0.0 \cdot 1.0) \cdot 148$$

$$= 445551.25 \text{ Kg}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$$

$$= (294.2388 + 0.0 + 1. + (173.745)) \cdot 148$$

$$= 707753.56 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (294.2388 + 0.0 + 1. + 0.0 + (173.745)) \cdot 148$$

$$= 707753.56 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 790.5756 \cdot 10.0 \cdot 0.49 \cdot 148$$

$$= 91830. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 321.0378) \cdot (150.0 - 1.5) \cdot 0.7 \cdot 148$$

$$= 1582176. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 790.5756 \cdot (58.0 - 0.0) \cdot 0.74 \cdot 148$$

$$= 804354. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (91830 + 1582176) = 1674006 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (91830 + 0 + 804354 + 0) = 896184 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (91830 + 804354 + 0) = 896184 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 1674005 Kgf, must exceed W = 428423 Kgf or W1 = 445551 Kgf
 Path 2-2 = 896183 Kgf, must exceed W = 428423 Kgf or W2 = 707753 Kgf
 Path 3-3 = 896183 Kgf, must exceed W = 428423 Kgf or W3 = 707753 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 193 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 93.5385 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 353.5385 mm.

Input Echo, WRC107/537 Item 1, Description: A1-Tube inlet :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.5000	mm.
Vessel Diameter	Dv	1764.000	mm.
Vessel Thickness	Tv	60.000	mm.
Design Temperature	T1	260.0	C
Vessel Material		SA-387 11	
Vessel UNS Number		K11789	
Vessel Cold S.I. Allowable	Smc	148.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	148.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	1.5000	mm.
Nozzle Diameter	Dn	490.576	mm.
Nozzle Thickness	Tn	150.000	mm.
Nozzle Material		SA-336 F11	
Nozzle UNS Number		K11572	
Nozzle Cold S.I. Allowable	SNmc	148.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	148.00	N./mm ²
Design Internal Pressure	Dp	191.122	bars
Include Pressure Thrust		No	

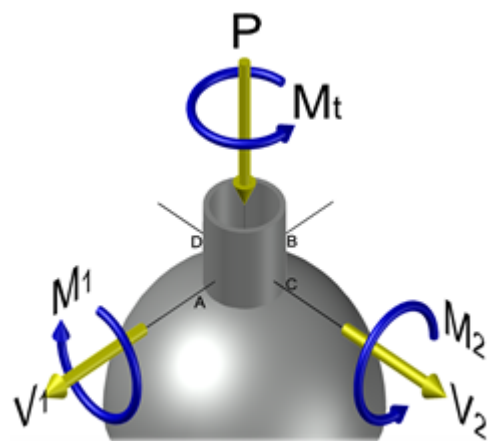
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-19579.9	Kgf
Longitudinal Shear (SUS)	(V1) V1	14684.9	Kgf
Circumferential Shear (SUS)	(Vc) V2	19579.9	Kgf
Circumferential Moment (SUS)	(Mc) M1	30.5	Kg-m.
Longitudinal Moment (SUS)	(Ml) M2	30.5	Kg-m.
Torsional Moment (SUS)	Mt	35.2	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(\text{t} - \text{ca}))} \\
 &= 790.576 + 2 \cdot 1.65 \cdot \sqrt{(912.75 (60.0 - 1.5))} \\
 &= 1553.125 \text{ mm.}
 \end{aligned}$$

WRC 107 Stress Calculation for SUStained loads:

Radial Load	P	-19579.9	Kgf
Circumferential Shear	(VC) V2	19579.9	Kgf
Longitudinal Shear	(VL) V1	14684.9	Kgf
Circumferential Moment	(MC) M1	30.5	Kg-m.
Longitudinal Moment	(ML) M2	30.5	Kg-m.
Torsional Moment	MT	35.2	Kg-m.

[illegible]



Long. P1 (SUS)	1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7
Long. Q (SUS)	1.9	-1.9	2.0	-2.0	1.9	-1.9	2.0	-2.0
Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear P1 (SUS)	2.6	2.6	-2.6	-2.6	-2.0	-2.0	2.0	2.0
Shear Q (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)	144.7	144.7	144.7	144.7	144.7	144.7	144.7	144.7
Pm+P1 (SUS)	148.9	148.9	148.9	148.9	148.2	148.2	148.2	148.2
Pm+P1+Q (Total)	154.0	145.6	154.1	145.6	153.5	145.2	153.7	145.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	144.74	148.00	Passed
Pm+P1 (SUS)	148.89	222.00	Passed
Pm+P1+Q (TOTAL)	154.14	443.99	Passed

*Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: B2-Shell outlet

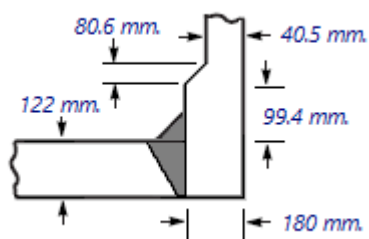
From: 20

Pressure for Reinforcement Calculations	P	191.034	bars
Temperature for Internal Pressure	Temp	260	°C
Parent Material		SA-387 11	+IT
Parent Allowable Stress at Temperature	Sv	148.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	148.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1700.00	mm.
Shell Finished (Minimum) Thickness	t	122.0000	mm.
Shell Internal Corrosion Allowance	c	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		950.00	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	+IT
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	148.00	N./mm ²
Allowable Stress At Ambient	Sna	148.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		400.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Hub Height of Integral Nozzle	h	99.4378	mm.
Height of Beveled Transition	L'	80.5622	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		180.0000	mm.
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	122.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: B2-Shell outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	325.476	mm.
Actual Thickness Used in Calculation	40.462	mm.



Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $191.03 \cdot 851.5001 / (148 \cdot 1.0 - 0.6 \cdot 191.03)$
= 119.1457 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $191.03 \cdot 164.24 / (148 \cdot 1.0 - 0.6 \cdot 191.03)$
= 22.9809 mm.

Intermediate Hub Nozzle Calculations:

The Diameter Limit is less than the Hub OD !

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 125.276/0.5773
= 216.9838 mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

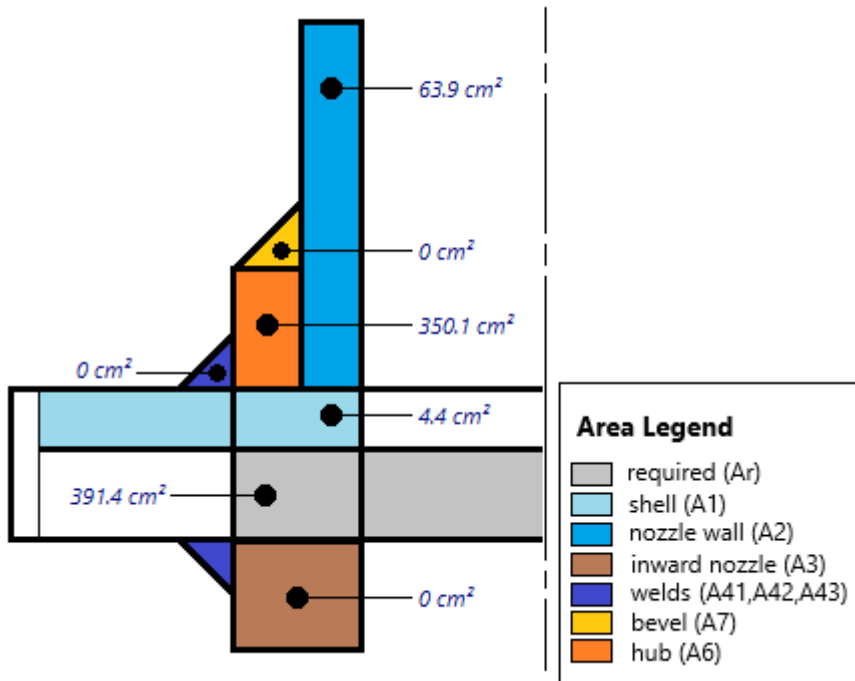
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	656.9512	mm.
Parallel to Vessel Wall, opening length	d	328.4756	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	277.4055	mm.

Weld Strength Reduction Factor [fr1]:
= min(1, S_n/S_v)
= min(1, 148.0/148.0)
= 1.000

Weld Strength Reduction Factor [fr2]:
= min(1, S_n/S_v)
= min(1, 148.0/148.0)
= 1.000

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(1.0, 1.0)
= 1.000



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	391.365	NA	NA
Area in Shell	A1	4.449	NA	NA
Area in Nozzle Wall	A2	63.925	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	350.067	NA	NA
TOTAL AREA AVAILABLE	Atot	418.441	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (328.4756 \cdot 119.1457 \cdot 1.0 + 2 \cdot 164.2378 \cdot 119.1457 \cdot 1.0 \cdot (1 - 1.0))$
 $= 391.365 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 328.476 (1.0 \cdot 120.5 - 1.0 \cdot 119.146) - 2 \cdot 164.238$
 $(1.0 \cdot 120.5 - 1.0 \cdot 119.1457) \cdot (1 - 1.0)$
 $= 4.449 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot ho) (tn - trn) fr2$
 $= (2 \cdot 200.0) (38.96 - 22.98) 1.0$
 $= 63.925 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= (Wo^2 - \text{Area Lost}) \cdot fr2 + (Wi - can / 0.707)^2 - \text{Area Lost} \cdot fr2$
 $= (10.0^2 - 1.) \cdot 1.0 + (0.0^2 - 0.0) \cdot 1.0$
 $= 0.000 \text{ cm}^2$



Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{lnp}, h_o, Hubht)) \cdot (Hubtk - t_n) \cdot fr2 \\ &= (2 \cdot \min(200.0, 200.0, 99.4)) \cdot (165.7 - 40.5) \cdot 1.0 \\ &= 350.067 \text{ cm}^2 \text{ Includes Bevel Area of } 100.925 \text{ cm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

$$\begin{aligned} \text{Wall Thickness for Internal/External pressures} & \quad ta = 24.4809 \text{ mm.} \\ \text{Wall Thickness per UG16(b),} & \quad tr16b = 3.0000 \text{ mm.} \\ \text{Wall Thickness, shell/head, internal pressure} & \quad trb1 = 120.6457 \text{ mm.} \\ \text{Wall Thickness} & \quad tb1 = \max(trb1, tr16b) = 120.6457 \text{ mm.} \\ \text{Wall Thickness} & \quad tb2 = \max(trb2, tr16b) = 3.0000 \text{ mm.} \\ \text{Wall Thickness per table UG-45} & \quad tb3 = 9.8312 \text{ mm.} \end{aligned}$$

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb3, \max(tb1, tb2)] \\ &= \min[9.831, \max(120.6457, 3.0)] \\ &= 9.8312 \text{ mm.} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(ta, tb) \\ &= \max(24.4809, 9.8312) \\ &= 24.4809 \text{ mm.} \end{aligned}$$

Available Nozzle Neck Thickness = $0.875 \cdot 40.462 = 35.404$ mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	33.2,	Allowable	:	148.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	336.8 N./mm ²	Passed
Occasional	:	36.0,	Allowable	:	196.8 N./mm ²	Passed
Shear	:	8.0,	Allowable	:	103.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -11 °C

Weld Size Calculations, Description: B2-Shell outlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned} &= \max(0, (A-A1+2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv) \\ &= \max(0, (391.3646 - 4.4485 + 2 \cdot 164.2378 \cdot 1.0 \cdot (1.0 \cdot 120.5 - 119.1457)) / 148) \\ &= 590616.69 \text{ Kg} \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A2+A4-(W_i \cdot Can/.707)^2 \cdot fr2) \cdot Sv \\ &= (413.9926 + 0.0 - 0.0 \cdot 1.0) \cdot 148 \\ &= 624765.12 \text{ Kg} \end{aligned}$$



Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S_v$$

$$= (413.9926 + 0.0 + 0.0 + (395.8131)) \cdot 148$$

$$= 1222095.25 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (413.9926 + 0.0 + 0.0 + 0.0 + (395.8131)) \cdot 148$$

$$= 1222095.25 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 656.9512 \cdot 10.0 \cdot 0.49 \cdot 148$$

$$= 76309. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 246.3567) \cdot (165.7378 - 1.5) \cdot 0.7 \cdot 148$$

$$= 1342795. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 656.9512 \cdot (122.0 - 1.5) \cdot 0.74 \cdot 148$$

$$= 1388660. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (76309 + 1342795) = 1419103 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (76309 + 0 + 1388660 + 0) = 1464969 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (76309 + 1388660 + 0) = 1464969 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 1419103 Kgf, must exceed W = 590616 Kgf or W1 = 624765 Kgf
 Path 2-2 = 1464968 Kgf, must exceed W = 590616 Kgf or W2 = 1222095 Kgf
 Path 3-3 = 1464968 Kgf, must exceed W = 590616 Kgf or W3 = 1222095 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 193 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 66.0333 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 388.0333 mm.

Input Echo, WRC297 Item 1, Description: B2-Shell outle:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	1700.000	mm.
Shell Thickness	122.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 11	
Vessel Cold S.I. Allowable	Smc	148.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	148.00 N./mm ²
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm.
Nozzle Diameter	325.476	mm.



Nozzle Thickness	180.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-336 F11	
Nozzle Cold S.I. Allowable	SNmc	148.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	148.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	191.03	bars
Radial Load	P	-13053.25	Kgf
Circumferential Shear	Vc	13053.25	Kgf
Longitudinal Shear	Vl	9789.91	Kgf
Circumferential Moment	Mc	13.57	Kg-m.
Longitudinal Moment	Ml	13.57	Kg-m.
Torsional Moment	Mt	15.66	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dv/Tv (15.934) is < 20
 check the limitations of WRC bulletin 297.*

*Warning - The ratio, Dn/Tn (3.808) is < 10
 check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 685.476 + 2 \cdot 1.65 \cdot \sqrt{(911.75(122.0 - 1.5))}$
 $= 1779.294 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1823.500 mm.
Nozzle Outside Diameter	(d) =	685.476 mm.
Vessel Thickness used	(T) =	120.500 mm.
Nozzle Thickness used	(t) =	178.500 mm.
T / t	=	0.675
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.462$$

Nr/P	=	0.090
Mr/P	=	0.165
M0/P	=	0.050
N0/P	=	0.112
MrD/Mc	=	0.286
NrDL/Mc	=	0.114
M0d/Mc	=	0.094
N0DL/Mc	=	0.054
MrD/Ml	=	0.149
NrDL/Ml	=	0.083
M0D/Ml	=	0.050
N0DL/Ml	=	0.133

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0



Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	126	145	126	145
Outplane Stress Summary	129	143	129	143

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	8	-8	8	-8
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	63	63	63	63
Inplane Stress Summary	72	55	72	55

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	0	0	0	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	129	143	129	143

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	63	63	63	63
Outplane Stress Summary	66	61	66	61

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	8	-8	8	-8
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	126	145	126	145
Inplane Stress Summary	135	137	135	137

Vessel Stresses (N./mm²):



CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)				
	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	0	0	0	0

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)				
	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	135	137	135	137

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)				
	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	13	36	13	36
Hoop Stress Summary	14	37	14	37

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)				
	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	6	6	6	6
Axial Stress Summary	10	4	10	4

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)				
	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)				
	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	14	37	14	37

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)				
	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0



Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	13	36	13	36

Hoop Stress Summary	14	37	14	37
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	6	6	6	6

Axial Stress Summary	10	4	10	4
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	0	0	0	0
----------------------	---	---	---	---

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Two * Max Shear Stress	14	37	14	37
------------------------	----	----	----	----

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	126	145	126	145	126	145	126	145
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	2	-2	2	-2	8	-8	8	-8
Long. Pm (SUS)	63	63	63	63	63	63	63	63
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	8	-8	8	-8	2	-2	2	-2
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	126.0	145.0	126.0	145.0	126.0	145.0	126.0	145.0
Pm+Pl (SUS)	126.0	145.0	126.0	145.0	126.0	145.0	126.0	145.0
Pm+Pl+Q (Total)	128.0	143.0	128.0	143.0	134.0	137.0	134.0	137.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
------------------------	-----------	----------------	--------



Pm (SUS)	145.00	148.00	Passed
Pm+Pl (SUS)	145.00	222.00	Passed
Pm+Pl+Q (TOTAL)	143.00	443.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	13	36	13	36	13	36	13	36
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	6	6	6	6	6	6	6	6
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	2	-2	2	-2	2	-2	2	-2
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	13.0	36.0	13.0	36.0	13.0	36.0	13.0	36.0
Pm+Pl (SUS)	13.0	36.0	13.0	36.0	13.0	36.0	13.0	36.0
Pm+Pl+Q (Total)	13.0	36.0	13.0	36.0	13.0	36.0	13.0	36.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	36.00	148.00	Passed
Pm+Pl (SUS)	36.00	222.00	Passed
Pm+Pl+Q (TOTAL)	36.00	443.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: B3-Shell outlet

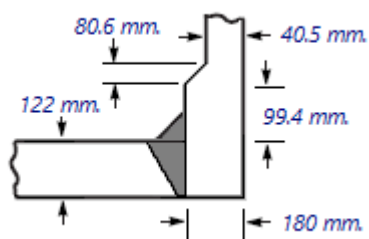
From: 20

Pressure for Reinforcement Calculations	P	191.034	bars
Temperature for Internal Pressure	Temp	260	°C
Parent Material		SA-387 11	+IT
Parent Allowable Stress at Temperature	Sv	148.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	148.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1700.00	mm.
Shell Finished (Minimum) Thickness	t	122.0000	mm.
Shell Internal Corrosion Allowance	c	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		950.00	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	+IT
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	148.00	N./mm ²
Allowable Stress At Ambient	Sna	148.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		360.00	deg
Diameter		400.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Hub Height of Integral Nozzle	h	99.4378	mm.
Height of Beveled Transition	L'	80.5622	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		180.0000	mm.
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	122.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: B3-Shell outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	325.476	mm.
Actual Thickness Used in Calculation	40.462	mm.



Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $191.03 \cdot 851.5001 / (148 \cdot 1.0 - 0.6 \cdot 191.03)$
= 119.1457 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $191.03 \cdot 164.24 / (148 \cdot 1.0 - 0.6 \cdot 191.03)$
= 22.9809 mm.

Intermediate Hub Nozzle Calculations:

The Diameter Limit is less than the Hub OD !

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / $\tan(30)$
= $125.276 / 0.5773$
= 216.9838 mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

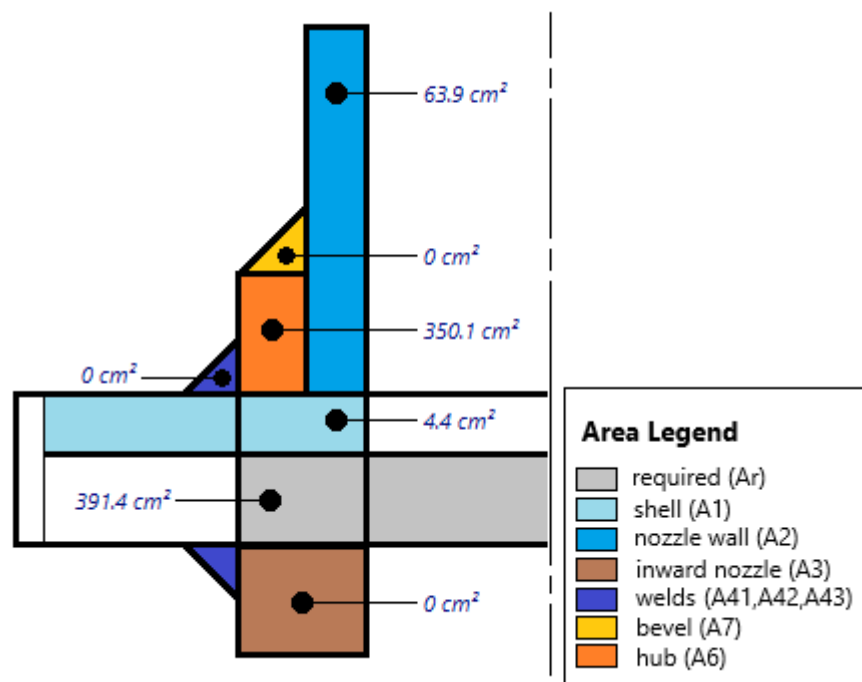
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	656.9512	mm.
Parallel to Vessel Wall, opening length	d	328.4756	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	277.4055	mm.

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n / S_v)$
= $\min(1, 148.0 / 148.0)$
= 1.000

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 148.0 / 148.0)$
= 1.000

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(1.0, 1.0)$
= 1.000



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	391.365	NA	NA
Area in Shell	A1	4.449	NA	NA
Area in Nozzle Wall	A2	63.925	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	350.067	NA	NA
TOTAL AREA AVAILABLE	Atot	418.441	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (328.4756 \cdot 119.1457 \cdot 1.0 + 2 \cdot 164.2378 \cdot 119.1457 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 391.365 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 328.476 (1.0 \cdot 120.5 - 1.0 \cdot 119.146) - 2 \cdot 164.238$$

$$(1.0 \cdot 120.5 - 1.0 \cdot 119.1457) \cdot (1 - 1.0)$$

$$= 4.449 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot ho) (tn - trn) fr2$$

$$= (2 \cdot 200.0) (38.96 - 22.98) 1.0$$

$$= 63.925 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= (Wo^2 - \text{Area Lost}) \cdot fr2 + (Wi - can / 0.707)^2 - \text{Area Lost} \cdot fr2$$

$$= (10.0^2 - 1.0) \cdot 1.0 + (0.0^2 - 0.0) \cdot 1.0$$

$$= 0.000 \text{ cm}^2$$



Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubt} - t_n) \cdot f_{r2} \\ &= (2 \cdot \min(200.0, 200.0, 99.4)) \cdot (165.7 - 40.5) \cdot 1.0 \\ &= 350.067 \text{ cm}^2 \text{ Includes Bevel Area of } 100.925 \text{ cm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

$$\begin{aligned} \text{Wall Thickness for Internal/External pressures} & \quad t_a = 24.4809 \text{ mm.} \\ \text{Wall Thickness per UG16(b),} & \quad t_{r16b} = 3.0000 \text{ mm.} \\ \text{Wall Thickness, shell/head, internal pressure} & \quad t_{rb1} = 120.6457 \text{ mm.} \\ \text{Wall Thickness} & \quad t_{b1} = \max(t_{rb1}, t_{r16b}) = 120.6457 \text{ mm.} \\ \text{Wall Thickness} & \quad t_{b2} = \max(t_{rb2}, t_{r16b}) = 3.0000 \text{ mm.} \\ \text{Wall Thickness per table UG-45} & \quad t_{b3} = 9.8312 \text{ mm.} \end{aligned}$$

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[t_{b3}, \max(t_{b1}, t_{b2})] \\ &= \min[9.831, \max(120.6457, 3.0)] \\ &= 9.8312 \text{ mm.} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(t_a, t_b) \\ &= \max(24.4809, 9.8312) \\ &= 24.4809 \text{ mm.} \end{aligned}$$

Available Nozzle Neck Thickness = $0.875 \cdot 40.462 = 35.404$ mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	33.2,	Allowable	:	148.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	336.8 N./mm ²	Passed
Occasional	:	36.0,	Allowable	:	196.8 N./mm ²	Passed
Shear	:	8.0,	Allowable	:	103.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -11 °C

Weld Size Calculations, Description: B3-Shell outlet

Intermediate Calc. for nozzle/shell Welds $T_{min} = 19.0000$ mm.

Results Per UW-16.1:

Required Thickness Actual Thickness
Nozzle Weld $6.0000 = \text{Min per Code } 7.0700 = 0.7 \cdot W_o \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned} &= \max(0, (A - A1 + 2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v) \\ &= \max(0, (391.3646 - 4.4485 + 2 \cdot 164.2378 \cdot 1.0 \cdot (1.0 \cdot 120.5 - 119.1457)) / 148) \\ &= 590616.69 \text{ Kg} \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A2 + A4 - (W_i - C_{an} / .707)^2 \cdot f_{r2}) \cdot S_v \\ &= (413.9926 + 0.0 - 0.0 \cdot 1.0) \cdot 148 \\ &= 624765.12 \text{ Kg} \end{aligned}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$$

$$= (413.9926 + 0.0 + 0.0 + (395.8131)) \cdot 148$$

$$= 1222095.25 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (413.9926 + 0.0 + 0.0 + 0.0 + (395.8131)) \cdot 148$$

$$= 1222095.25 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 656.9512 \cdot 10.0 \cdot 0.49 \cdot 148$$

$$= 76309. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 246.3567) \cdot (165.7378 - 1.5) \cdot 0.7 \cdot 148$$

$$= 1342795. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 656.9512 \cdot (122.0 - 1.5) \cdot 0.74 \cdot 148$$

$$= 1388660. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (76309 + 1342795) = 1419103 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (76309 + 0 + 1388660 + 0) = 1464969 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (76309 + 1388660 + 0) = 1464969 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 1419103 Kgf, must exceed W = 590616 Kgf or W1 = 624765 Kgf
 Path 2-2 = 1464968 Kgf, must exceed W = 590616 Kgf or W2 = 1222095 Kgf
 Path 3-3 = 1464968 Kgf, must exceed W = 590616 Kgf or W3 = 1222095 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 193 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 66.0333 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 388.0333 mm.

Input Echo, WRC297 Item 1, Description: B3-Shell outle:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	1700.000	mm.
Shell Thickness	122.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 11	
Vessel Cold S.I. Allowable	Smc	148.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	148.00 N./mm ²
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm.
Nozzle Diameter	325.476	mm.



Nozzle Thickness	180.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-336 F11	
Nozzle Cold S.I. Allowable	SNmc	148.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	148.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	191.03	bars
Radial Load	P	-13053.25	Kgf
Circumferential Shear	Vc	13053.25	Kgf
Longitudinal Shear	Vl	9789.91	Kgf
Circumferential Moment	Mc	13.57	Kg-m.
Longitudinal Moment	Ml	13.57	Kg-m.
Torsional Moment	Mt	15.66	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dv/Tv (15.934) is < 20
 check the limitations of WRC bulletin 297.
 Warning - The ratio, Dn/Tn (3.808) is < 10
 check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 685.476 + 2 \cdot 1.65 \cdot \sqrt{(911.75(122.0 - 1.5))}$
 $= 1779.294 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1823.500 mm.
Nozzle Outside Diameter	(d) =	685.476 mm.
Vessel Thickness used	(T) =	120.500 mm.
Nozzle Thickness used	(t) =	178.500 mm.
T / t	=	0.675
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.462$$

Nr/P	=	0.090
Mr/P	=	0.165
M0/P	=	0.050
N0/P	=	0.112
MrD/Mc	=	0.286
NrDL/Mc	=	0.114
M0d/Mc	=	0.094
N0DL/Mc	=	0.054
MrD/Ml	=	0.149
NrDL/Ml	=	0.083
M0D/Ml	=	0.050
N0DL/Ml	=	0.133

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0



Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	126	145	126	145
Outplane Stress Summary	129	143	129	143

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	8	-8	8	-8
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	63	63	63	63
Inplane Stress Summary	72	55	72	55

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	0	0	0	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	129	143	129	143

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	63	63	63	63
Outplane Stress Summary	66	61	66	61

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	8	-8	8	-8
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	126	145	126	145
Inplane Stress Summary	135	137	135	137

Vessel Stresses (N./mm²):



CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	0	0	0	0

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	135	137	135	137

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	13	36	13	36
Hoop Stress Summary	14	37	14	37

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	6	6	6	6
Axial Stress Summary	10	4	10	4

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	14	37	14	37

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0

Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	13	36	13	36

Hoop Stress Summary	14	37	14	37
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	6	6	6	6

Axial Stress Summary	10	4	10	4
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	0	0	0	0
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Two * Max Shear Stress	14	37	14	37
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WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	126	145	126	145	126	145	126	145
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	2	-2	2	-2	8	-8	8	-8
Long. Pm (SUS)	63	63	63	63	63	63	63	63
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	8	-8	8	-8	2	-2	2	-2
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	126.0	145.0	126.0	145.0	126.0	145.0	126.0	145.0
Pm+Pl (SUS)	126.0	145.0	126.0	145.0	126.0	145.0	126.0	145.0
Pm+Pl+Q (Total)	128.0	143.0	128.0	143.0	134.0	137.0	134.0	137.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
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Pm (SUS)	145.00	148.00	Passed
Pm+Pl (SUS)	145.00	222.00	Passed
Pm+Pl+Q (TOTAL)	143.00	443.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	13	36	13	36	13	36	13	36
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	6	6	6	6	6	6	6	6
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	2	-2	2	-2	2	-2	2	-2
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	13.0	36.0	13.0	36.0	13.0	36.0	13.0	36.0
Pm+Pl (SUS)	13.0	36.0	13.0	36.0	13.0	36.0	13.0	36.0
Pm+Pl+Q (Total)	13.0	36.0	13.0	36.0	13.0	36.0	13.0	36.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	36.00	148.00	Passed
Pm+Pl (SUS)	36.00	222.00	Passed
Pm+Pl+Q (TOTAL)	36.00	443.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: A2-Shell inlet

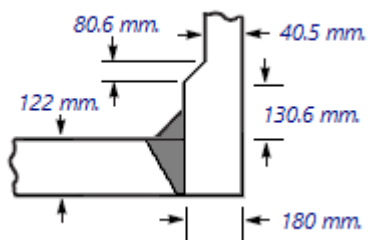
From: 40

Pressure for Reinforcement Calculations	P	191.008	bars
Temperature for Internal Pressure	Temp	260	°C
Parent Material		SA-387 11	+IT
Parent Allowable Stress at Temperature	Sv	148.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	148.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1700.00	mm.
Shell Finished (Minimum) Thickness	t	122.0000	mm.
Shell Internal Corrosion Allowance	c	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		12350.00	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	+IT
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	148.00	N./mm ²
Allowable Stress At Ambient	Sna	148.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		400.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Hub Height of Integral Nozzle	h	130.5620	mm.
Height of Beveled Transition	L'	80.5622	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		180.0000	mm.
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	122.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: A2-Shell inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	325.476	mm.
Actual Thickness Used in Calculation	40.462	mm.



Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $191.01 \cdot 851.5001 / (148 \cdot 1.0 - 0.6 \cdot 191.01)$
= 119.1286 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $191.01 \cdot 164.24 / (148 \cdot 1.0 - 0.6 \cdot 191.01)$
= 22.9776 mm.

Intermediate Hub Nozzle Calculations:

The Diameter Limit is less than the Hub OD !

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 125.276/0.5773
= 216.9838 mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

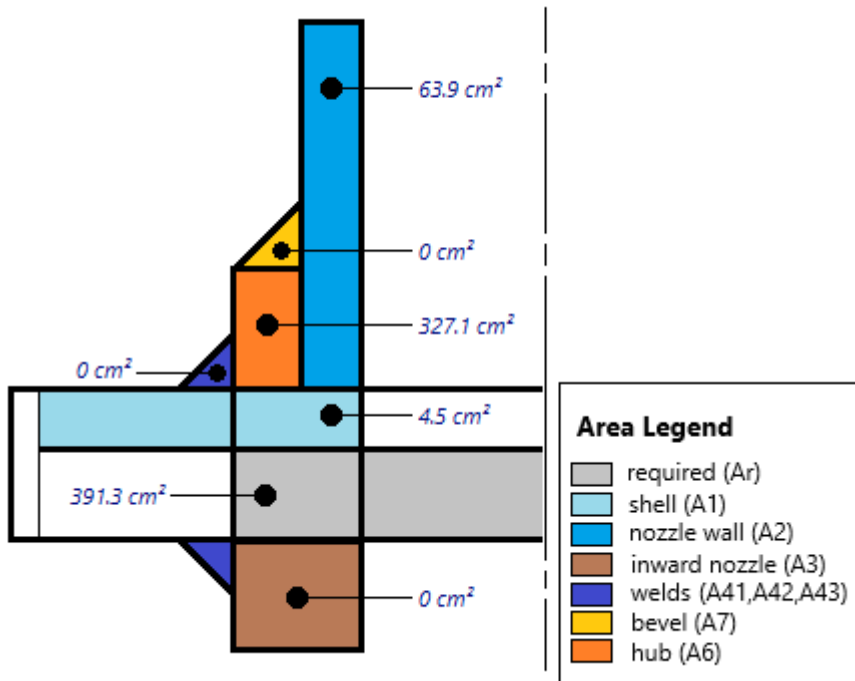
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	656.9512	mm.
Parallel to Vessel Wall, opening length	d	328.4756	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	301.2500	mm.

Weld Strength Reduction Factor [fr1]:
= min(1, S_n/S_v)
= min(1, 148.0/148.0)
= 1.000

Weld Strength Reduction Factor [fr2]:
= min(1, S_n/S_v)
= min(1, 148.0/148.0)
= 1.000

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(1.0, 1.0)
= 1.000



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	391.308	NA	NA
Area in Shell	A1	4.505	NA	NA
Area in Nozzle Wall	A2	63.938	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	327.125	NA	NA
TOTAL AREA AVAILABLE	Atot	395.568	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (328.4756 \cdot 119.1286 \cdot 1.0 + 2 \cdot 164.2378 \cdot 119.1286 \cdot 1.0 \cdot (1 - 1.0))$
 $= 391.308 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 328.476 (1.0 \cdot 120.5 - 1.0 \cdot 119.129) - 2 \cdot 164.238$
 $(1.0 \cdot 120.5 - 1.0 \cdot 119.1286) \cdot (1 - 1.0)$
 $= 4.505 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot ho) (tn - trn) fr2$
 $= (2 \cdot 200.0) (38.96 - 22.98) 1.0$
 $= 63.938 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= (Wo^2 - \text{Area Lost}) \cdot fr2 + (Wi - can / 0.707)^2 - \text{Area Lost} \cdot fr2$
 $= (10.0^2 - 1.) \cdot 1.0 + (0.0^2 - 0.0) \cdot 1.0$
 $= 0.000 \text{ cm}^2$



Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, Hubht)) \cdot (Hubtk - t_n) \cdot fr_2$$

$$= (2 \cdot \min(200.0, 200.0, 130.6)) \cdot (165.7 - 40.5) \cdot 1.0$$

$$= 327.125 \text{ cm}^2 \text{ Includes Bevel Area of } 0.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 24.4776 \text{ mm.}$
 Wall Thickness per UG16(b), $tr_{16b} = 3.0000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb_1 = 120.6286 \text{ mm.}$
 Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 120.6286 \text{ mm.}$
 Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 3.0000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb_3 = 9.8312 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[9.831, \max(120.6286, 3.0)]$$

$$= 9.8312 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(24.4776, 9.8312)$$

$$= 24.4776 \text{ mm.}$$

Available Nozzle Neck Thickness = $0.875 \cdot 40.462 = 35.404 \text{ mm.} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	33.2,	Allowable	:	148.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	336.8 N./mm ²	Passed
Occasional	:	36.0,	Allowable	:	196.8 N./mm ²	Passed
Shear	:	8.0,	Allowable	:	103.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -11 °C

Weld Size Calculations, Description: A2-Shell inlet

Intermediate Calc. for nozzle/shell Welds $T_{min} = 19.0000 \text{ mm.}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot fr_1 \cdot (E1 \cdot t - tr)) \cdot Sv)$$

$$= \max(0, (391.3083 - 4.5048 + 2 \cdot 164.2378 \cdot 1.0 \cdot (1.0 \cdot 120.5 - 119.1286)) / 148)$$

$$= 590531.81 \text{ Kg}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A4-(W_i \cdot Can / .707)^2 \cdot fr_2) \cdot Sv$$

$$= (391.0632 + 0.0 - 0.0 \cdot 1.0) \cdot 148$$

$$= 590161.81 \text{ Kg}$$



Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$$

$$= (391.0632 + 0.0 + 0.0 + (395.8131)) \cdot 148$$

$$= 1187491.88 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (391.0632 + 0.0 + 0.0 + 0.0 + (395.8131)) \cdot 148$$

$$= 1187491.88 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 656.9512 \cdot 10.0 \cdot 0.49 \cdot 148$$

$$= 76309. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 246.3567) \cdot (165.7378 - 1.5) \cdot 0.7 \cdot 148$$

$$= 1342795. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 656.9512 \cdot (122.0 - 1.5) \cdot 0.74 \cdot 148$$

$$= 1388660. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (76309 + 1342795) = 1419103 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (76309 + 0 + 1388660 + 0) = 1464969 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (76309 + 1388660 + 0) = 1464969 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 1419103 Kgf, must exceed W = 590531 Kgf or W1 = 590161 Kgf
 Path 2-2 = 1464968 Kgf, must exceed W = 590531 Kgf or W2 = 1187491 Kgf
 Path 3-3 = 1464968 Kgf, must exceed W = 590531 Kgf or W3 = 1187491 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 192 bars

The Drop for this Nozzle is : 66.0333 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 388.0333 mm.

Input Echo, WRC297 Item 1, Description: A2-Shell inlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	1700.000	mm.
Shell Thickness	122.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 11	
Vessel Cold S.I. Allowable	Smc 148.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 148.00	N./mm ²
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm.
Nozzle Diameter	325.476	mm.
Nozzle Thickness	180.0000	mm.
Nozzle Stress Concentration Factor	1.000	



Nozzle Material SA-336 F11
 Nozzle Cold S.I. Allowable SNmc 148.00 N./mm²
 Nozzle Hot S.I. Allowable SNmh 148.00 N./mm²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	191.01	bars
Radial Load	P	-13053.25	Kgf
Circumferential Shear	Vc	13053.25	Kgf
Longitudinal Shear	Vl	9789.91	Kgf
Circumferential Moment	Mc	13.57	Kg-m.
Longitudinal Moment	Ml	13.57	Kg-m.
Torsional Moment	Mt	15.66	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dv/Tv (15.934) is < 20
 check the limitations of WRC bulletin 297.*
*Warning - The ratio, Dn/Tn (3.808) is < 10
 check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 685.476 + 2 \cdot 1.65 \cdot \sqrt{(911.75(122.0 - 1.5))}$
 $= 1779.294 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1823.500 mm.
Nozzle Outside Diameter	(d) =	685.476 mm.
Vessel Thickness used	(T) =	120.500 mm.
Nozzle Thickness used	(t) =	178.500 mm.
T / t	=	0.675
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.462$$

Nr/P	=	0.090
Mr/P	=	0.165
M0/P	=	0.050
N0/P	=	0.112
MrD/Mc	=	0.286
NrDL/Mc	=	0.114
M0d/Mc	=	0.094
N0DL/Mc	=	0.054
MrD/Ml	=	0.149
NrDL/Ml	=	0.083
M0D/Ml	=	0.050
N0DL/Ml	=	0.133

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	0	0	0	0
Outplane Bending (Ml)	0	0	0	0



Normal Pressure Stress	126	145	126	145
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Outplane Stress Summary	129	143	129	143
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	8	-8	8	-8
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	63	63	63	63

Inplane Stress Summary	72	55	72	55
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0

Shear Stress Summary	0	0	0	0
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Two * Max Shear Stress	129	143	129	143
------------------------	-----	-----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	63	63	63	63

Outplane Stress Summary	66	61	66	61
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	8	-8	8	-8
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	126	145	126	145

Inplane Stress Summary	135	137	135	137
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to	Cu Left	Cl Left	Du Right	Dl Right
--	------------	------------	-------------	-------------

circumferential plane)	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	0	0	0	0

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	135	137	135	137

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	13	36	13	36
Hoop Stress Summary	14	37	14	37

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	6	6	6	6
Axial Stress Summary	10	4	10	4

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	14	37	14	37

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0

Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	13	36	13	36
Hoop Stress Summary	14	37	14	37

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	6	6	6	6
Axial Stress Summary	10	4	10	4

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	14	37	14	37

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	126	145	126	145	126	145	126	145
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	2	-2	2	-2	8	-8	8	-8
Long. Pm (SUS)	63	63	63	63	63	63	63	63
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	8	-8	8	-8	2	-2	2	-2
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	126.0	145.0	126.0	145.0	126.0	145.0	126.0	145.0
Pm+Pl (SUS)	126.0	145.0	126.0	145.0	126.0	145.0	126.0	145.0
Pm+Pl+Q (Total)	128.0	143.0	128.0	143.0	134.0	137.0	134.0	137.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	145.00	148.00	Passed
Pm+Pl (SUS)	145.00	222.00	Passed



Pm+Pl+Q (TOTAL)	143.00	443.99	Passed
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Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	13	36	13	36	13	36	13	36
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	6	6	6	6	6	6	6	6
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	2	-2	2	-2	2	-2	2	-2
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	13.0	36.0	13.0	36.0	13.0	36.0	13.0	36.0
Pm+Pl (SUS)	13.0	36.0	13.0	36.0	13.0	36.0	13.0	36.0
Pm+Pl+Q (Total)	13.0	36.0	13.0	36.0	13.0	36.0	13.0	36.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	36.00	148.00	Passed
Pm+Pl (SUS)	36.00	222.00	Passed
Pm+Pl+Q (TOTAL)	36.00	443.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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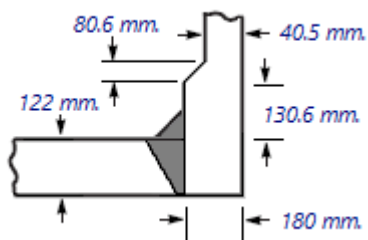
Input, Nozzle Desc: A3-Shell inlet From: 40

Pressure for Reinforcement Calculations	P	191.008	bars
Temperature for Internal Pressure	Temp	260	°C
Parent Material		SA-387 11	+IT
Parent Allowable Stress at Temperature	Sv	148.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	148.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1700.00	mm.
Shell Finished (Minimum) Thickness	t	122.0000	mm.
Shell Internal Corrosion Allowance	c	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		12350.00	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	+IT
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	148.00	N./mm ²
Allowable Stress At Ambient	Sna	148.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		360.00	deg
Diameter		400.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Hub Height of Integral Nozzle	h	130.5620	mm.
Height of Beveled Transition	L'	80.5622	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		180.0000	mm.
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	122.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: A3-Shell inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	325.476	mm.
Actual Thickness Used in Calculation	40.462	mm.



Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $191.01 \cdot 851.5001 / (148 \cdot 1.0 - 0.6 \cdot 191.01)$
= 119.1286 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $191.01 \cdot 164.24 / (148 \cdot 1.0 - 0.6 \cdot 191.01)$
= 22.9776 mm.

Intermediate Hub Nozzle Calculations:

The Diameter Limit is less than the Hub OD !

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 125.276/0.5773
= 216.9838 mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

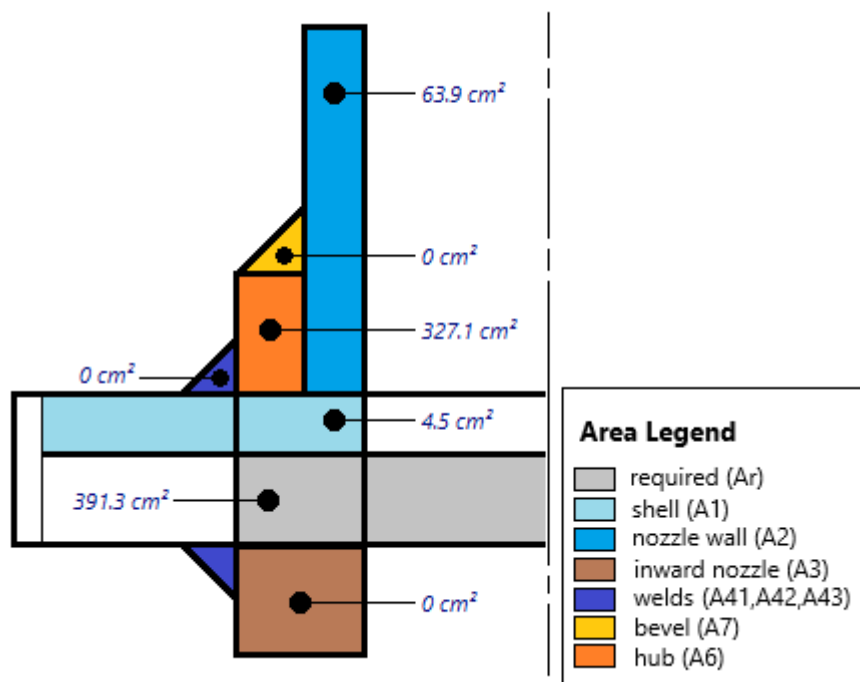
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	656.9512	mm.
Parallel to Vessel Wall, opening length	d	328.4756	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	301.2500	mm.

Weld Strength Reduction Factor [fr1]:
= min(1, S_n/S_v)
= min(1, 148.0/148.0)
= 1.000

Weld Strength Reduction Factor [fr2]:
= min(1, S_n/S_v)
= min(1, 148.0/148.0)
= 1.000

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(1.0, 1.0)
= 1.000



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	391.308	NA	NA
Area in Shell	A1	4.505	NA	NA
Area in Nozzle Wall	A2	63.938	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds A41+A42+A43		0.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	327.125	NA	NA
TOTAL AREA AVAILABLE	Atot	395.568	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (328.4756 \cdot 119.1286 \cdot 1.0 + 2 \cdot 164.2378 \cdot 119.1286 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 391.308 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 328.476 (1.0 \cdot 120.5 - 1.0 \cdot 119.129) - 2 \cdot 164.238$$

$$(1.0 \cdot 120.5 - 1.0 \cdot 119.1286) \cdot (1 - 1.0)$$

$$= 4.505 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot ho) (tn - trn) \cdot fr2$$

$$= (2 \cdot 200.0) (38.96 - 22.98) \cdot 1.0$$

$$= 63.938 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= (Wo^2 - \text{Area Lost}) \cdot fr2 + (Wi - can / 0.707)^2 - \text{Area Lost} \cdot fr2$$

$$= (10.0^2 - 1.) \cdot 1.0 + (0.0^2 - 0.0) \cdot 1.0$$

$$= 0.000 \text{ cm}^2$$



Area Available in the Hub Section [A6]:
 $= (2 \cdot \min(T_{lnp}, h_o, Hubht)) \cdot (Hubtk - t_n) \cdot fr2$
 $= (2 \cdot \min(200.0, 200.0, 130.6)) \cdot (165.7 - 40.5) \cdot 1.0$
 $= 327.125 \text{ cm}^2$ Includes Bevel Area of 0.000 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 24.4776 \text{ mm.}$
 Wall Thickness per UG16(b), $tr16b = 3.0000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb1 = 120.6286 \text{ mm.}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 120.6286 \text{ mm.}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb3 = 9.8312 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[9.831, \max(120.6286, 3.0)]$
 $= 9.8312 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(24.4776, 9.8312)$
 $= 24.4776 \text{ mm.}$

Available Nozzle Neck Thickness = $0.875 \cdot 40.462 = 35.404 \text{ mm.} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	33.2,	Allowable	:	148.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	336.8 N./mm ²	Passed
Occasional	:	36.0,	Allowable	:	196.8 N./mm ²	Passed
Shear	:	8.0,	Allowable	:	103.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -11 °C

Weld Size Calculations, Description: A3-Shell inlet

Intermediate Calc. for nozzle/shell Welds $T_{min} = 19.0000 \text{ mm.}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$
 $= \max(0, (391.3083 - 4.5048 + 2 \cdot 164.2378 \cdot 1.0 \cdot (1.0 \cdot 120.5 - 119.1286)) / 148)$
 $= 590531.81 \text{ Kg}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (391.0632 + 0.0 - 0.0 \cdot 1.0) \cdot 148$
 $= 590161.81 \text{ Kg}$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$$

$$= (391.0632 + 0.0 + 0.0 + (395.8131)) \cdot 148$$

$$= 1187491.88 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (391.0632 + 0.0 + 0.0 + 0.0 + (395.8131)) \cdot 148$$

$$= 1187491.88 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 656.9512 \cdot 10.0 \cdot 0.49 \cdot 148$$

$$= 76309. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 246.3567) \cdot (165.7378 - 1.5) \cdot 0.7 \cdot 148$$

$$= 1342795. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 656.9512 \cdot (122.0 - 1.5) \cdot 0.74 \cdot 148$$

$$= 1388660. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (76309 + 1342795) = 1419103 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (76309 + 0 + 1388660 + 0) = 1464969 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (76309 + 1388660 + 0) = 1464969 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 1419103 Kgf, must exceed W = 590531 Kgf or W1 = 590161 Kgf
 Path 2-2 = 1464968 Kgf, must exceed W = 590531 Kgf or W2 = 1187491 Kgf
 Path 3-3 = 1464968 Kgf, must exceed W = 590531 Kgf or W3 = 1187491 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 192 bars

The Drop for this Nozzle is : 66.0333 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 388.0333 mm.

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 328.48 mm.

Input Echo, WRC297 Item 1, Description: A3-Shell inlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	1700.000	mm.
Shell Thickness	122.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 11	
Vessel Cold S.I. Allowable	Smc	148.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	148.00 N./mm ²
Diameter Basis for Nozzle	ID	



Nozzle Corrosion Allowance	1.5000	mm.
Nozzle Diameter	325.476	mm.
Nozzle Thickness	180.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-336 F11	
Nozzle Cold S.I. Allowable	SNmc	148.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	148.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUStained loads.

Design Internal Pressure	Dp	191.01	bars
Radial Load	P	-13053.25	Kgf
Circumferential Shear	Vc	13053.25	Kgf
Longitudinal Shear	Vl	9789.91	Kgf
Circumferential Moment	Mc	13.57	Kg-m.
Longitudinal Moment	Ml	13.57	Kg-m.
Torsional Moment	Mt	15.66	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dv/Tv (15.934) is < 20
 check the Limitations of WRC bulletin 297.*

*Warning - The ratio, Dn/Tn (3.808) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 685.476 + 2 \cdot 1.65 \cdot \sqrt{911.75(122.0 - 1.5)}$$

$$= 1779.294 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1823.500	mm.
Nozzle Outside Diameter	(d) =	685.476	mm.
Vessel Thickness used	(T) =	120.500	mm.
Nozzle Thickness used	(t) =	178.500	mm.
T / t	=	0.675	
d / t	=	10.000	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.462$$

Nr/P	=	0.090
Mr/P	=	0.165
M0/P	=	0.050
N0/P	=	0.112
MrD/Mc	=	0.286
NrDL/Mc	=	0.114
M0d/Mc	=	0.094
N0DL/Mc	=	0.054
MrD/Ml	=	0.149
NrDL/Ml	=	0.083
M0D/Ml	=	0.050
N0DL/Ml	=	0.133

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	2	-2	2	-2



Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	126	145	126	145

Outplane Stress Summary	129	143	129	145
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	8	-8	8	-8
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	63	63	63	63

Inplane Stress Summary	72	55	72	55
------------------------	----	----	----	----

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0

Shear Stress Summary	0	0	0	0
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	129	143	129	143
------------------------	-----	-----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	63	63	63	63

Outplane Stress Summary	66	61	66	61
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	8	-8	8	-8
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	126	145	126	145

Inplane Stress Summary	135	137	135	137
------------------------	-----	-----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	0	0	0	0

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	135	137	135	137

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	13	36	13	36
Hoop Stress Summary	14	37	14	37

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	6	6	6	6
Axial Stress Summary	10	4	10	4

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	14	37	14	37

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	13	36	13	36

Hoop Stress Summary	14	37	14	37
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	6	6	6	6

Axial Stress Summary	10	4	10	4
----------------------	----	---	----	---

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	0	0	0	0
----------------------	---	---	---	---

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	14	37	14	37
------------------------	----	----	----	----

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	126	145	126	145	126	145	126	145
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	2	-2	2	-2	8	-8	8	-8
Long. Pm (SUS)	63	63	63	63	63	63	63	63
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	8	-8	8	-8	2	-2	2	-2
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	126.0	145.0	126.0	145.0	126.0	145.0	126.0	145.0
Pm+Pl (SUS)	126.0	145.0	126.0	145.0	126.0	145.0	126.0	145.0
Pm+Pl+Q (Total)	128.0	143.0	128.0	143.0	134.0	137.0	134.0	137.0

Vessel Stress Summation Comparison (N./mm²):

Type of	Max. S.I.	S.I. Allowable	Result
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Stress Int.			
Pm (SUS)	145.00	148.00	Passed
Pm+Pl (SUS)	145.00	222.00	Passed
Pm+Pl+Q (TOTAL)	143.00	443.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	13	36	13	36	13	36	13	36
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	6	6	6	6	6	6	6	6
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	2	-2	2	-2	2	-2	2	-2
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	13.0	36.0	13.0	36.0	13.0	36.0	13.0	36.0
Pm+Pl (SUS)	13.0	36.0	13.0	36.0	13.0	36.0	13.0	36.0
Pm+Pl+Q (Total)	13.0	36.0	13.0	36.0	13.0	36.0	13.0	36.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	36.00	148.00	Passed
Pm+Pl (SUS)	36.00	222.00	Passed
Pm+Pl+Q (TOTAL)	36.00	443.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: B1-Tube outlet

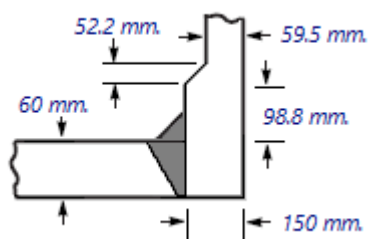
From: 50

Pressure for Reinforcement Calculations	P	191.000	bars
Temperature for Internal Pressure	Temp	260	°C
Parent Material		SA-387 11	+IT
Parent Allowable Stress at Temperature	Sv	148.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	148.00	N./mm ²
Inside Diameter of Hemispherical Head	D	1764.00	mm.
Head Finished (Minimum) Thickness	t	60.0000	mm.
Head Internal Corrosion Allowance	c	1.5000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	+IT
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	148.00	N./mm ²
Allowable Stress At Ambient	Sna	148.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		600.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Hub Height of Integral Nozzle	h	98.7569	mm.
Height of Beveled Transition	L'	52.2431	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		150.0000	mm.
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	58.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: B1-Tube outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	490.576	mm.
Actual Thickness Used in Calculation	59.512	mm.



Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
= $P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
= $191.0 \cdot 883.5001 / (2 \cdot 148 \cdot 1.0 - 0.2 \cdot 191.0)$
= 57.7596 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $191.0 \cdot 246.79 / (148 \cdot 1.0 - 0.6 \cdot 191.0)$
= 34.5250 mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / $\tan(30)$
= 90.488/0.5773
= 156.7295 mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

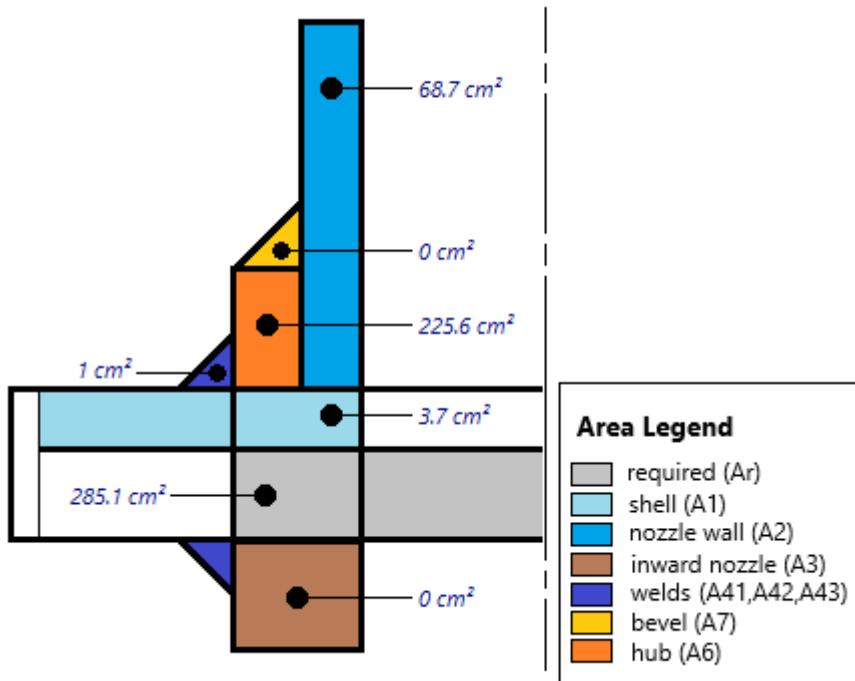
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	987.1512	mm.
Parallel to Vessel Wall, opening length	d	493.5756	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	146.2500	mm.

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n/S_v)$
= $\min(1, 148.0/148.0)$
= 1.000

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n/S_v)$
= $\min(1, 148.0/148.0)$
= 1.000

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(1.0, 1.0)$
= 1.000



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	285.087	NA	NA
Area in Shell	A1	3.655	NA	NA
Area in Nozzle Wall	A2	68.700	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	225.609	NA	NA
TOTAL AREA AVAILABLE	Atot	298.963	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (493.5756 \cdot 57.7596 \cdot 1.0 + 2 \cdot 148.5 \cdot 57.7596 \cdot 1.0 \cdot (1 - 1.0))$
 $= 285.087 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 493.576(1.0 \cdot 58.5 - 1.0 \cdot 57.76) - 2 \cdot 148.5$
 $(1.0 \cdot 58.5 - 1.0 \cdot 57.7596) \cdot (1 - 1.0)$
 $= 3.655 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 146.25)(58.01 - 34.53)1.0$
 $= 68.700 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 1.000 \text{ cm}^2$



Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{lp}, h_o, H_{ubht})) \cdot (H_{ubt} - t_n) \cdot fr_2 \\ &= (2 \cdot \min(146.3, 200.0, 98.8)) \cdot (150.0 - 59.5) \cdot 1.0 \\ &= 225.609 \text{ cm}^2 \text{ Includes Bevel Area of } 46.883 \text{ cm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

$$\begin{aligned} \text{Wall Thickness for Internal/External pressures} & \quad ta = 36.0250 \text{ mm.} \\ \text{Wall Thickness per UG16(b),} & \quad tr_{16b} = 3.0000 \text{ mm.} \\ \text{Wall Thickness, shell/head, internal pressure} & \quad trb_1 = 59.2596 \text{ mm.} \\ \text{Wall Thickness} & \quad tb_1 = \max(trb_1, tr_{16b}) = 59.2596 \text{ mm.} \\ \text{Wall Thickness} & \quad tb_2 = \max(trb_2, tr_{16b}) = 3.0000 \text{ mm.} \\ \text{Wall Thickness per table UG-45} & \quad tb_3 = 9.8312 \text{ mm.} \end{aligned}$$

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb_3, \max(tb_1, tb_2)] \\ &= \min[9.831, \max(59.2596, 3.0)] \\ &= 9.8312 \text{ mm.} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(ta, tb) \\ &= \max(36.025, 9.8312) \\ &= 36.0250 \text{ mm.} \end{aligned}$$

Available Nozzle Neck Thickness = $0.875 \cdot 59.512 = 52.073 \text{ mm.} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	34.5,	Allowable	:	148.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	335.5 N./mm ²	Passed
Occasional	:	36.4,	Allowable	:	196.8 N./mm ²	Passed
Shear	:	5.4,	Allowable	:	103.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -11 °C

Weld Size Calculations, Description: B1-Tube outlet

Intermediate Calc. for nozzle/shell Welds $T_{min} \quad 19.0000 \text{ mm.}$

Results Per UW-16.1:

Required Thickness Actual Thickness
Nozzle Weld $6.0000 = \text{Min per Code } 7.0700 = 0.7 \cdot W_o \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned} &= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - tr)) \cdot S_v) \\ &= \max(0, (285.0872 - 3.6546 + 2 \cdot 148.5 \cdot 1.0 \cdot (1.0 \cdot 58.5 - 57.7596)) / 148) \\ &= 428034.66 \text{ Kg} \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A_2 + A_4 - (W_i - Can / .707)^2 \cdot fr_2) \cdot S_v \\ &= (294.3087 + 1. - 0.0 \cdot 1.0) \cdot 148 \\ &= 445656.72 \text{ Kg} \end{aligned}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$$

$$= (294.3087 + 0.0 + 1. + (173.745)) \cdot 148$$

$$= 707859.00 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (294.3087 + 0.0 + 1. + 0.0 + (173.745)) \cdot 148$$

$$= 707859.00 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 790.5756 \cdot 10.0 \cdot 0.49 \cdot 148$$

$$= 91830. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 321.0378) \cdot (150.0 - 1.5) \cdot 0.7 \cdot 148$$

$$= 1582176. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 790.5756 \cdot (58.0 - 0.0) \cdot 0.74 \cdot 148$$

$$= 804354. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (91830 + 1582176) = 1674006 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (91830 + 0 + 804354 + 0) = 896184 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (91830 + 804354 + 0) = 896184 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 1674005 Kgf, must exceed W = 428034 Kgf or W1 = 445656 Kgf
 Path 2-2 = 896183 Kgf, must exceed W = 428034 Kgf or W2 = 707859 Kgf
 Path 3-3 = 896183 Kgf, must exceed W = 428034 Kgf or W3 = 707859 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 193 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 93.5385 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 353.5385 mm.

Input Echo, WRC107/537 Item 1, Description: B1-Tube outlet:

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.5000	mm.
Vessel Diameter	Dv	1764.000	mm.
Vessel Thickness	Tv	60.000	mm.
Design Temperature	T1	260.0	C
Vessel Material		SA-387 11	
Vessel UNS Number		K11789	
Vessel Cold S.I. Allowable	Smc	148.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	148.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.



Attachment Type	Type	Round
WRC107 Attachment Classification	Holsol	Hollow
Diameter Basis for Nozzle	Nbasis	ID
Corrosion Allowance for Nozzle	Can	1.5000 mm.
Nozzle Diameter	Dn	490.576 mm.
Nozzle Thickness	Tn	150.000 mm.
Nozzle Material	SA-336 F11	
Nozzle UNS Number	K11572	
Nozzle Cold S.I. Allowable	SNmc	148.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	148.00 N./mm ²
Design Internal Pressure	Dp	191.000 bars
Include Pressure Thrust	No	

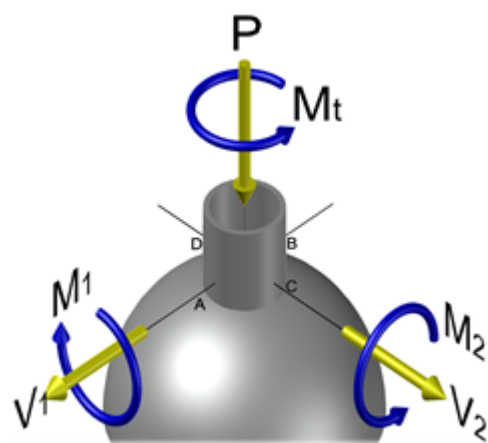
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-19579.9	Kgf
Longitudinal Shear (SUS)	(V1) V1	14684.9	Kgf
Circumferential Shear (SUS)	(Vc) V2	19579.9	Kgf
Circumferential Moment (SUS)	(Mc) M1	30.5	Kg-m.
Longitudinal Moment (SUS)	(Ml) M2	30.5	Kg-m.
Torsional Moment (SUS)	Mt	35.2	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))} \\
 &= 790.576 + 2 \cdot 1.65 \cdot \sqrt{(912.75 (60.0 - 1.5))} \\
 &= 1553.125 \text{ mm.}
 \end{aligned}$$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load	P	-19579.9	Kgf
Circumferential Shear	(VC) V2	19579.9	Kgf
Longitudinal Shear	(VL) V1	14684.9	Kgf
Circumferential Moment	(MC) M1	30.5	Kg-m.
Longitudinal Moment	(ML) M2	30.5	Kg-m.
Torsional Moment	MT	35.2	Kg-m.



Dimensionless Param: U = 1.71 TAU = 5.00 (2.16) RHO = 0.39

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SP 1	0.02219	(A,B,C,D)
M(x) / P	SP 1	0.02041	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SM 1	0.02065	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SM 1	0.02391	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SM 1	0.02065	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SM 1	0.02391	(A,B,C,D)
N(y) * T / P	SP 1	0.03090	(A,B,C,D)
M(y) / P	SP 1	0.00576	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SM 1	0.02557	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SM 1	0.00760	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SM 1	0.02557	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SM 1	0.00760	(A,B,C,D)

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress Load		Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2
Rad.	Bend. P	6.9	-6.9	6.9	-6.9	6.9	-6.9	6.9	-6.9
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-0.1	0.1	0.1	-0.1
Rad.	Memb. ML	-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-0.1	0.1	0.1	-0.1	0.0	0.0	0.0	0.0
Tot.	Rad. Str.	8.1	-5.6	8.2	-5.7	8.1	-5.6	8.2	-5.7
Tang.	Memb. P	1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7
Tang.	Bend. P	1.9	-1.9	1.9	-1.9	1.9	-1.9	1.9	-1.9
Tang.	Memb. MC	0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Tang.	Bend. MC	0.0	0.0	0.0	0.0	-0.0	0.0	0.0	-0.0
Tang.	Memb. ML	-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tang.	Bend. ML	-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Tang. Str.		3.6	-0.2	3.7	-0.2	3.6	-0.2	3.7	-0.2
	Shear VC	2.6	2.6	-2.6	-2.6	0.0	0.0	0.0	0.0
	Shear VL	0.0	0.0	0.0	0.0	-2.0	-2.0	2.0	2.0
	Shear MT	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tot. Shear		2.6	2.6	-2.6	-2.6	-2.0	-2.0	2.0	2.0
Str. Int.		9.3	7.6	9.4	7.6	8.8	6.7	8.9	6.8

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

[illegible]



Long. P1 (SUS)	1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7
Long. Q (SUS)	1.9	-1.9	2.0	-2.0	1.9	-1.9	2.0	-2.0
Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear P1 (SUS)	2.6	2.6	-2.6	-2.6	-2.0	-2.0	2.0	2.0
Shear Q (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)	144.6	144.6	144.6	144.6	144.6	144.6	144.6	144.6
Pm+P1 (SUS)	148.8	148.8	148.8	148.8	148.1	148.1	148.1	148.1
Pm+P1+Q (Total)	153.9	145.5	154.0	145.5	153.5	145.1	153.6	145.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	144.65	148.00	Passed
Pm+P1 (SUS)	148.80	222.00	Passed
Pm+P1+Q (TOTAL)	154.05	443.99	Passed

*Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.*

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Input Echo, Tubesheet Number 1, Description:

Shell Data:

Main Shell Description: Main Shell-2

Shell Maximum Design Pressure	Psd,max	191.00 bars
Shell Maximum Operating Pressure	Psox,max	191.00 bars
Shell Minimum Operating Pressure	Psox,min	0.00 bars
Shell Thickness	ts	122.0000 mm.
Shell Internal Corrosion Allowance	cas	1.5000 mm.
Shell External Corrosion Allowance	casext	0.0000 mm.
Inside Diameter of Shell	Ds	1700.000 mm.
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	260.00 °C
Shell Material		SA-387 11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K11789
Shell Allowable Stress at Temperature	Ss	148.00 N./mm ²
Shell Allowable Stress at Ambient		148.00 N./mm ²

Channel Data:

Channel Description: Bottom Dish end

Channel Type:		Hemispherical Head
Channel Maximum Design Pressure	Ptd,max	191.00 bars
Channel Maximum Operating Pressure	Ptox,max	191.00 bars
Channel Minimum Operating Pressure	Ptox,min	0.00 bars
Channel Thickness	tc	71.0000 mm.
Channel Internal Corrosion Allowance	cac	1.5000 mm.
Channel External Corrosion Allowance	cacext	0.0000 mm.
Inside Diameter of Channel	Dc	1764.000 mm.
Channel Design Temperature	TEMPC	260.00 °C
Channel Flange Matl		SA-387 11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K11789
Channel Allowable Stress at Temperature	Sc	148.00 N./mm ²
Channel Allowable Stress at Ambient		148.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	5139
Tube Wall Thickness	et	1.6500 mm.
Tube Outside Diameter	D	12.7000 mm.
Total Straight Tube Length	Lt	12910.00 mm.
Straight Tube Length (bet. inner tubsht faces) L		12610.00 mm.
Design Temperature of the Tubes		260.00 °C
Tube Material		SA-213 T11
Tube Material UNS Number		K11597
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		111.40 N./mm ²
Tube Allowable Stress At Ambient		118.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	167.80 N./mm ²
Tube Pitch (Center to Center Spacing)	P	19.0500 mm.
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	1.6500 mm.
Groove Weld Leg	ag	1.6500 mm.
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	733.019 mm.
Largest Center-to-Center Tube Distance	Ul	0.0000 mm.
Length of Expanded Portion of Tube	ltx	0.0000 mm.
Tube-side pass partition groove depth	hg	8.0000 mm.
Area of Tube Layout	Ap	15789.6 cm ²
Perimeter of Tube Layout	Cp	4706.0 mm.

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature T 260.00 °C
 Tubesheet Material SA-336 F11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K11572
Tubesheet Allowable Stress at Temperature	S	148.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	148.00 N./mm ²
Thickness of Tubesheet	h	150.0000 mm.
Tubesheet Corr. Allowance (Shell side)	Cats	1.5000 mm.
Tubesheet Corr. Allowance (Channel side)	Catc	1.5000 mm.
Tubesheet Outside Diameter	A	1944.000 mm.

Area of the Untubed Lanes	AL	0.0 cm ²
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Additional Data for Fixed/Floating Tubesheet Exchangers:

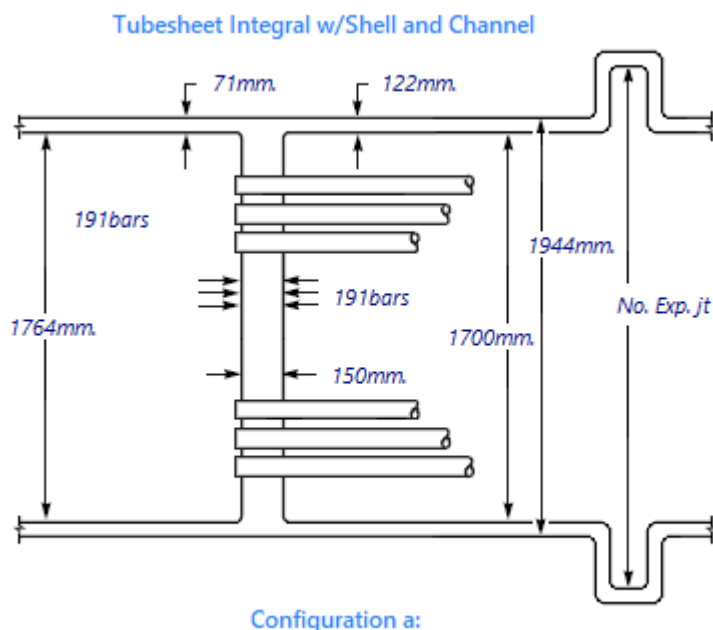
Unsupported Tube Span under consideration	l	535.000 mm.
Tube End condition corresponding to Span (l)	k	1.00

Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		Yes
Differential Design Pressure	Pdif	20.0000 bars
Run Multiple Load Cases		Yes

Mean Shell Metal Temp. along Shell len.	Tsm	0.00 °C
Mean Tube Metal Temp. along Tube length	Ttm	0.00 °C
Junction Stress Reduction option		Perform Plastic Calculation

Tubesheet Integral with		Both
Tubesheet Extended as Flange		No

Is Exchanger in Creep range (skip EP, Use 3S for Sps)		No
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ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:



Shell - TE-1 Carbon & Low Alloy Steels, Group 1
 Shell - TM-1 Material Group C

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	260.0 °C	0.18891E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20547E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
 Channel - TM-1 Material Group C

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	260.0 °C	0.18891E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
 Tubes - TM-1 Material Group C

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	260.0 °C	0.18891E+09 KPa.
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20547E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	260.0 °C	0.18891E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
 TubeSheet - TM-1 Material Group C

Elastic Mod. at Design Temperature	260.0 °C	0.18891E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
 Please insure these properties are consistent with the
 type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Differential pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (20.0 \cdot (12.7/2 - 0)) / (111.4 \cdot 1.0 + 0.4 \cdot 20.0)$
 $= 0.1132 + 0.0000 = 0.1132 \text{ mm.}$

Tube Required Thickness under External Pressure (Differential pressure) :

External Pressure Chart CS-2 at 260.00 °C
 Elastic Modulus for Material 186156960.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.6500	12.70	12610.00	7.70	50.0000	0.0185675	110.97
EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 219.9734 \text{ bars}$						

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.3108	12.70	12610.00	40.87	50.0000	0.0006586	61.31
EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 61.3063) / (3 \cdot 40.8678) = 20.0004 \text{ bars}$						

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.3108 mm.
Allowable Internal Pressure at Corroded thickness	323.02 bars
Required Internal Design Pressure	20.00 bars
Allowable External Pressure at Corroded thickness	219.97 bars
Required External Design Pressure	20.00 bars
Required Thickness due to Shell Side pressure	0.3108 mm.

Note: The tubes were evaluated using the specified differential pressure.

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)



Note: Shell/Channel Design Pressures modified by Input differential pressure:
Pdif = 20.000 Ps = 191.000 Pt = 171.000 bars

Results for ASME Fixed Tubesheet Calculations for Configuration a:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio [p]:

$$\begin{aligned} &= l_{tx} / h \text{ (modified for corrosion if present)} \\ &= 0/150.0 \\ &= 0.0000 \text{ (must be } 0 \leq p \leq 1 \text{)} \end{aligned}$$

Effective tube hole diameter [d*]:

$$\begin{aligned} &= \max(dt - 2 \cdot tt(Et/E) \cdot (StT/S) \cdot (\rho); dt - 2 \cdot tt) \\ &= \max(12.7 - 2 \cdot 1.65 (0.18891E+09/1.8891E+09) \cdot \\ &\quad (111/147) \cdot (0); 12.7 - 2 \cdot 1.65) \\ &= 12.7000 \text{ mm.} \end{aligned}$$

Equivalent outer tube limit circle diameter [Do]:

$$\begin{aligned} &= 2 \cdot ro + dt \\ &= 2 \cdot 733.019 + 12.7 \\ &= 1478.738 \text{ mm.} \end{aligned}$$

Determine the basic ligament efficiency for shear [μ]:

$$\begin{aligned} &= (p - dt) / p \\ &= (19.05 - 12.7) / 19.05 \\ &= 0.333 \end{aligned}$$

Equivalent outer tube limit radius [ao]:

$$\begin{aligned} &= Do / 2 \\ &= 1478.738/2 \\ &= 739.3690 \text{ mm.} \end{aligned}$$

Effective tube pitch [p*]:

$$\begin{aligned} &= p / \sqrt{ (1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) } \\ &= 19.05 / \sqrt{ (1 - 4 \cdot \min(0 \cdot 100.0 ; 4 \cdot 1478.738 \cdot 19.05) / (\pi \cdot 1478.738^2)) } \\ &= 19.0500 \text{ mm.} \end{aligned}$$

Effective ligament efficiency for bending [μ*]:

$$\begin{aligned} &= (p* - d*) / p* \\ &= (19.05 - 12.7) / 19.05 \\ &= 0.3333 \end{aligned}$$

Ratio [ps]:

$$\begin{aligned} &= as / ao \\ &= 850.0/739.369 \\ &= 1.1496 \end{aligned}$$

Ratio [pc]:

$$\begin{aligned} &= ac / ao \\ &= 882.0/739.369 \\ &= 1.1929 \end{aligned}$$

Compute parameter [xt]:

$$\begin{aligned} &= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot ao))^2 \\ &= 1 - 5139 ((12.7 - 2 \cdot 1.65) / (2 \cdot 739.369))^2 \\ &= 0.7923 \end{aligned}$$

Determine parameter [xs]:

$$\begin{aligned} &= 1 - Nt(dt / (2 \cdot ao))^2 \\ &= 1 - 5139 (12.7 / (2 \cdot 739.369))^2 \\ &= 0.6209 \end{aligned}$$

Determine the value [h'g]:



= max((hg - CATC); 0) (For pressure only cases)
 = max((8.0 - 0); 0)
 = 8.000 mm.

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

= $\pi \cdot ts (Ds + ts) Es / L$
 = $\pi \cdot 122.0 (1700.0 + 122.0) \cdot 1.8891E+09 / 12610.0$
 = 1066849.1250 Kg/mm.

Axial Tube Stiffness [Kt]:

= $\pi \cdot tt (Dt - tt) Et / L$
 = $\pi \cdot 1.65 (12.7 - 1.65) \cdot 1.8891E+09 / 12610.0$
 = 87.5066 Kg/mm.

Stiffness Factor [Ks,t]:

= $Ks / (Nt \cdot Kt)$
 = $1066849 / (5139 \cdot 87.507)$
 = 2.37238

Rigidity Ratio [J]:

= $1 / (1 + Ks/Kt)$
 = $1 / (1 + 1066849/0)$
 = 1.0000 (= 1 if No Exp. Jt.)

Shell coefficient [β_s]:

= $((12(1 - \nu_s^2))^{1/4}) / ((Ds + ts) ts)$
 = $((12(1 - 0.3^2))^{1/4}) / ((1700.0 + 122.0) 122.0)$
 = 0.0039 1/mm.

Shell coefficient [ks]:

= $\beta_s \cdot Es \cdot ts^3 / (6(1 - \nu_s^2))$
 = $0.004 \cdot 0.18891E+09 \cdot 122.0^3 / (6(1 - 0.3^2))$
 = 2422433280.0000 bars·mm.²

Shell coefficient [λ_s]:

= $6(Ds + ts) \cdot ks / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$
 = $6(1700.0 + 122.0) \cdot 0.24224E+1 / 150.0^3 \cdot (1 + 150.0 \cdot 0.004 + (150.0 \cdot 0.004)^2 / 2)$
 = 13696895.0000 bars

Shell coefficient [δ_s]:

= $Ds^2 / (4 \cdot Es \cdot ts) \cdot (1 - Ds \cdot \nu_s / (2(Ds + ts)))$
 = $1700.0^2 / (4 \cdot 0.18891E+09 \cdot 122.0) \cdot (1 - 1700.0 \cdot 0.3 / (2(1700.0 + 122.0)))$
 = 0.26959E-01 mm./N./mm²

Channel coefficient [β_c]:

= $((12(1 - \nu_c^2))^{1/4}) / ((Dc + tc) tc)$
 = $((12(1 - 0.3^2))^{1/4}) / ((1764.0 + 71.0) 71.0)$
 = 0.0050 1/mm.

Channel coefficient [kc]:

= $\beta_c \cdot Ec \cdot tc^3 / (6(1 - \nu_c^2))$
 = $0.005 \cdot 0.18891E+09 \cdot 71.0^3 / (6(1 - 0.3^2))$
 = 623669248.0000 bars·mm.²

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

CqDelta = CqTheta = CmTheta = CmDelta = 1

Channel coefficient [λ_c]:

= $6(Dc + tc / \cos(\theta)) \cdot kc / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta)$
 = $6(1764.0 + 71.0 / \cos(0)) \cdot 0.62367E+09 / 150.0^3 \cdot (1.0 + 150.0 \cdot 0.005 \cdot (1.0 + 1.0) / 2 + (150.0 \cdot 0.005)^2 / 2 \cdot 1.0)$
 = 4152072.5 bars

Channel coefficient [δ_c]:

= $Dc^2 / (4 \cdot Ec \cdot Tc) \cdot (1/2 - Dc \cdot \nu_c / (2(Dc + tc)))$
 = $1764.0^2 / (4 \cdot 0.18891E+09 \cdot 71.0) \cdot (1/2 - 1764.0 \cdot 0.3 / (2(1764.0 + 71.0)))$
 = 0.20635E-01 mm./N./mm²



Channel coefficient [zetac]:
zetac = 0

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$h/p = 7.874016$; $\mu^* = 0.333333$; $E^* = 61414216$ KPa.
 $E^*/E = 0.325091$; $\nu^* = 0.334167$

Note: As h/p (7.874) is > 2 , data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor [Xa]:
 $= ((24(1 - \nu^{*2})Nt \cdot Et \cdot tt(dt - tt)ao^2) / (E^* \cdot L \cdot H^3))^{1/4}$
 $= ((24(1 - 0.334^2)5139 \cdot 0.18891E+09 \cdot 1.65$
 $(12.7 - 1.65)739.369^2) / (61414216 \cdot 12610.0 \cdot 150.0^3))^{1/4}$
 $= 2.9806$

Values from Table 4.18.3

$Zd = 0.056202$; $Zv = 0.108292$; $Zm = 0.510753$
 $Za = 0.208360E+01$; $Zw = 0.108292$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:
 $= A / Do$
 $= 1944.0/1478.738$
 $= 1.3146$

Compute coefficient [F]:
 $= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$
 $= (1 - 0.33) / 61414216 \cdot (13696895 + 4152072 +$
 $0.18891E+09 \cdot \ln(1.31))$
 $= 19.9116$

Compute parameter [ϕ]:
 $= (1 + \nu^*) F$
 $= (1 + 0.3342) 19.9116$
 $= 26.5654$

Compute parameter [Q1]:
 $= (ps - 1 - \phi \cdot Zv) / (1 + \phi \cdot Zm)$
 $= (1.1496 - 1 - 26.5654 \cdot 0.1083) / (1 + 26.5654 \cdot 0.5108)$
 $= -0.187200$

Compute Parameter [Qz1]:
 $= (Zd + Q1 \cdot Zw) / 2 \cdot Xa^4$
 $= (0.0562 + -0.1872 \cdot 0.10829) / 2 \cdot 2.98064^4$
 $= 1.4180$

Compute parameter [Qz2]:
 $= (Zv + Q1 \cdot Zm) / 2 \cdot Xa^4$
 $= (0.10829 + -0.1872 \cdot 0.51075) / 2 \cdot 2.98064^4$
 $= 0.5004$

Compute parameter [U]:
 $= (Zw + (ps - 1) Zm) Xa^4 / (1 + \phi \cdot Zm)$
 $= (0.1083 + (1.1496 - 1) 0.5108) 2.98064^4 / (1 + 26.5654 \cdot 0.5108)$
 $= 1.0008$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [yb]:
 $= 0$ (config a)

Compute Parameter [γ]:
 $= 0.000$ mm. (For Pressure only cases)

Calculate parameter [ω_s]:
 $= ps \cdot ks \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$
 $= 1.1496 \cdot 0.24224E+1 \cdot 0.0039 \cdot 0.026959 (1 + 150.0 \cdot 0.0039)$



$$= 45693.0781 \text{ mm}^2$$

Calculate parameter [$\omega*s$]:

$$\begin{aligned} &= ao^2(\rho_s^2 - 1) \cdot (\rho_s - 1) / 4 - \omega s \\ &= 739.369^2(1.15^2 - 1) \cdot (1.15 - 1) / 4 - 45693.078 \\ &= -39115.6289 \text{ mm}^2 \end{aligned}$$

Parameter [ωc]:

$$\begin{aligned} &= \rho_c \cdot kc[\delta C(CmTheta \cdot \beta C + Cq\Delta \cdot h \cdot \beta C^2) - \\ &\quad \zeta c(CmTheta + h \cdot CqTheta \cdot h \cdot \beta C / 2)) \\ &= 1.193 \cdot 0.62367E+09[0.020635 (1.0 + 0.005 \cdot 1.0 \cdot 150.0) - \\ &\quad 0 (1.0/150.0 + 1.0 \cdot 0.005/2)) \\ &= 13573.1934 \text{ mm}^2 \end{aligned}$$

Parameter [$\omega*c$]:

$$\begin{aligned} &= ao^2[(\rho_c^2 + 1) \cdot (\rho_c - 1) / 4 - (\rho_s - 1) / 2] - \omega c \\ &= 739.36902^2[(1.19291^2 + 1) \cdot (1.19291 - 1) / 4 - (1.14963 - 1) / 2] - 13573.19336 \\ &= 9409.6270 \text{ mm}^2 \end{aligned}$$

Pressure [$P*S$]:

$$= 0 \text{ For Pressure only cases or Configurations d,e,f,A,B,C,D}$$

Pressure [$P*C$]:

$$= 0 \text{ For Pressure only cases or Configurations b,c,d,B,C,D}$$

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [$P's$]:

$$\begin{aligned} &= Ps\{ xs + 2(1 - xs) vt + [2/Kst \cdot (Ds/Do)^2] vs - \\ &\quad [(\rho_s^2 - 1) / (J \cdot Kst)] - [(1 - J) / (2J \cdot Kst)] [(Dj^2 - (Ds)^2) / Do^2] \} \\ &= 191.0 \{ 0.621 + 2(1 - 0.621) 0.3 + [2/2.372 \cdot \\ &\quad (1700.0/1478.738)^2] 0.3 - [(1.15^2 - 1) / (1.0 \cdot 2.372)] - \\ &\quad [(1 - 1.0) / (2 \cdot 1.0 \cdot 2.372)] [(0^2 - (1700.0)^2) / 1478.738^2] \} \\ &= 199.9878 \text{ bars} \end{aligned}$$

Pressure [$P't$]:

$$\begin{aligned} &= [xt + 2(1 - xt) vt + 1 / (J \cdot Kst)] Pt \\ &= [0.792 + 2(1 - 0.792) 0.3 + 1 / (1. \cdot 2.372)] 171.0 \\ &= 228.8758 \text{ bars} \end{aligned}$$

Pressure [P_Y]:

$$\begin{aligned} &= Nt \cdot Kt \cdot \gamma / (\pi \cdot ao^2) \\ &= 5139 \cdot 87.507 \cdot 0 / (3.142 \cdot 739.369^2) \\ &= 0.0000 \text{ bars} \end{aligned}$$

Pressure [P_w]:

$$\begin{aligned} &= -\gamma b \cdot U \cdot W* / (2 \cdot \pi \cdot ao^2) \\ &= -0 \cdot 1.001 \cdot 0 / (2 \cdot \pi \cdot 739.369^2) \\ &= -0.0000 \text{ bars} \end{aligned}$$

Calculate the pressure [$Prim$]:

$$\begin{aligned} &= -(U/ao^2) \cdot (\omega*s \cdot Ps - \omega*c \cdot Pt) \\ &= -(1.001/739.369^2) \cdot (-39115.629 \cdot 191.0 - 9409.627 \cdot 171.0) \\ &= 16.6228 \text{ bars} \end{aligned}$$

Calculate the pressure [P_w]:

$$\begin{aligned} &= (U/ao^2) \cdot (\omega s \cdot P*s - \omega c \cdot P*c) \\ &= (1.001/739.369^2) \cdot (45693.0781 \cdot 0 - 13573.1934 \cdot 0) \\ &= 0.0000 \text{ bars} \end{aligned}$$

Determine the effective pressure [Pe]:

$$\begin{aligned} &= J \cdot Kst / (1 + J \cdot Kst(Qz1 + (\rho_s - 1) Qz2)) \cdot \\ &\quad (P's - P't + P_Y + P_w + P_w + Prim) \\ &= 0.1000E+01 \cdot 2.372 / (1 + 1.0 \cdot 2.372 (1.418 + (1.15 - \\ &\quad 1) 0.5)) \cdot (199.988 - 228.876 + 0 + 0 + 0 + 16.623) \\ &= -6.4070 \text{ bars} \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [$Q2$]:



$$= [((\omega * s \cdot P_s - \omega * c \cdot P_t) - (\omega_s \cdot P * s - \omega_c \cdot P * c)) + W * \cdot y_b / (2 \cdot \pi)] / (1 + \phi \cdot Z_m)$$

$$= [((-39115.629 \cdot 191.0 - 9409.627 \cdot 171.0) - (45693.078 \cdot 0 - 13573.193 \cdot 0)) \cdot 0.01 + 0 \cdot 0 / (2 \cdot \pi)] / (1 + 26.56536 \cdot 0.51075)$$

$$= -6355.9233 \text{ Kg}$$

Calculate factor [Q3]:

$$= Q1 + 2 \cdot Q2 / (Pe \cdot a_o^2)$$

$$= -0.187 + 2 \cdot -6355.923 / (-6.407 \cdot 739.369^2)$$

$$= 0.16871$$

Fm Value from Table 4.18.3 = 0.148566

Tubesheet Bending Stress - original thickness [σ]:

$$= (1.5 \cdot F_m / \mu *) \cdot (2 \cdot a_o / (H - h'g))^2 \cdot Pe$$

$$= (1.5 \cdot 0.1486 / 0.3333) \cdot (2 \cdot 739.369 / (150.0 - 8.0))^2 \cdot -6.41$$

$$= -46.4532 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot S = 1.5 \cdot 148.0 = 222.0 \text{ N./mm}^2$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$= (1.5 \cdot F_m / \mu *) \cdot (2 \cdot a_o / (h - h'g))^2 \cdot Pe$$

$$= (1.5 \cdot 0.0213 / 0.3333) \cdot (2 \cdot 739.369 / (25.412 - 8.0))^2 \cdot -3.21$$

$$= -221.9922 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 25.4116 + 0 + 0$$

$$= 25.4116 \text{ mm.}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [Tlimit]:

$$= (2 \cdot \mu \cdot h / a_o) \cdot \min(0.8S; 0.533S_y)$$

$$= (2 \cdot 0.333 \cdot 150.0 / 739.369) \cdot 118.398$$

$$= 16.0134 \text{ N./mm}^2$$

Shear Stress is not required to be computed; [Pe] <= Taulimit

Tubesheet Shear Stress is probably low, use the following req. thk:

$$\text{Tubesheet thickness (Incl. Corr.)} = 3.8100 \text{ mm.}$$

$$\text{Tubesheet Shear Stress} = 28.4145 \text{ N./mm}^2$$

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd] :

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(25.4116 ; 3.81)$$

$$= 25.4116 \text{ mm.}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The Ftmin and Ftmax Coefficients from Table 4.18.4:

$$F_{tmin} = -0.8338, F_{tmax} = 2.939$$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - Pe \cdot F_{tmin}) / (X_t - X_s)$$

$$= ((191.0 \cdot 0.6209 - 171.0 \cdot 0.7923) - (-6.407) \cdot -0.834) / (0.7923 - 0.6209)$$

$$= -12.9720 \text{ N./mm}^2$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - Pe \cdot F_{tmax}) / (X_t - X_s)$$

$$= ((191.0 \cdot 0.6209 - 171.0 \cdot 0.7923) - (-6.407) \cdot 2.939) / (0.7923 - 0.6209)$$

$$= 1.1320 \text{ N./mm}^2$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

$$= \max(| \sigma_{t,1} | ; | \sigma_{t,2} |) = 12.972 \text{ N./mm}^2$$



Allowable Tube Stress [σ_t, A]:
= Sot = 111.3993 N./mm²

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:
= max((3.25 - 0.25(Z_d + Q₃ • Z_w)X_a⁴); 1.25)
= max((3.25 - 0.25(0.056 + 0.169 • 0.108)2.981⁴); 1.25)
= 1.7805 (Should be ≤ 2)

Determine the factor [rt]:
= $(\sqrt{ \frac{dt^2 + (dt - 2 \cdot tt)^2 }{4} })$
= $(\sqrt{ \frac{12.7^2 + (12.7 - 2 \cdot 1.65)^2 }{4} })$
= 3.9501 mm.

Determine the factor [Ct]:
= $\sqrt{ \frac{2 \cdot \pi^2 \cdot Et}{Syt} })$
= $\sqrt{ \frac{2 \cdot \pi^2 \cdot 0.18891E+09}{167} })$
= 149.0780

Determine the factor [Ft]:
= k • L/r
= 1.0 • 535.0/3.95
= 135.4403

Buckling Allowable Stress [Stb]:
= (Sy/t/Fs) • (1 - Ft/(2 • Ct))
= (167/1.78) • (1 - 135.44/(2 • 149.078))
= 51.432 N./mm² (Never greater than Sot)

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:
= $\sigma_{t,max} \cdot \text{Tube Area}$
= 12.97 • 0.5728
= 75.77 Kgf

Tube Metal Area [At]:
= $\pi(d_o - t)t$
= $\pi(12.7 - 1.65)1.65$
= 0.573 cm²

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:
= $\pi \cdot t(d_o - t)S$
= $\pi \cdot 1.65 (12.7 - 1.65)111.399$
= 650.654 Kgf

Groove Weld Strength [Fg]:
= $0.85 \cdot \pi \cdot a_g(d_o + 0.67 \cdot a_g)S_w$ (but not > Ft)
= $0.85 \cdot \pi \cdot 1.65 (12.7 + 0.67 \cdot 1.65)111.4$
= 650.6538 Kgf

Weld Strength Factor [fw]:
= S / S_w where S_w = min(S and St)
= 111.399/111.399
= 1.000

Design Strength Ratio [fd]:
= 1 for full strength welds

Fillet Weld Strength [Ff]:
= $0.55 \cdot \pi \cdot a_f(d_o + 0.67 \cdot a_f)S_w$ (but not > Ft)
= $0.55 \cdot \pi \cdot 1.65 (12.7 + 0.67 \cdot 1.65)111.4$
= 447.0977 Kgf

Minimum Length of Weld [ar]:
= $2(\sqrt{ (0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t)fw \cdot fd }) - 0.75 \cdot d_o$



$$= 2 \cdot \sqrt{(0.75 \cdot 12.7)^2 + 1.07 \cdot 1.65 \cdot (12.7 - 1.65) \cdot 1.0 \cdot 1.0} - 0.75 \cdot 12.7$$

$$= 1.949 \text{ mm.}$$

Minimum Required Fillet Weld Leg [afr]: 0.9900 mm.
 Minimum Required Groove Weld Leg [agr]: 0.9900 mm.

Tube-Tubesheet Jt allowable, 650.65 is >= tube strength 650.65 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:
 $= S_s \cdot E_{sw} = 148.0 \cdot 1.0 = 148.0 \text{ N./mm}^2$

Axial Membrane Stress in Shell [σ_s, m]:
 $= a_o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$
 $= 739.369^2 / ((1700.0 + 122.0) 122.0) \cdot [-6.41 + (1.15^2 - 1)(191.0 - 171.0)] + 850.0^2 \cdot 171.0 / ((1700.0 + 122.0) 122.0)$
 $= 55.5906 \text{ N./mm}^2$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:
 For a given Shell thickness of 122.0 mm., the minimum Shell length adjacent to the tubesheet should be 819.741 mm.

Shell Membrane Stress due to Joint Interaction [σ_s, m]:
 $= a_o^2 / ((D_s + t_s) t_s) [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$
 $= 739.369^2 / ((1700.0 + 122.0) 122.0) [-6.41 + (1.15^2 - 1)(191.0 - 171.0)] + 850.0^2 \cdot 171.0 / ((1700.0 + 122.0) 122.0)$
 $= 55.5906 \text{ N./mm}^2$

Shell Bending Stress due to Joint Interaction [σ_s, b]:
 $= 6 \cdot k_s / t_s^2 \{ \beta_s [\delta_s \cdot P_s + a_s^2 \cdot P \cdot s / (E_s \cdot t_s)] + 6(1 - \nu^2) / (E \cdot h) (a_o / h)^3 (1 + h \cdot \beta_s / 2) [P_e (Z_v + Z_m \cdot Q_1) + 2 / a_o^2 \cdot Z_m \cdot Q_2] \}$
 $= 6 \cdot 0.24224 \text{E}+1 / 122.0^2 \{ 0.004 [0.026959 \cdot 191.0 + 850.0^2 \cdot 0 / (.23047 \text{E}+11)] + 6(1 - 0.33^2) / (61414216) (739.37 / 150.0)^3 (1 + 150.0 \cdot 0. / 2) [-6.4 (0.108 + 0.511 \cdot -0.187) + 2 / 739.37^2 \cdot 0.511 \cdot -6355.923] \}$
 $= 30.8688 \text{ N./mm}^2$

Shell Stress Summation vs. Allowable
 $:: |\sigma_{s,m}| + |\sigma_{s,b}| \leq 1.5 \cdot S_s$
 $:: |55.59| + |30.87| \leq 222.0 \text{ N./mm}^2$
 $:: 86.46, \text{ must be } \leq 222.0 \text{ N./mm}^2$

Channel Membrane Stress due to Joint Interaction [σ_c, m]:
 $= a_c^2 \cdot P_t / ((D_c + t_c) t_c \cdot \cos(\theta))$
 $= 882.0^2 \cdot 171.0 / ((1764.0 + 71.0) 71.0 \cdot \cos(0))$
 $= 102.109 \text{ N./mm}^2$

Channel Bending Stress due to Joint Interaction [σ_c, b]:
 $= 6 \cdot k_c / t_c^2 \cdot \{ (\delta_c \cdot C_m \Delta + \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t - 6((1 - \nu^2) / E \cdot h) (a_o / h)^3 \cdot (C_m \theta + h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot [P_e (Z_v + Z_m \cdot Q_1) + 2 / a_o^2 \cdot Z_m \cdot Q_2] \}$
 $= 6 \cdot 0.62367 \text{E}+09 / 71.0^2 \cdot \{ (0.020635 \cdot 1.0 \cdot 0.005 - 0 \cdot 1.0) \cdot 171.0 - 6((1 - 0.334^2) / 61414216) \cdot (739.369 / 150.0)^3 \cdot (1.0 + 150.0 \cdot 0.005 / 2 \cdot 1.0) \cdot [-6.4 (0.10829 + 0.51075 \cdot -0.187) + 2 / 739.369^2 \cdot 0.51075 \cdot -6355.923] \}$
 $= 264.3724 \text{ N./mm}^2$

Channel Stress Summation vs. Allowable
 $:: |\sigma_{c,m}| + |\sigma_{c,b}| \leq 1.5 \cdot S_c$
 $:: |102.11| + |264.37| \leq 222.0 \text{ N./mm}^2$
 $:: 366.48 \text{ must be } \leq 222.0 \text{ N./mm}^2 \text{ (See Elast-Plast iteration below)}$



Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	46.5 <=	222.0 N./mm ²	Pass
Tubesheet Shear Stress	4.3 <=	118.4 N./mm ²	Pass
Maximum Tube Stress	13.0 <=	111.4 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-13.0 <=	-51.4 N./mm ²	Pass
Maximum Force on any one Tube	75.8 <=	650.7 Kg ^f	Pass
Axial Membrane Stress in Shell	55.6 <=	148.0 N./mm ²	Pass
Shell Stress (jt. inter.)	86.5 <=	222.0 N./mm ²	Pass
Channel Stress (jt. inter.)	366.5 <=	222.0 N./mm ²	See below

Additional Results for the 2nd Elastic-Plastic Iteration.



These calculations are not intended to be used in the creep range. Check Part 4.18 in Section VIII Division 2.

Compute Allowable [Sstars]:
 = min(Sys, SPS,s/2)
 = min(251, 502/2) = 251.41 N./mm²

Compute Allowable [S*c]:
 = min(Syc, SPS,c/2)
 = min(251, 502/2) = 251.41 N./mm²

Compute Factor [facts]:
 = min([1.4 - 0.4* abs(σs,b) / Sstars], 1)
 = min([1.4 - 0.4* abs(30.87)/251.41], 1) = 1.0

Compute Factor [factc]:
 = min([1.4 - 0.4* abs(σc,b) / Sstarc], 1)
 = min([1.4 - 0.4* abs(264.37)/251.41], 1) = 0.979

Compute Modified Modulus [Ec]:
 = Ec • factc
 = 0.18891E+09 • 0.9794 = 0.18502E+09 KPa.

Tubesheet Bending Stress - final thickness [σf]:
 = (1.5 • Fm / μ*) • (2 • ao/(h - h'g))² • Pe
 = (1.5 • 0.0212/0.3333) • (2 • 739.369/(25.319 - 8.0))² • -3.19
 = -221.9898 N./mm²

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:
 = h + Cats + Catc
 = 25.3186 + 0 + 0
 = 25.3186 mm.

For the Elastic-Plastic iteration, Shell Allowable = SPS,s
 ShellAllow = 502.8217 N./mm²

ASME Section VIII-2 Part 4.18.8.4 Step 12:

For the Elastic-Platic iteration, the channel allowable stress = SPS,c
 Challow = 502.8217 N./mm²

Computations completed for ASME tubesheet configuration: a

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	46.6 <=	222.0 N./mm ²	Pass
Tubesheet Shear Stress	4.3 <=	118.4 N./mm ²	Pass



Maximum Tube Stress	13.0	<=	111.4 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-13.0	<=	-51.4 N./mm ²	Pass
Maximum Force on any one Tube	75.8	<=	650.7 Kgf	Pass
Axial Membrane Stress in Shell	55.6	<=	148.0 N./mm ²	Pass
Shell Stress (jt. inter.)	86.5	<=	502.8 N./mm ²	Pass
Channel Stress (jt. inter.)	366.5	<=	502.8 N./mm ²	Pass

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 6 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend	Stresses Allwd	Shear Allwd	Case Type	Pass/ Fail
D3uc	25.412	...	47	222	4	118 Ps+Pt-Th	D3 Ok
O3uc	4.269	...	42	503	4	118 Ps+Pt+Th	O3 Ok
O4uc	3.810	...	...	503	...	118 Ps+Pt+Th	O4 Ok
D3c	26.146	...	47	222	4	118 Ps+Pt-Th-c	D3 Ok
O3c	7.258	...	43	503	4	118 Ps+Pt+Th-c	O3 Ok
O4c	6.810	...	...	503	...	118 Ps+Pt+Th-c	O4 Ok
Max:	26.1459	... mm.	0.212	0.036	(Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+)-Th]:
 With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:
 Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Shell Axial Membrane Stress Summary:

Case#	----	Shell Stresses Ten Allwd Cmp	----	:	----	Shell Band Stress Ten Allwd Cmp	----	:	Pass Fail
D3uc	56	147	...	...	...	...	...	...	Ok
O3uc	56	502	...	...	...	...	...	...	Ok
O4uc	...	502	...	...	...	...	...	...	Ok
D3c	56	147	...	...	...	...	...	...	Ok
O3c	56	502	...	...	...	...	...	...	Ok
O4c	...	502	...	...	...	...	...	...	Ok
Max	RATIO	0.382	...	...	...	...	...	...	

Tube, Shell and Channel Stress Summary:

Case#	----	Tube Stresses Ten Allwd Cmp	Tube Loads Allwd	Shell Stress Allwd	Channel Stress Allwd	Pass Fail
D3uc	13	111	-13	-51	76 650	86 502 366 502 Ok
O3uc	13	222	-13	-51	76 1301	86 502 366 502 Ok
O4uc	...	222	...	...	...	1301 502 366 502 Ok
D3c	13	111	-13	-52	77 650	86 502 375 502 Ok



03c	13	222	-13	-52	77	1301	86	502	375	502	Ok
04c	...	222	...	...	...	1301	...	502	...	502	Ok
Max RATIO	0.118		0.252		0.118		0.172		0.746		

Summary of Thickness Comparisons for 6 Load Cases:

Thickness (mm.)	Required	Actual	P/F
Tubesheet thickness :	26.146	150.000	Ok
Tube thickness :	0.311	1.650	Ok
Tube-tubesheet fillet weld leg :	0.990	1.650	Ok
Tube-tubesheet groove weld leg :	0.990	1.650	Ok

Min Shell length of thk, (122.000) adj. to tubesheet: 819.741 mm.

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm.
 Due to Pressure Shell Compresses by : -1.348 mm.
 Due to Pressure + Thermal Shell Compresses by : -1.348 mm.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	71.726	1.000	76.280	1.000
Tubesheet Shear Stress	253.575	1.000	299.326	1.000
Tube Tensile Stress	324.006	1.000	466.204	1.000
Tube Compressive Stress	133.173	1.000	472.095	1.000
Tube-Tubesheet Joint load	324.007	1.000	466.206	1.000
Shell Stress (Axial, Junction)	71.726	0.203	76.280	0.240
Tubesheet-Channel Junction	71.726	0.355	76.280	0.040
Tube Pressure Stress	323.019	1.000	219.972	1.000
Minimum MAWP	71.726		76.280	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	71.726	1.000	76.312	1.000
Tubesheet Shear Stress	255.629	1.000	301.917	1.000
Tube Tensile Stress	356.180	1.000	515.994	1.000
Tube Compressive Stress	159.933	1.000	548.123	1.000
Tube-Tubesheet Joint load	356.180	1.000	515.994	1.000
Shell Stress (Axial, Junction)	71.726	0.160	76.312	0.191
Tubesheet-Channel Junction	71.726	0.281	76.312	0.031
Tube Pressure Stress	342.152	1.000	243.286	1.000
Minimum MAPnc	71.726		76.312	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm.)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.2639	0.2640	...	...	...	...
2	-1.3476	-1.3480	...	...	...	...
3	-1.1113	-1.1116	...	...	...	...
4	...	...	...	...	...	...
5	...	...	...	...	...	...
6	...	...	...	...	...	...
7	-1.1113	-1.1116	...	...	...	...



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Tubesheet MDMT Calculations:

Shell Side MDMT calculation:

Governing thickness on the shell side per figure UCS-66.3 (e):

$$\begin{aligned} &= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{shell thickness})) \\ &= \max(150.0/4, \min(150.0, 122.0)) \\ &= 122.000 \text{ mm.} \end{aligned}$$

MDMT reduction ratio per UCS 66 (b)(1)(b):

$$\begin{aligned} &= \min(\text{ps}/\text{Shellside MAPnc}, 1) \\ &= \min(191.0/76.31, 1) \\ &= 1.000 \end{aligned}$$

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm.. This component must be impact tested.

Channel Side MDMT calculation:

Governing Channel side thickness per figure UCS-66.3 (e):

$$\begin{aligned} &= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{channel thickness})) \\ &= \max(150.0/4, \min(150.0, 71.0)) \\ &= 71.000 \text{ mm.} \end{aligned}$$

MDMT reduction ratio per UCS 66 (b)(1)(b):

$$\begin{aligned} &= \min(\text{pt}/\text{Tubeside MAPnc}, 1) \\ &= \min(191.0/71.73, 1) \\ &= 1.000 \end{aligned}$$

Thickness Ratio = 1.0, Temperature Reduction per Fig. UCS 66.1 = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 24 °C

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